OIL INDUSTRY SAFETY DIRECTORATE

तरलीकृत प्राकृतिक गैस के प्रबंधन (एलएनजी), भंडारण और पुनर्गैसीकरण के लिए मानक

ओ आई एस डी–मानक-194

STANDARD FOR HANDLING, STORAGE & REGASIFICATION OF LIQUEFIED NATURAL GAS (LNG)

OISD-STANDARD-194

Inception	August 2000
1st Revised Edition	April 2016
2nd Revised Edition	*****2025

FOR RESTRICTED CIRULATION ONLY

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PREFACE

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for sovereignty and prosperity of the country. As the industry essentially deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as only safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January, 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (MoP&NG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders (including the public at large), drawing inputs from international standards and adapting them to Indian conditions by leveraging the experience of the constituents. These standards cover inbuilt design safety,asset integrity and best operating practices in the field of production, processing, storage and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD, are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

Executive Director
Oil Industry Safety Directorate

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The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed or withdrawn at least every ten years.

Suggestions for revision are invited from the oil and gas industry and should be submitted to OISD at **ps-ed.oisd@gov.in**.

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OISD recognize the support & cooperation of several other experts from industry, who contributed to the preparation, review and finalization of this document.

[1] Sh. Pradeep Bansal joined Dhamra LNG Terminal Pvt. Ltd. on 26.09.2024 as Chief Operating Officer
[2] Sh. Jamunalal Rout repatriated to PESO

STANDARD FOR HANDLING, STORAGE AND REGASIFICATION OF LIQUEFIED NATURAL GAS (LNG)

INDEX

1.0 INTRODUCTION

The standard provides for safety and design aspects of all the major components of LNG receiving terminals and LNG facilities including unloading, storage, re-gasification & truck loading.

This standard also outlines the operating practices & guidelines for protection of property & persons involved in handling the operations of LNG. A number of international codes and standards exist pertaining to design, operations and safety of LNG terminals and LNG facilities. This standard deals with the design and operating practices for LNG handling in India. In the interest of safety, it is important that persons engaged in handling LNG, understand the properties & the associated hazards.

2.0 SCOPE

This standard lays down minimum safety requirements for the layout, design and operation of LNG terminals, covering unloading, storage, regasification, send-out, and truck loading facilities. This standard also specifies the minimum design criteria and fire protection facilities required for LNG terminals, along with some engineering considerations for the design and installation of fire protection and safety systems.
Safety requirements for small scale LNG facilities are included in a separate chapter, while general safety provisions required for LNG road transportation and Floating Storage & Regasification Unit (FSRU)/ Floating Storage Unit (FSU) facilities are included as Annexure-II.

The following are excluded from the scope of the standard:

a) Upstream gas gathering, treating system and natural gas liquefaction facilities are not covered in the standard.
b) Portable storage containers of LNG are not a part of the standard.

3.0 DEFINITIONS

3.1. Boil Off Gas (BOG):

The vapor/ gas generated by evaporation of LNG due to heat ingress in equipment and piping containing LNG or rapid evaporation of LNG due to a temperature rise or a pressure drop.

3.2. Bunkering:

The loading of a ship's bunker or tank with fuel for use in connection with propulsion or auxiliary equipment.

3.3. Container:

A vessel, tank, portable tank, or cargo tank used for or capable of holding, storing, or transporting liquid or gas.

3.4. Container, Frozen Ground:

A container in which the maximum liquid level is below the normal surrounding grade and that is constructed essentially of natural materials, such as earth and rock, is dependent upon the freezing of water-saturated earth materials, and that has appropriate methods for maintaining its tightness or is impervious by nature.

3.5. Container, Pre-stressed Concrete:

A concrete container is considered to be pre-stressed when the stresses created by the different loading or loading combinations do not exceed allowable stresses.

3.6. Deriming:

Deriming, synonymous with defrosting or de-icing refers to the removal, by heating/evaporation/sublimation/solution, of accumulated constituents that form solids, such as water, carbon dioxide, etc. from the low-temperature process equipment.

3.7. Design Pressure:

The pressure used in the design calculations of vessel for the purpose of determining the minimum thickness of the various component parts of the vessels.

3.8. Dyke or Bund Wall:

Raised impermeable structure, able to withstand the hydrostatic pressure and temperature of the spilled liquid, around the perimeter of an impounding area for the confinement of LNG or other hazardous liquid spills, usually associated with storage areas.

3.9. Emergency Release Coupler (ERC):

The coupler fitted in each hard arm together with quick-acting flanking valves so that a dry-break release can be achieved in emergency situations.

3.10. Emergency Release System (ERS):

The system which triggers the ERC process.

3.11. Emergency Shutdown System (ESD):

A system that safely and effectively stops whole plant or individual unit during abnormal situation or in emergency.

3.12. Failsafe:

Design features which will result in a safe condition in the event of a malfunction or failure of power, instrument air, components, or control devices.

3.13. Fired Equipment:

Any equipment in which the combustion of fuels takes place. Included among others are fired boilers, fired heaters, internal combustion engines, certain integral heated vaporisers, the primary heat source for remote heated vaporisers, gas-fired oil foggers, fired regeneration heaters and flared vent stacks.

3.14. Fixed-Length Dip Tube:

A pipe that has a fixed open end fitted inside a container at a designated elevation that is intended to show a liquid level.

3.15. Floating Storage and Regasification Unit (FSRU):

A Floating Storage and Regasification Unit (FSRU), is an LNG regasification terminal whose main structure is a special ship which is capable of transporting, storing, and re-gasifying LNG on board. Floating regasification also requires either an offshore terminal, which typically includes a buoy and connecting undersea pipelines to transport regasified LNG to shore, or an onshore dockside receiving terminal.

3.16. Floating Storage Unit (FSU):

FSUs are similar to FSRUs, except FSU vessels do not contain regasification capabilities.

3.17. Hazardous Fluid:

Any liquid or gas that is flammable or toxic or corrosive.

3.18. Ignition Source:

Any item or substance capable of an energy release of type and magnitude sufficient to ignite any flammable mixture of gases or vapours.

3.19. Impounding Area:

Area that may be defined through the use of dykes or the topography at the site for the purpose of containing any accidental spill of LNG or other hazardous liquid.

3.20. Impounding Basin:

Container or leak-tight area connected to an impounding area or spill collection area where LNG or other hazardous liquid spills can be collected and safely confined and controlled.

3.21. Liquefied Natural Gas (LNG):

Colourless and odourless cryogenic fluid in the liquid state composed predominantly of methane and can contain minor quantities of ethane, propane, butane, nitrogen, or other components normally found in natural gas.

3.22. LCNG:

LCNG means Compressed Natural Gas (CNG) produced at the fuelling station from LNG by pumping and vaporisation.

3.23. LNG Facility:

Facility used for unloading, storage, vaporization, transfer or dispensation or handling of LNG/ LCNG.

3.24. LNG Terminal:

A group of one or more units/ facilities handling LNG i.e., unloading, storage, re-gasification, pipeline transfer, loading or liquid dispatch etc. along with associated systems like utilities, blow down, flare system, fire water storage and fire water network, control room and administration service buildings like workshop, fire station, laboratory, canteen etc.

3.25. Maximum Allowable Working Pressure:

The maximum pressure permissible at the top of an equipment, a container or a pressure vessel while operating at design temperature.

3.26. Operating Basis Earthquake (OBE)

Maximum earthquake event for which no damage occurs, and the restart and safe operation can continue. The event should result in no loss to the operational integrity of the facility.

3.27. Primary Components:

Primary components include those whose failure would permit leakage of the LNG being stored, those exposed to a temperature between (-51°C) and (-168°C) and those subject to thermal shock. Primary components include but are not limited to the following parts of a single-wall tank or of the inner tank in a double-wall tank; shell plates, bottom plates, roof plates, knuckle plates, compression rings, shell stiffeners, manways, and nozzles including reinforcement, shell anchors, pipe tubing, forging, and bolting. These are the parts of LNG containers that are stressed to a significant level.

3.28. Process Plant:

The systems required to condition, liquefy, or vaporise natural gas in all areas of application.

3.29. Roll-over:

Rapid uncontrolled mass movement of stored liquid, correcting an unstable state of stratified liquids of different densities and resulting in significant evolution of product vapour.

3.30. Safe Shutdown Earthquake (SSE):
Maximum earthquake event for which the essential fail-safe functions and mechanisms are designed to be preserved.

3.31. Secondary Components:
Secondary components include those that will not be stressed to a significant level and their failure will not result in leakage of the LNG being stored or those exposed to the boil-off gas and having a design metal temperature of (-51°C) or higher.

3.32. Shall:
"Shall" are mandatory; any deviation shall require relaxation of rules from the Board of Directors.

3.33. Should:
"Should" indicates that the provision is recommended as a good engineering practice.

3.34. Storage Tank[1]:
A low-pressure container designed for an internal gas pressure of 15 psi or less, in accordance with API Std 620 or API Std 650.

3.35. Stratification:
When LNG of different densities is received in the same tank, there is a possibility that layers are created with a less dense liquid overlaying a heavier one. This is called stratification.

3.36. Suspended Deck:
Structure suspended from the fixed roof for supporting the internal insulation above the primary liquid container.

3.37. Truck:
Vehicle used for transporting/ carrying liquified natural gas in bulk.

3.37.1 Tank Truck:
Tank truck means a single self-propelled vehicle with a tank mounted thereon.

3.37.2 Tank Trailer:
Tank trailer means a vehicle with a tank mounted thereon or built as integral part thereof and constructed in such a manner that it has at least two axles and all its load rests on its own wheels.

3.37.3 Tank Semi-Trailer:
Tank semi-trailer" means a tank trailer constructed in such a manner that when it is drawn by a tractor by means of fifth wheel connection, some part of the load rests on the towing vehicle.

3.37.4 ISO Container:
ISO Tank Container means a tank container that includes two basic elements, the vessel and the framework, suitable for the carriage of compressed gas for conveyance by road, rail and sea, including interchange between these forms of transport and complies with requirements of ISO 1496.

3.38. Thermal Corner Protection:
A thermally insulating and liquid tight structure in the bottom annular section of a storage tank to protect the secondary liquid container against low temperatures in the event of leakage from the primary container.

[1] *Underground cryogenic concrete and spherical LNG storage tanks are not covered in this standard.*

3.39. Transfer Area:

That portion of an LNG plant containing piping systems where LNG, flammable liquids, or flammable refrigerants are introduced into or removed from the facility, such as ship unloading areas, or where piping connections are routinely connected or disconnected. Transfer areas do not include product sampling devices or permanent plant piping.

3.40. Transition Joint:

A connector fabricated of two or more metals used to effectively join piping sections or two different materials that are not amenable to usual welding or joining techniques.

3.41. Transfer System:

Includes transfer piping and cargo transfer system.

3.42. Vaporisation:

Means an addition of thermal energy for changing a liquid or semi-solid to vapour or gaseous state.

3.43. Vaporiser:

Equipment designed to introduce thermal energy in a controlled manner for changing a liquid to a vapour or gaseous state.

3.43.1 Ambient Vaporizer:

A vaporizer that derives its heat from naturally occurring heat sources, such as the atmosphere, seawater, or geothermal waters.

3.43.2 Heated Vaporizer:

A vaporizer that derives heat for vaporization from the combustion of fuel, electric power, or waste heat, such as from boilers or internal combustion engines.

3.43.3 Integral Heated Vaporizer:

A vaporizer, including submerged combustion vaporizers, in which the heat source is integral to the actual vaporizing exchanger.

3.43.4 Remote Heated Vaporizer:

A heated vaporizer in which the primary heat source is separated from the actual vaporizing exchanger, and an intermediate fluid (e.g., water, steam, isopentane, glycol) is used as the heat transport medium.

3.43.5 Process Vaporizer:

A vaporizer that derives its heat from another thermodynamic or chemical process to utilize the refrigeration of the LNG.

4.0 TERMINAL PROCESS SYSTEM

4.1 LIQUEFIED NATURAL GAS (LNG)

4.1.1 Natural gas is liquefied at a temperature in the range of -162°C to -168°C at atmospheric pressure to facilitate transportation in the form of Liquefied Natural Gas (LNG) in cryogenic tankers. After regasification it is used by consumers.

LNG is a colourless, odourless, low density and slightly viscous liquid. The main characteristic of LNG is that its specific volume is nearly 600 times less than that of natural gas. Owing to this characteristic, greater quantities can be stored/ transferred in liquid phase than in gaseous phase.

4.1.2 Upon release from containment to the atmosphere, LNG will vaporise and release gas which, at ambient temperature, will have about 600 times the volume of liquid vaporised. Generally, at temperature below -112°C, this gas is heavier than ambient air at 15.6°C. However, as its temperature rises, it becomes lighter than air. The predominant component of LNG is methane.

Typical LNG Composition Range (mole %)

COMPONENTS	
C_1	83 to 99
C_2	0.3 to 12.0
C_3	0.2 to 3.6
C_4	0.1 to 1
C_5	0 to 0.1
N_2	0.03 to 1.40

Typical values of molecular weight and calorific value of LNG

Molecular weight	16 to 20
Gross calorific value	8800 kcal/scm to 10800 kcal/scm

4.1.3 Liquefied natural gas chain consists of

- Production of natural gas from fields and transportation to liquefaction plant.
- Natural gas liquefaction plant and ship loading facility.
- Ships for carrying LNG between the liquefaction plant & receiving terminal.
- LNG receiving, storage and regasification.
- Regasification & dispatch into pipeline system including distribution network.
- LNG loading facilities and transportation of LNG by means of trucks including ISO containers via. roadways, railways, seaways, including inland waterways to the consumer's end subject to regulatory approvals.

4.2 LNG RECEIVING TERMINAL:

The purpose of LNG receiving terminal is to unload LNG tankers, store, re-gasify and send it out through the pipeline transmission network. LNG terminals can also have the facility to re-load the LNG. The LNG receiving terminal facilities are divided into four sections namely receiving, storage, regasification and send out sections (Refer typical flow scheme at the Annexure–III).

- RECEIVING SECTION
- STORAGE SECTION
- REGASIFICATION SECTION
- SEND OUT SECTION

In addition to the above, the terminal consists of various utilities, flare system, firefighting facilities, control room, loading, administrative block, laboratory, workshop, warehouse, sub-station and other associated infrastructures.

4.3 RECEIVING SECTION

Conceptual studies should be undertaken to evaluate geotechnical, navigation manoeuvring and mooring operations to determine the risks and provide for safeguards while finalizing the location of the LNG jetty facility.

The LNG ship tankers are moored and berthed along the jetty specially designed for LNG handling. LNG is pumped out of the ship tanks to the land-based storage with the help of unloading arms connected to the ship, through an insulated cryogenic pipeline.

4.3.1 THE JETTY

The jetty consists of berthing facility, unloading arms and other associated facilities.

4.3.2 BERTHS

i) The number and size of the berths are determined by the cargo quantity delivered, the size of the ships, time intervals between two ships & site conditions. The berths may be installed either parallel or perpendicular to the bank at the end of the jetty. A docking aid and piloting system helps aid the pilot and vessel master to monitor speed and distance accurately so as to manoeuvre safely alongside the jetty terminal.

ii) The berth normally includes breasting and mooring dolphins and unloading platform having unloading arms and other facilities such as drain vessels, fire and gas detection and firefighting system etc. Land access to the moored ships shall be provided.

iii) Breakwater, if required, based on the model studies and downtime analysis shall also be provided.

iv) Exclusion of ignition sources**:** No uncontrolled ignition source should be within a predetermined safe area, centred on the LNG carrier's cargo manifold. The minimum area from which all ignition sources must be excluded should be determined from the design considerations and dispersion studies envisaged in the risk analysis report.

v) Mooring layout: The jetty should provide mooring points of strength and in an array that would permit all LNG carriers using the terminal to be held alongside in all conditions of wind and currents. Facility for mooring line load monitoring is recommended to ensure that loads are balanced, and vessel safely secured.

vi) Quick Release Hooks**:** All mooring points should be equipped with quick release hooks. Multiple hook assemblies should be provided at those points where multiple mooring lines are deployed so that no more than one mooring line is attached to a single hook. A remote release system should be provided to release each hook from a safe location.

vii) The design of approach trestle, shall cover the following: -

a) Deflection for the piles
b) Placement of different pipes (fire water piping to be separately routed)
c) Gap between approach road and process piping

4.3.2.1 UNLOADING ARMS

i) Unloading arm consist of pipe length connected to each other by swivel joints, moved by hydraulic actuators. The connection of the arm end to the ship crossovers flange shall be provided with a special automatic ERC device.
During emergency this automatic device will come into operation and de-coupling system gets activated.

ii) Emergency Release System (ERS): Each unloading arm shall be fitted with an ERS system, able to be interlinked to the ship's Emergency Shut Down (ESD) system. This system shall operate in two stages; the first stage shall stop LNG pumping and close block valves in the pipelines; the second stage shall trigger automatic activation of the dry-break coupling at the ERC together with its quick-acting flanking valves.
The ERS system should conform to an accepted industry standard. The volume between flanking valves should be as low as possible. Site specific drift study shall be taken into consideration for finalizing timings for each stage.

iii) No drain shall be open to atmosphere. Provision should be given to collect the LNG from the unloading arm and associated piping to a closed system by way of providing adequately sized blow down vessel or any other suitable arrangement. Facility to send the liquid from the blow down vessel to unloading line shall be provided.

iv) The size of the arms should depend on the unloading flow rate. Usual sizes are 10" and 12" for LNG tankers upto 75,000 m^3 capacity and 16" to 20" for larger vessels of 120,000 m^3 to 266,000 m^3 (Q-max) capacity.

4.3.2.2 GENERAL:

i) General cargo, other than ships' stores for the LNG tanker, shall not be handled within 30 m of the point of transfer connection while LNG is being transferred through piping systems. Ship bunkering shall not be permitted during LNG unloading operations.

ii) Vehicle traffic shall be prohibited on the berth within 30 m of the loading and unloading manifold while transfer operations are in progress. Warning signs or barricades shall be used to indicate that transfer operations are in progress.

iii) Prior to transfer, the officer in charge of vessel cargo transfer and the officer in charge of the shore terminal shall inspect their respective facilities to ensure that transfer equipment is in proper operating condition. Following this inspection, they shall meet and determine the transfer procedure, verify that adequate ship-to-shore communications exist, and review emergency procedure.

iv) Interlocking between ship and terminal control room shall be established and the control of unloading operations shall be monitored from the terminal control room.

v) Terminal Security: An effective security regime should be in place to enforce the designated ignition exclusion zone and prevent unauthorised entry of personnel into the terminal and jetty area, whether by land or by sea.

vi) Operating Limits: Operating criteria, expressed in terms of wind speed, wave height and current should be established for each jetty. Such limits should be developed according to ship size, mooring restraint and loading arm operating envelope. Separate sets of limits should be established for (a) berthing, (b) stopping cargo transfer, (c) loading arm disconnection and (d) departure from the berth.

vii) The ships should be berthed in such a way that in case of emergency the ship can sail out head-on immediately. All other instructions and procedures of Port Regulatory Authority are to be observed.

4.3.2.3 UNLOADING LINE

i) The unloading and transfer lines for LNG should have minimum number of flange joints. Consideration should be given to provide cold sensors for flanges of size 200 mm and above as well as where there are clusters of flanges.

ii) Length of the unloading line is to be kept to a minimum. In case it is not feasible, alternative options available are:

- To have additional line running parallel
- Increase size of line

iii) The unloading lines need to be kept in cold condition to avoid stress and cyclic fatigue due to frequent warm-up and cooling down operation. This should be done by one of the following methods.

- Continuous circulation of LNG (LNG goes through the unloading line and sent back to the recondenser/ vaporisation section through a special small diameter line).
- Alternatively, two unloading lines may be installed. When unloading is not taking place, this loop should be used for re-circulation for keeping the lines in cool-down condition.
- Line should be fully filled with LNG and the boil off formed is sent to the tank or to the recondenser/ vaporiser section.
- The unloading shall not be commenced till the healthiness of ERS and ESD is ensured. The hot and cold ESD testing should be carried out prior to unloading every time.

4.4 STORAGE SECTION

The storage section consists of LNG storage tanks, in-tank pumps, BOG system and BOG re-condensing facility.

4.4.1 STORAGE TANK

The primary function of storage is to receive and store LNG for providing continuous supply to regasification section. An LNG tank is designed to ensure the following functions:

4.4.1.1 LIQUID RETENTION

The storage tank shall be capable of withstanding the hydrostatic load of the liquid and low temperature of LNG. In order to meet these conditions, materials such as low carbon austenitic stainless steels, aluminium magnesium alloy, 9% Nickel ferritic steel, Invar (36% Ni steel) and pre-stressed concrete (as outer tank) can be used.

4.4.1.2 GAS TIGHTNESS

Tanks should be tight enough to prevent any evaporation losses and also to avoid ingress of air and moisture.

For concrete outer tanks, a seal coating is generally provided, to prevent natural porosity of the concrete.

4.4.1.3 THERMAL INSULATION

Thermal insulation shall be provided to:
- Limit boil-off rates (usually within 0.1 % of total volume per day).
- Avoid condensation and cold spots on the outer shell.
- Ensure personnel protection wherever required.

4.4.1.4 THERMAL STRESSES

Under normal operating conditions, the tank is subjected to variation in the temperature. Also, during start up, tank temperature is required to be brought down from ambient to cryogenic temperature. Sometimes the tank may require de-riming for various reasons like repair of internals, modifications etc. Hence, the tanks shall be capable of withstanding the temperature variation.

4.5 PIPING

4.5.1 All nozzles for the piping requirements for an LNG tank shall be from the top. Side penetration shall be avoided to minimise the risk of serious leakage. The minimum piping requirements are:

- Fill lines
- Withdrawal line (In-tank pump column)
- Boil-off line to remove LNG vapour.
- Cool down line for initial cooling of tanks during commissioning of the tank.
- Nitrogen purge lines to purge the inner tank and annular space.
- Nitrogen purging line for pump column.
- Instrument nozzles.
- Pressure make-up line.
- Pump re-circulation line.
- Purge release vent line.
- Pressure relief valve line.
- Vacuum relief line

4.5.2 LNG lines are normally fully filled lines. However, specific operating conditions can result in differential temperatures at the top and bottom of the pipe causing bowing of pipes and potential spills. Piping design should include stress analysis, expansion loops and supports as well as proper piping and equipment cool down procedures to address differential contraction covering all anticipated operation and upset conditions. Use of bellows expansion joints should be avoided in LNG lines.

4.5.3 Physical phenomenon such as surge pressure in LNG receipt and transfer lines, flashing and two-phase flow shall be addressed in the piping and equipment design. ESD valves shall be fail-safe and fire safe.

4.5.4 Piping loads and thermal expansion/ contraction of piping transferred to the tank nozzle connections should be within the maximum allowable loads, as recommended in the applicable design code/ manufacturer. Bellow expansion joints should be avoided.

4.5.5 Valves shall be designed and manufactured for cryogenic service. Extended bonnet valves used in cryogenic service shall have their stems near to vertical position.

4.5.6 LNG heats up and expands, if confined to a fixed volume. Hence any potentially blocked piping or equipment should be provided with thermal expansion relief valves with discharge to closed system.

4.5.7 Inlet piping shall be designed to minimize stratification/ layering of LNG [Stratification occurs when heavier LNG has been added at the bottom of a tank with partially filled lighter LNG or lighter LNG added at the top of the heavier LNG or due to ageing (storing for long duration) of LNG. This leads to sudden and rapid release of vapour, called Roll-Over].

This can be prevented by having two fill lines one ending at the top of the tank and other extending to the bottom, to inject denser LNG at the top and lighter LNG at the bottom. Mixing nozzles may also be used to avoid stratification.

Roll-Over conditions shall be prevented by active management of stored LNG that includes monitoring temperatures and densities, mixing the tank contents by appropriate top and bottom filling or by circulation.

Other operational preventive measures that can be considered are:

- avoiding storing significantly different qualities of LNG in the same tank.
- cycle tank usage to prevent stagnation of LNG inventories.
- adopting appropriate filling procedure considering the respective densities of the LNG.
- adopting specific processing for LNG that contain nitrogen molar fraction higher than 1%.

4.5.8 Vaporiser piping involves high flow rates, pressures as well as transition from cryogenic piping materials to carbon steel material. This could result in embrittlement failure if cold gas or liquid were to come in contact with carbon steel, in case of failure of process interlocks.

Other features of the LNG tanks are covered under **Section-6.0**

4.6 BOIL- OFF GAS & RELIQUEFACTION

4.6.1 Boil-off Gas (BOG) primarily comprises of the gas produced by the process of vaporisation of LNG in the storage tank by the heat ingress through the insulation, from the surrounding atmosphere, connected piping, pumping energy, displaced vapour due to incoming liquid, on account of run-down liquid flashing due to pressure changes and due to barometric pressure changes. BOG system consists of boil–off gas recovery from the tanks (compressor and re-condenser) and diversion to the LNG regasification/ send out system or injection into the pipeline transmission network. BOG is also used for vapour return to the ship tanks during unloading thereby avoiding pressure drop in the ship tanks. If vapour return to the ship tanks is not considered, the BOG system should be designed to handle this additional quantity. BOG system includes provision to route BOG to flare also.

4.6.2 During roll-over condition, the instantaneous boil-off gas generation is substantially high and necessary provision shall be provided to protect the tank from overpressure as well as to take care of the safe discharge.

4.6.3 BOG Recovery/ Utilisation Options:

i) Re-liquefaction & Recycle to Storage: Liquefaction process used in the LNG production plant may be used for re-liquefaction. Re-liquefaction process is less favourable compared to other facilities due to higher energy consumption.

ii) Pressurisation & Mixing with gas discharged from the Terminal: The boil–off gas is compressed to the network pressure and mixed with the re-gasified product. But while mixing, low calorific value of the boil–off gas may reduce the heating value of the network gas.

iii) Re-condensation & incorporation into the re-gasified LNG: The re-condensation of the Boil off gas is carried out by utilizing cold energy of subcooled LNG. This condensed BOG along with LNG is pressurized up to pressure above gas grid pressure before regasification. Pressurisation of recondensed BOG instead of compression of BOG in gaseous phase leads to energy savings & safer operation.

iv) As a fuel gas in power generation process or internal use.

v) The receiving terminal shall be provided with flare system to enhance the plant safety. The flaring of BOG should be done only as a final solution when the normal BOG handling system is not available.

4.7 LNG PUMPING

4.7.1 IN-TANK (PRIMARY) PUMPS

The tanks are provided with in-tank submerged pumps (retractable type), which are also known as primary pumps. These are provided in storage tanks with nozzles only at the top. Pumps as well as the electric motor are submerged in LNG. Lubrication and the cooling of the pump are done by LNG itself. In-tank pumps shall be provided with minimum continuous by-pass flow line (MCF) as a safeguard to the pumps. These pumps are installed in wells, equipped with foot valves, which can be isolated to enable pump removal for maintenance. When foot valve is closed, the column shall be purged with nitrogen gas to allow the safe removal of the pump even with liquid in the tank. Arrangement for well purge, well-draining and venting should be provided. The pumps are supplied by electrical power through nitrogen purged glands at the top of the tank. The tank should have necessary facility to lift / handle the in-tank pump at the tank top (Jib crane, Hoist etc). Pump OEM's operation and maintenance manual should be followed for safe, smooth & trouble-free operation.

4.7.2 If the network pressure is not too high, in-tank pumps alone may be sufficient to bring up to the network pressure through vaporisers. If the pipeline network pressure is high, two stage pumping may be needed which also helps in BOG re-liquefaction at intermediate pressure instead of compressing BOG vapours to the line pressure.

4.7.3 The discharge pressure of the in-tank pump is usually guided by the re-condenser pressure wherever provided. The design pressure of the pump would also consider the cool-down requirements of the ship unloading line.

4.8 SEND OUT SECTION

In send out section, LNG is pumped and brought to a pressure slightly higher than the network pressure through secondary pumps and vaporised & warmed to a temperature above 0^0C and metered before it is sent for distribution.

4.8.1 HIGH PRESSURE (SECONDARY) PUMPS

These pumps are used for pumping the LNG from the intermediate pressure to the network pressure through vaporisers. These pumps shall be provided with a minimum continuous by-pass flow line (MCF) as a safeguard to the pumps. The MCF line design should take care of erosion caused by flashing/eddies, if any. Pump OEM's operation and maintenance manual should be followed for safe, smooth & trouble-free operation.

4.8.2 VAPORISATION

4.8.2.1 Vaporisation is accomplished by the transfer of heat to LNG from water/ ambient air / process stream. In the vaporisation process, LNG is heated to its bubble point, vaporised, and then warmed up to the required temperature.

4.8.2.2 LNG vaporizers are to be designed based on the quantity of heat to be exchanged with LNG for its vaporization, maximum LNG flow rate, amount of heat available in the heating medium, lowest temperature of the heating medium, operating pressure range as well as LNG composition.

4.8.2.3 Vaporizers shall be designed for a working pressure at least equal to the maximum discharge pressure of the HP pump or the pressurized container system supplying them, whichever is greater.

4.8.2.4 Vaporiser tubes are generally fitted with fins for better heat transfer. Owing to the light weight, good conductivity, corrosion resistant strength of aluminium alloy, fin tubes are generally made of aluminium alloy.

4.8.2.5 Regasified LNG (R-LNG) outlet temperature should be monitored and controlled carefully in order to avoid any LNG or cold vapour passing into the network in view of the metallurgy and mechanical integrity of downstream pipeline.

4.8.2.6 In the case of vaporisers, where water is used as a medium, water outlet temperature shall be maintained higher than water freezing point.

4.8.2.7 Major types of vaporisers are:

i) HEATED VAPORISER

These vaporisers derive heat from the combustion of fuel, electric power, or waste heat.

a) INTEGRAL HEATED VAPORISER

They are classified as those heated vaporisers in which the heat source is integral to the actual vaporising exchanger. Submerged combustion vaporisers come under this classification.

b) REMOTE HEATED VAPORISERS

In these types of vaporisers, the primary heat source is separated from the actual vaporising exchanger and an intermediate fluid (e.g., water, steam, iso-pentane, glycol, brine etc.) is used as the heat transport medium.

ii) AMBIENT VAPORISERS

These are classified as those vaporisers, which derive heat from naturally occurring sources such as atmosphere, seawater, or geothermal water.

iii) PROCESS VAPORISERS

These vaporisers derive heat from another thermodynamic or chemical process or in such a fashion as to conserve or utilise the refrigeration from the LNG.

4.8.2.8 The four types of vaporisers which are predominantly used in LNG Terminals are:

a) OPEN RACK VAPORISER
It is a heat exchanger that uses water as the source of heat. They are generally constructed out of finned aluminium alloy tubes. Corrosion protection is provided for surfaces that come in contact with water that is sprayed on the outside of the finned tubes.

The minimum water flow for ORV should be ensured as per design during running of the ORV. A well-designed safeguarding system shall be provided to protect downstream carbon steel piping from cold breakthrough.

b) INTERMEDIATE FLUID VAPORIZERS (IFV)
Intermediate fluid vaporizer uses an Intermediate Heat Transfer Fluid (IHTF) in a closed loop to transfer heat from a heat source to the LNG for vaporization. Heat transfer for LNG Vaporization occurs in a shell and tube exchanger. Typically, Intermediate HTFs can be Glycol-water, Methanol-water, etc.

Intermediate Heat Transfer Fluid flows through the intermediate fluid vaporizers (shell and tube exchanger) where it rejects heat to vaporize LNG. Cooled IHTF is circulated using pumps. Expansion drum placed upstream of the circulating pumps is used to account for system volume contraction and expansion.

There are several options to warm the IHTF solution for recycling back into the shell and tube vaporizer, for example, a waste heat recovery system, air heater etc. These intermediate fluid vaporizers have a very compact design due to high heat transfer coefficients thus reducing space requirements.

c) SUBMERGED COMBUSTION VAPORISER

In this type, LNG flows through a tube coil fabricated from stainless steel that is submerged in a water bath. Water contained in the bath is heated by direct contact with hot effluent gases from submerged gas burner.

Submerged combustion vaporiser shall not be located in an enclosed structure/ building to avoid accumulation of hazardous products of combustion.

d) AMBIENT AIR VAPORISER
Ambient air vaporizers are the heat exchangers which vaporize LNG by using heat absorbed directly from the ambient air. These vaporizers can be of natural-draft type or forced-draft type.

4.8.2.9 SAFETY FEATURES OF VAPORISERS AND CONNECTED PIPING.

a) Vaporisers shall be designed for working pressure at least equal to the maximum discharge pressure of the LNG pump or pressurised container system supplying them, whichever is greater.
b) Manifold vaporisers shall have both inlet and discharge block valves at each vaporiser.
c) The outlet valve of each vaporiser, piping components and relief valves installed upstream of each vaporiser outlet valve shall be suitable for operation at LNG temperature.
d) Suitable automatic equipment shall be provided to prevent the discharge of either LNG or vaporised gas into a distribution system at a temperature below the minimum design temperature of the send out system. Such automatic equipment shall be independent of all other flow control systems and shall incorporate shut down valves used only for contingency purposes.
e) Isolation of an idle manifold vaporiser to prevent leakage of LNG into that vaporiser shall be accomplished with two inlet valves with safe bleed arrangement in between.
f) Each heated vaporiser shall be provided with safety interlock to shut off the heat source from a location at least 15 m distant from the vaporiser. The device shall also be operable at its installed location.
g) A fire safe shutoff valve should be installed on the LNG line inlet to a heated vaporiser at least 15 m away from the vaporiser. This shutoff valve shall be operable either at installed location or from a remote location and the valve shall be protected from becoming inoperable due to external icing conditions.
h) If a flammable intermediate fluid is used with a remote heated vaporiser, shutoff valves shall be provided on both the hot and cold lines of the intermediate fluid system. The controls for these valves shall be located at least 15 m from the vaporiser.
i) The vaporisers shall be fitted with local as well as control room indications for pressure and temperature of both fluid streams at inlet and outlet.
j) Instrumentation for storage, pumping and vaporisation facilities shall be designed for failsafe condition in case of power or instrument air failure.

4.8.2.10 RELIEF DEVICES ON VAPORIZERS

a) Each vaporiser shall be provided with safety relief valves sized in accordance with the following as applicable:

i) The relief valve capacity of heated or process vaporisers shall be such that the relief valves will discharge 110 percent of rated vaporiser natural gas flow capacity without allowing the pressure to rise more than 10 percent above the vaporiser maximum allowable working pressure.

ii) The relief valve capacity of ambient vaporisers shall be such that the relief valves will discharge at least 150 percent of rated vaporiser natural gas flow capacity without allowing the pressure to rise more than 10 percent above the vaporiser maximum allowable working pressure.

b) Relief valves on heated vaporisers shall be so located that they are not subjected to temperature exceeding 60°C during normal operation unless designed to withstand higher temperature.

c) The discharges from the relief valves should be routed to the flare. If not connected to flare, vapours may safely be disposed to atmosphere, provided that this can be accomplished without creating problems like, formation of flammable mixture at ground level or on elevated structure where personnel are likely to be present. The height of release to be based on the dispersion analysis and risk analysis so that the integrity of the facility is maintained.

4.9 LNG COLD RECOVERY

4.9.1 LNG cold recovery system may be used in an LNG Terminal. It aims at recovering the part of the potential cold energy available in LNG so as to use it effectively in cold utilising plants.

4.9.2 LNG cold utilisation process is divided into:

i) Cold is used directly to cool down another element, by simple heat transfer. Some of the schemes under this class are:

- Re-liquefaction/ re-condensation of BOG
- Cooling of industrial fluids
- Air separation and liquefaction plants
- Food freezing
- Power plant cooling
- Cold warehouse
- HVAC of plant buildings

ii) Only a portion of the cold is utilised in the receiving terminal for boil–off gas re-liquefaction / re-condensation. Remaining cold may be utilised in the nearby industries for heat exchange with industrial fluids or for power generation units.

iii) In the case of LNG cold recovery facility at the terminal, all the safety features provided on the LNG vaporisers shall be applicable.

4.10 TRUCK LOADING FACILITY

4.10.1 BRIEF DESCRIPTION

The purpose of LNG loading system is to transfer LNG from the storage tanks to truck to deliver LNG to other sites via road transportation. LNG loading operation shall be done in the presence of trained personnel. LNG for loading into the truck shall be tapped off from the LNG in-tank pump discharge. LNG vapour return from the truck shall be routed to the BOG suction header. When no loading is in progress the recirculation should be maintained in the LNG filling line, through a recirculation line for maintaining chilled condition. The loading facility should be provided with the LNG liquid loading arm and vapour return arm.
For monitoring of uniform chilling of the LNG feed line during no-loading situation, a suitable number of skin temperature indications with alarm shall be provided in the control room.

The LNG truck shall be double-walled vacuum-insulated, cryogenic vessel suitable for transporting cryogenic fluids. The truck for road movement shall be designed, constructed and tested in accordance with the Static and Mobile Pressure Vessels (Unfired) Rules as amended from time to time.
Typical LNG loading facility consists of the following:

- LNG filling line, vapor return line & a recirculation line with adequate instrumentation
- Liquid/ vapour loading arm, batch flow meter & control valve
- Weigh bridge/ flowmeter for custody transfer.
- Sick tanker unloading facility.

4.10.2 DESIGN CONSIDERATIONS OF LNG LOADING/ UNLOADING FACILITIES

a) Loading Facility

Each loading station shall consist of the following:

- Automatic flow control valve or suitable control mechanism shall be provided in LNG loading lines.
- A vapour return line with an isolation valve connected back to the storage vessel/ BOG line shall be provided.
- Proper tanker earthing connection with START permissive shall be provided.
- Properly designed loading arm shall be provided at the end of filling and vapour return lines for connecting to the truck. The loading arms shall be provided with cryogenic breakaway couplings. These arms shall be of approved type and tested as per OEM recommendations.
- Weigh bridges of suitable capacity for tanker weighment shall be provided.
- Breakaway coupling should be tested in both forward and reverse direction once a year.

b) Sick Tanker Unloading Facility

A sick tanker unloading facility shall be provided. The truck unloading shall be done by vapors using pressure differential method of liquid transfer by any of the following methods:

- Utilising vapours from BOG compressors discharge.
- Utilizing the gas from the pressurized gas (send out) network with proper pressure control.

- Utilizing truck pressure building coil.
- Dedicated unloading compressors/pumps.

Suitable protection shall be considered for truck system over-pressurization with due considerations of impact of high temperature.

c) Others

All drains, vents and safety valve discharges shall be routed to the closed system/flare system. In case of non-availability of flare system, the discharge from safety valve shall be vented to atmosphere at a safe location minimum at an elevation of 3 meter above the nearest working platform within a radius of 15 meters for effective dispersion of hydrocarbons. A dedicated drain and vent connection should be provided for the loading and vapour arms. Operational requirement is that after every loading operation, before disconnection, the liquid holdup in the LNG liquid and vapor arms are required to be drained, depressurized, and purged. The provision of drain-pipe should be considered. The liquids collected in the drain-pipe would gradually vaporize. Nitrogen connection can be provided to facilitate the vaporization process.

4.10.3 SAFETY SYSTEM AND SAFETY PRECAUTIONS

a) Safety System

The gantry shall have gas and spill detector at potential spillage and gas emission locations. There shall be flame detectors which shall cover the entire gantry and detect any fire. In addition, there shall be manual call points at appropriate locations.
The fire and gas detection shall be considered as follows:

- Fire detectors
- Gas detectors
- Low temperature (spill) detector

The signals generated from the detectors shall be connected to F&G system PLC which shall be integrated with the ESD system.

The shutdown valves should be provided for all the process incoming and outgoing lines to/from loading gantry and shall be located at least 15 meters away from the loading gantry at an easily accessible location.

The Emergency Shutdown (ESD) philosophy shall be designed to initiate appropriate shutdown action on detection of any emergency situation or through detector signal.

Emergency push button or hand switch shall be provided in control room and also in field at safe location (at least 15m away) for manual actuation of ESD system and fire water spray system by operator in case of emergency. The field related button/ switch to be provided on either end of the gantry at easily accessible locations.

ESD shall be caused in case of either of the following:

- Signals from two detectors of different types of gas or spill or flame
- Initiation of manual call points
- Hand switches actuation (provided in the field as well as in the operator console & control room)

In addition, the following logic shall also be incorporated to stop individual truck filling operation:

- Gas detection will stop filling operation.
- Earth relay contact indicating inadequate earthing of the truck will stop filling operation.

In addition to above, fire water spray with deluge system shall be provided in the gantry which shall be activated automatically (either flame/ fire detector or quartzoid bulb assembly). Apart from this, provision for actuation by manual switches shall be given.

b) Safety Precautions

Following precaution should be taken due to associated hazards during transfer of LNG to or from a truck.

1. No source of ignition must be allowed in the area where product transfer operations are carried out.
2. At least 02 no. of 9 Kg DCP fire extinguishers shall be available near the truck during transfer operations.
3. The first operation after positioning the truck shall be to provide proper earthing connection of the tanker. Earthing shall be disconnected just before the release of the truck.
4. While disconnecting arm, connections shall be loosened slightly at first to allow release of trapped pressure, if any.
5. Appropriate personal protective equipment (cryogenic suits, flame retardant coveralls, etc.) shall be used while making or breaking the connections to avoid cold burns.
6. The master switch shall be put off immediately after parking the truck in position. No electrical switch on the truck shall be turned “on" or “off" during the transfer operation.
7. No repair of the truck shall be allowed while it is in the loading area.
8. Use of wheel chokes shall be ensured.
9. Filling/transfer operations shall be stopped immediately in the event of -
   (i) Uncontrolled leakage occurring
   (ii) A fire in the vicinity
   (iii) Lightning and thunderstorm

Provision to stop the filling/transfer operations shall be available from field as well as remote location. Stop switch in field shall be located at a safe distance (minimum 15 meters away) from the source of hazard to be protected.

4.10.4 PROCEDURES FOR LOADING OPERATION

(a) PRE-FILLING ACTIVITIES

1. Verification of the truck is necessary as per standard procedure before it is allowed to enter the premises. The following shall be checked:

   - Availability of two safety valves, level gauge, excess flow protection and control valve on liquid and vapour lines, pressure gauge, on the vessel / bullet.

- Availability of fire screen between cabin and vessel is provided. For this purpose, cabins with metallic back cover with suitable view glass will be considered as fire screen.
- Availability of 02 nos. of 9 Kg DCP fire extinguishers.
- Fitment of PESO approved spark arrestors at the exhaust.
- No leakage in exhaust silencer pipe exists.
- Manufacturer's name plate with date of testing with next due date of testing is fitted on the vessel.
- Availability of valid explosive licence and RTO registration certificate.
- Blind flanges/caps are provided on vessel.
- Availability of earthing cable.
- Bonding between vessel and chassis and between flanges is satisfactory.
- Earthing / bonding point is available.
- Third party inspection/test certificates for vessel/fittings are available.
- Liquid / vapour line valves are in good condition.
- Check for presence of loose battery terminals; cellphones, flammable chemicals etc. in the tanker cabin.

2. Move truck to the loading bay/weigh bridge and record the empty weight.
3. When the truck enters the loading bay, following steps shall be followed by the operator:
   i) Check for residual positive pressure and LNG (Heel) in the truck. If heel is present, then the truck is considered ready for filling. Then proceed to step (iv).

   ii) If the heel is not present, then the vapour space shall be checked for moisture content and oxygen with portable analysers. The moisture content should be such that the dew point shall be less than -20°C and oxygen less than 1000 ppm.

   iii) If any of the parameters mentioned at point no i) and ii) above, are not complied, then the truck shall be purged with nitrogen.

   iv) Once the oxygen and moisture are within limit, the operator shall prepare a 'Filling Advice' as per procedure. The purpose of the Filling Advice is to advise the operator about the amount of LNG to be loaded, empty weight of the truck, destination, vehicle identification number, driver particulars and other relevant details.

   v) After issuing the filling advice, the truck shall be earthed.

   vi) The filling arm shall be connected to the truck and the vapour return arm shall be connected to the vapour return line.

For nitrogen purging, it shall be ensured that the valve downstream of flow control valve is closed. The nitrogen line shall be slowly opened, and oxygen and the moisture content shall be checked at the vents. The procedure for venting and purging shall be individually developed meeting all the safety requirements.

b) FILLING OPERATION

Before starting the filling operation, following shall be checked by the operator:

- The truck is in right location on the ramp.
- The wheels are choked, ignition is OFF, and the key is with the truck loading operator.
- Truck is earthed.
- The filling and vapour return arm is under positive nitrogen pressure.
- All arms and the valves are as per operating condition.
- None of the thermal relief valves are relieving and do not show any sign of tampering.

- All instrumentations are in the normal condition and there is no alarm being indicated.
- Truck cabinet door is open and secured and all components are in working order.

Start the filling operation as per the 'standard operating procedure' developed for each filling.

- Filling the truck should be simultaneously monitored and confirmed by level instrument on the truck and batch flow controller.
- Recirculation should be started after the filling is complete.

c) POST-FILLING OPERATION

- The line between arms and the nearest ball valve in the filling line shall be drained into the drain header for both filling and vapour return line. Once all the liquid is drained, both the arms should be allowed to warm up to near about ambient temperature.
- The liquid arm shall be purged with nitrogen and vented out through the vent valve.
- Once all the vapour is purged out, then the filling arm should be disconnected from the truck and stowed at its original position.
- Start the draining and filling operation as per the standard operating procedure to be individually developed.
- Chokes should be removed from the truck and the keys handed over to the driver.
- The driver should sign and takes a copy of the ticket and then drive away.

4.10.5 BLACK START (FIRST FILL)

The black start will be applicable when the truck loading system is being started for the first time or after a long shutdown. The important precaution for the black start is to maintain the slow rate of cooling to bring down the temperature of the pipe from ambient to -160°C.

The line from the storage tanks to the loading gantry shall be flushed and purged with nitrogen prior to the black start. The black start procedure shall be individually developed, meeting the entire safety requirement.

4.10.6 DRAIN AND VENTS

Drain and vents shall be provided to meet all the requirement of draining, purging, venting etc. Appropriate system shall be provided to handle the discharge from the TSV's also.

4.10.7 FLEXIBLE HOSES

a) Flexible hoses may be used to make small temporary connections for the transfer of LNG and other cryogenic liquids such as refrigerant and liquid nitrogen, for example when emptying or filling trucks of LNG. The use of flexible hoses shall be in accordance with the hazard assessment as per EN 1473.
b) Flexible hoses shall not exceed 15 m in length and 0.5 m³ in volume. It is intended that the hose be designed and tested to satisfy the generally accepted rated pressure i.e. at least PR 40. Hoses may be then selected with a rated pressure equal to or greater than the maximum allowable pressure of the equipment to which it is to be attached.
c) Flexible hoses shall not be used for the transfer of LNG between LNG carriers and shore at conventional LNG terminals.
d) Flexible hoses shall be designed in accordance with relevant codes and/ or standards, such as EN 12434/ EN 21012.

e) Hoses for LNG transfer shall be tested at least annually to the maximum pump pressure or relief valve setting and shall be inspected visually before each use for damage or defects.

5.0 TERMINAL LAYOUT

5.1 PHILOSOPHY

Terminal lay out philosophy must consider location of the facilities at a site of suitable size, topography, and configuration. Due consideration to be given to minimise the hazards to persons and property due to leaks and spills of LNG and other hazardous fluids at site. Before selecting a site, all site related characteristics which could affect the integrity and security of the facility shall be determined. A site must provide ease of access so that personnel, equipment, materials from offsite locations can reach the site for firefighting or controlling spill associated hazards or for the evacuation of the personnel.

OISD–STD-118 covers the layout consideration for the oil and gas installations. The above standard is also generally applicable for consideration of layout of LNG Terminal. However specific points related to LNG standards are brought out here.

5.2 BASIC INFORMATION

5.2.1 Information on following items should be collected before proceeding with the development of overall plot plan.

- Terminal capacity
- Process units and capacities
- Process flow diagram indicating flow sequence
- Utility requirements
- Unloading system along with tanker berthing system with capacity
- LNG storage tanks, sizes and type of storage tanks
- Other storage tanks
- LNG transfer and vaporisation
- Number of flares
- Provision for spill containment and leak control
- Inter distances between the equipment
- Operating and maintenance philosophy for grouping of utilities
- Plant and non-plant buildings
- Environmental considerations
- Scrap yards and dumping ground
- Fire station
- Chemical storage
- Ware house and open storage areas.

5.2.2 Information related to each item should include, but not limited to, following:

- Extreme temperatures and pressures for normal operations as well as emergency conditions.
- Concrete structures subject to cryogenic temperatures
- Fail safe design
- Structural requirement
- Requirement of dyke and vapour barrier.
- Shut off valves and relief devices.

5.2.3 Data on the following infrastructure facilities should be identified and collected before detailed layout activity is taken up. Due consideration should be given for the same while deciding/ finalizing terminal layout.

- Site location map
- Seismic characteristics and investigation report.
- Soil characteristics
- Prevailing wind speed and direction over a period
- Meteorological data including corrosive characteristics of the air and frequency of lightning
- Area topography contour map
- High flood level in the area and worst flood occurrence.
- Source of water supply and likely entry / exit point
- Electric supply source and direction of entry point
- LNG entry point/ Gas exit point
- Minimum inter distances between facilities as well as between facilities & boundaries
- Storm water disposal point and effluent disposal point
- Approach roads to main Terminal areas
- Surrounding risks
- Air routes and the proximity of the Airports.

5.3 BLOCKS

5.3.1 In addition to points indicated in OISD-STD-118, as applicable, containment of potential spills of LNG or other hazardous liquid, especially in case of LNG storage and jetty area should also be considered.

5.3.2 LAY OUT OF BLOCKS / FACILITIES

The LNG Terminal may consist of the following basic blocks/ facilities.

- The Jetty for berthing of ship and unloading of LNG.
- Unloading line from Jetty to shore terminal.
- LNG Storage

- Re-gasification consisting of pumping and vaporisation.
- Send-Out
- Utility Block
- Fire Station
- Flare system
- Control Room
- Administrative Block
- Workshop
- Warehouse
- Electrical Substation.
- Laboratory
- Truck Loading
- Occupational Health Centre
- Emergency Control Centre
- Independent communication infrastructure

5.4 ROADS

OISD-STD-118 is to be followed as applicable. In addition, land access to the moored ships shall be provided. If necessary, a separate road may lead to the berths in order to provide the crew with a free access to the ship.

5.5 LOCATION

OISD-STD-118 shall be followed as applicable. In addition, the receiving terminal should be as close as possible to the unloading jetty.

5.6 ERECTION & MAINTENANCE

OISD-STD-118 shall be followed as applicable.

5.7 FUTURE EXPANSION

Future expansion requirement shall be assessed and provision of space for the same should be made.

5.8 GENERAL CONSIDERATIONS

Minimum inter-distances between blocks / facilities shall be maintained as per Table-1 of OISD-STD-118 or as per the risk analysis studies whichever is higher. Inter distances between specific equipment as mentioned below are to be maintained.

5.9 SPACING REQUIREMENT OF LNG TANKS AND PROCESS EQUIPMENT

5.9.1 LNG TANK SPACING

LNG tanks with capacity more than 265 m^3 should be located at minimum distance of 0.7 times the container diameter from the property line but not less than 30 meters. Minimum distance between adjacent LNG tanks should be 1/4 of sum of diameters of each tank. The distance between tanks should be further reviewed in accordance with hazard assessment, but in no case, it shall be less than the criteria mentioned above.

However, any LNG storage/ process equipment of capacity more than $0.5m^3$ shall not be located in buildings.

5.9.2 VAPORIZER SPACING

Vaporisers and their primary heat sources, unless the intermediate heat transfer fluid is non-flammable, shall be located at least 15 m from any other source of ignition. In multiple vaporiser installations, an adjacent vaporiser or primary heat source is not considered to be a source of ignition. Integral heated vaporisers shall be located at least 30 m from a property line that may be built upon and at least 15 m from any impounded LNG, flammable liquid, flammable refrigerant or flammable gas storage containers or tanks. Remote heated, ambient and process vaporisers shall be located at least 30 m from a property line that can be built upon. Remote heated and ambient vaporisers may be located within impounding area. The inter distances in multiple heated vaporisers a clearance of at least 2 m shall be maintained. The types of heaters are as mentioned in section 4.8.2.

5.9.3 PROCESS EQUIPMENT SPACING

i) For process equipment spacing Table-2 of OISD-STD-118 as applicable shall be followed.

ii) Fired equipment and other sources of ignition shall be located at least 15 m from any impounding area or container drainage system.

5.9.4 CONTROL ROOM AND SUBSTATION

i) Control Room shall be constructed as per OISD-STD-163.

ii) The minimum distance of 60 m shall be maintained between LNG storage tank and substation.

5.9.5 UNLOADING FACILITY SPACING

i) A pier or dock used for pipeline transfer of LNG shall be located so that any marine vessel being loaded or unloaded is at least 30 m from any bridge crossing a navigable waterway. The loading or unloading manifolds shall be at least 60m from such a bridge.

ii) LNG and flammable refrigerant loading and unloading connections shall be at least 15m from uncontrolled sources of ignition, process areas, storage containers, control room and important plant structures. This does not apply to structures or equipment directly associated with the transfer operation.

5.9.6 LNG TANKER LOADING /UNLOADING FACILITIES

i) The layout of the LNG facilities including the arrangement and location of plant roads, walkways, doors, and operating equipment shall be designed to permit personnel and equipment to reach any area affected by fire rapidly and effectively. The layout shall permit access from at least two directions.

ii) LNG truck loading gantry shall be covered and located in a separate block and shall not be grouped with other facilities.

iii) Adequate space for turning of truck shall be provided, which is commensurate with the capacities of the trucks. However, the space for turning of truck with minimum radius of 20 M shall be provided.

iv) Maximum number of LNG tank lorry bays shall be restricted to 8 in one group. Separation distance between the two groups shall not be less than 30 M.

v) The layout of the loading location shall be such that truck being unloaded shall be in drive out position.

vi) The weigh bridge of adequate capacity shall be provided and with adequate approach way for proper manoeuvrability for vehicles.

vii) Due consideration should be given for the dedicated parking area for LNG trucks with controlled access of other vehicles. The parking area shall be located in a secured area and provided with adequate number of hydrants/ monitors to cover the entire parking area. Suitable arrangement for safe venting of vapor generated during waiting period in the parking area, preferably to a closed system, should be considered.

viii) Escape routes shall be specified and marked in LNG plants for evacuation of employees in emergency. Properly laid out roads around various facilities shall be provided within the installation area for smooth access of fire tenders, etc. in case of emergency.

5.9.7 ELECTRICAL CLASSIFICATION

Classification of areas for Electrical installations in LNG Terminal shall be as per OISD-STD-113 as applicable.

5.9.8 BUILDINGS AND STRUCTURES

Buildings or structural enclosures in which LNG, flammable refrigerant and gases are handled shall be of lightweight, non-combustible construction with non-load-bearing walls.

6.0 LNG STORAGE TANK

The Liquefied Natural gas is stored at about -158°C to -168°C. LNG tanks shall be designed to ensure proper liquid retention, gas tightness, thermal insulation and environment safety.

6.1 CLASSIFICATION OF STORAGE SYSTEM

6.1.1 GENERAL

The most common type of refrigerated storage tanks was “Single Containment Tank” meaning it is having a single retaining compartment, surrounded by a low bund wall. The failure of a single containment tank would result in an immediate release of liquid as well as vapour to the surrounding environment resulting in potentially major hazard. To decrease the probability of failure of a single containment tank more stringent requirements related to material selection, design, construction, inspection, and testing are to be considered.

6.1.2 Even though the probability of failure reduces to a very low level with the above requirements, the consequences of a failure may be so serious that a secondary protection system is required to eliminate the risk of large inventory release around the tank in case of leakage or failure of the primary/inner container/tank. This can be achieved by double containment or full containment tanks. Membrane type tanks are also considered to reduce the risk; however, external/internal damage scenarios should carefully be examined for membrane type tanks as part of risk analysis as internal membrane is not designed for independent mechanical/structural integrity.

Four types of LNG Storage Tanks are considered here.

- Single Containment
- Double Containment
- Full Containment
- Membrane

The selection of storage tanks shall be decided based on the location, adjacent installations, habitation in the surrounding, operational, environmental, safety and reliability considerations.

6.1.3 SINGLE CONTAINMENT TANK

The single containment storage for liquefied natural gas is usually dome roof, flat bottomed tanks. In the past, the most common type of storage system consisted of a tank with a single liquid retaining container – referred as "*Single Containment tank*" – surrounded by a bund wall/ dyke (Ref. Figure-1). The outer wall (if any) of a single containment storage system is primarily for the retention and protection of insulation and is not designed to contain liquid in the event of product leakage from the inner container.

A single containment tank shall be surrounded by a bund wall/ dyke to contain any leakage.

6.1.4 DOUBLE CONTAINMENT TANK

A double containment LNG storage tank is designed and constructed so that both inner and outer wall shall be independently capable of containing the LNG stored. The LNG is normally stored within the inner tank, but the outer tank shall be able to contain the LNG product leakage from the inner tank. The outer tank is not designed to contain product vapour in the event of liquid leakage from the inner tank (Ref. Figure-2). The outer tank, if it is made of metal, shall be of cryogenic grade. If outer tank is made of pre-stressed concrete, the same shall be suitable for withstanding temperature and hydrostatic head. To minimise the pool of escaping liquid in case of failure of inner tank the outer tank should not be located at a distance exceeding 6 m from the inner tank.

6.1.5 FULL CONTAINMENT TANK

A full containment storage tank is one meeting all the requirements of a double containment storage plus the additional requirement that it shall avoid the uncontrolled release of product vapour in the event of liquid leakage from the inner tank (Ref. Figure-3).

A pre-stressed concrete outer tank with / without an earth embankment can be used as the secondary liquid container for a double, full containment, and membrane tank. The construction of a full height retaining wall of concrete has the added advantage of protection against blast overpressure and impact of projectiles. (Ref. Figure-4). If the outer tank is made of metal, it shall be of cryogenic grade. If the outer tank is made of pre-stressed concrete, the same shall be suitable for withstanding temperature and hydrostatic head.

6.1.6 MEMBRANE TANK:

Membrane tank technology has predominantly been used in LNG carriers (ships) and for in-ground LNG tanks. Membrane tanks have also been used for above ground LNG storage in some cases.

The in-ground membrane tanks have been in use for LNG storage and are also considered as acceptable option. However, the design of the in-ground membrane tanks shall be as per the available international standards. This standard covers only above ground Membrane tanks.

Though membrane tank technology is proprietary in nature, design of membrane shall be governed by applicable standards.

i) The main characteristic of this type of tank is the separation of the tightness and mechanical strength functions. Tightness is ensured by a membrane not subjected to any stress. Stresses are taken up by the concrete wall through the load bearing insulation provided between membrane and concrete wall.

ii) The primary container, constituted by a membrane together with thermal insulation and a concrete tank jointly forming an integrated, composite structure. This composite structure shall provide the liquid containment. The concrete tank shall have special provisions so that in case of a major leak through the membrane & insulation system, it is capable of containing the liquid LNG & vapor both, till controlled venting of vapor can occur.

iii) The vapour of the primary container is contained by a steel roof liner, which forms, with the membrane, an integral gas tight containment.

iv) The insulation space, between the membrane and the concrete tank, is isolated from the vapour space of the tank. A nitrogen breather system operates on the space to control the pressure of the nitrogen within normal operating limits and to monitor the gas concentration in order to check continuously the integrity of the primary container. In case of a methane leak, the nitrogen sweeping mode is activated to maintain the resulting gas concentration within normal operating limits.

v) The membrane containment tank systems shall also meet the additional requirements as specified in NFPA 59A.

6.2 SELECTION CRITERIA

6.2.1 Safety and reliability are the most important aspects for LNG storage tanks, holding large inventory of flammable gas. Majority of technical improvements on LNG storage tanks, to date, have therefore been directed towards improvements in both safety and reliability.

6.2.2 The following list summarises a number of loading conditions and considerations that have influence on the selection of the type of storage tank.

i) The factors which are not subjected to control:

- Earthquake
- Wind

- Cyclone, floods, tsunami
- Snow, Climate
- Objects flying from outside the plant.

ii) The factors that are subjected to limited control.

- In plant flying objects
- Maintenance Hazards
- Pressure waves from internal plant explosions.
- Fire in bund or at adjacent tank or plant.
- Overfill, Overpressure (process), block discharge.
- Roll-Over
- Major metal failure e.g., brittle failure
- Minor metal failure e.g., leakage
- Metal fatigue, Corrosion
- Failure of pipe work attached to bottom, shell or roof.
- Foundation collapse.

iii) The factors that are subjected to full control.

- Proximity of other plant
- Proximity of control rooms, offices, and other buildings within plant
- Proximity of habitation outside plant
- National or local authority requirements
- Requirements of the applied design codes.

6.2.3 The main criteria for selection of the type of tank shall be decided based on the risk analysis study and the level of risk it is posing on the surrounding.

6.2.4 There is no limit on the height of the tank envisaged other than engineering considerations.

6.2.5 No capacity restriction for LNG tank is envisaged considering the technological developments in this area.

6.3 BASIC DESIGN CONSIDERATIONS

This section describes the basic design considerations for single, double, full containment and membrane tanks.

6.3.1 Material of construction:

The parts of LNG container which will be in contact with LNG or cold vapour shall be physically and chemically compatible with LNG. Any of the materials authorised for service at -168°C by the ASME Boiler and Pressure Vessel Code shall be permitted. Normally, for single containment tank, improved 9% Ni steel/ Austenitic stainless steel/ Aluminium Magnesium alloy are used. For double, or full containment tanks, 9% Ni steel with impact testing is used. In case of membrane tanks, normally austenitic stainless steel is used as material of construction for membranes.

6.3.2 Liquid loading*:*

i) The maximum filling volume of LNG container must take into consideration the expansion of the liquid due to change in pressure/ temperature to avoid overfilling.

ii) For double containment and full containment, the primary container shall be designed for a liquid load at the minimum design temperature specified. The design level shall be the maximum liquid level specified or 0.5 m below the top of the shell, whichever is lower.

iii) The outer tank for double containment, full containment and membrane tanks shall be designed to contain the maximum liquid content of the primary container at the minimum design temperature specified. The outer concrete tank shall have 9% nickel steel secondary bottom and 9% nickel steel insulated 'Thermal Corner Protection' (TCP). These should be linked together. The top of the TCP shall be anchored into pre-stressed concrete wall, approximately 5 meters above the base slab.

6.3.3 Insulation:

i) The refrigerated storage tanks for LNG shall be adequately insulated in order to minimise the boil-off gas generation due to heat leak from ambient. Proper insulation shall be ensured in tank base, tank shell, tank roof, suspended deck etc.

ii) The extent of insulation depends on boil-off considerations for which the storage tank is designed. Normally boil-off rate of within 0.1 % of hold up liquid volume per day is considered.

iii) The possibility of an adjacent tank fire must be taken into consideration when designing insulation for LNG storage tanks. Tank spacing, water deluge systems, quantity and hazard index of LNG contents must be considered when specifying insulating materials.

6.3.4 Soil protection*:*

i) The soil under the LNG storage tank shall not be allowed to become cold. If the soil becomes too cold, frost penetrates into the ground. Consequently, ice lenses form in the soil (mainly in clay types of soil) and the growth of these ice lenses can result in high expansion forces, which can lift and damage the tank or parts of the tank. To prevent such occurrence heating system shall be incorporated in the foundation. An automatic on/off switch system should activate the electrical heating system and ensure that the tank foundation, at its coldest location, should be within acceptable temperature range i.e. +5°C to +10°C.

ii) As an alternative to electrical bottom heating system, free ventilated tank bottom by elevated structure may also be used. In such a setup, slope shall be ensured for the paved portion below the tank from centre to periphery to avoid accumulation of liquid. Gas detector shall be provided for detection of any leakage and accumulation below the tank.

iii) Electrical heating system shall consist of a number of independent parallel circuits so designed that electrical failure of any one circuit does not affect power supply to the remaining circuits. Electrical heating shall be so designed that, in the case of electrical failure of a main power supply cable or a power transformer, sufficient time is available to repair before damage occurs due to excessive cooling. Alternatively, provision for connecting a standby heating power source should be made.

6.3.5 Leak Detection in annular space*:*

i) Leak detection facility shall be provided in the annular space between primary container and secondary container. Liquid may be present in the annular space due to spillage from inner tank or leak of the inner tank. If liquid is detected in the annular space, it should be removed carefully. For tanks with a perlite filled annular space, liquid can be removed by evaporation. Temperature sensors may be used for leak detection. A system alarm may be provided if there is a malfunction in the monitoring system.

ii) Provision for Nitrogen purging of the annular space should be considered. This will also be useful in leak detection by monitoring hydrocarbons levels in the Nitrogen purge.

6.3.6 Pressure and Vacuum relief system:

The following guidelines for the design of pressure and vacuum relief system of cryogenic LNG tanks shall be provided.

1) Pressure relief valve shall be entirely separate from the vacuum relief valve. Pressure relief valve shall relieve from inner tank. In order to take care of malfunction of any of the relief valves, due to blockage in the sensor line, one extra relief valve (n+1) shall be installed. Further sensor lines for each relief valve can form a common ring with individual tapping from the sensor ring for each of the pilots. Pilot operated pressure relief valves are preferred over pallet operated relief valves.

2) Vacuum relief valves (n+1 philosophy) shall relieve into the space between the outer roof and suspended roof. Pilot operated vacuum relief valves are not acceptable for vacuum protection as the valve action is not fail safe against main valve diaphragm or bellows rupture. Conventional pallet type vacuum relief valves shall only be used.

3) Relief valves to atmosphere should be adequately sized to relieve the worst-case emergency flows, assuming that all outlets from the tank are closed, including the outlet to flares and to boil-off gas system.
Relief valves so designed shall also be capable of discharging flow rates from any likely combination of the following:

a) evaporation due to heat input to tank, equipment, and recirculation lines

b) displacement due to filling at maximum possible flowrate or return gas from carrier during loading

c) flash during filling

d) variations in atmospheric pressure

e) vapourised LNG in de-super heaters

f) recirculation from a submerged pump

g) roll-over

Vapours may safely be disposed to the atmosphere, provided that can be accomplished without creating problems like, formation of flammable mixture at ground level or on elevated structure where personnel are likely to be present. Such disposal shall comply to all the stipulations as mentioned from time to time in Environmental Acts & Rules.

4) Provision shall be made to inject nitrogen or dry chemical powder at the mouth of pressure safety relief valve discharge.

5) Vacuum relief should be based on withdrawal of liquid at the maximum rate, withdrawal of vapour at the maximum compressor suction rate, variation in atmospheric pressure etc.

6) The flare stack should be continuously purged in order to avoid air ingress and shall be provided with pilot burner.

7) Provision shall be made to maintain the internal pressure of LNG container within the limits set by the design specification by releasing to flare via a pressure control valve installed in the BOG line from tank to compressor. Factors that shall be considered in sizing of flare system shall include the following:

   - Operational upsets, such as failure of control device / BOG compressor tripping etc.
   - Loss of refrigeration
   - Vapour displacement and flash vaporisation during filling.
   - Drop in barometric pressure.
   - Depressurisation load from Emergency depressurisation vents from vaporisers/RLNG headers.

8) Safety relief system shall suitably be designed to take care of pressure rise in annular space due to leakage from inner tank and subsequent vaporisation of the liquid. One of the following two options shall be considered to allow safe venting and relief:

   a) Independent safety hatch/rupture disc on outer tank.
   b) Safety relief valves connected to inner tank including the suspended deck vents and perlite retaining barrier to be adequately sized and specified.

9) For the pressurised systems, the safety relief valve vent shall be so positioned to release the hydrocarbon at safe height.

10) Relief from tank PSV shall not form a cloud on the tank and the PSV discharge shall be routed to safe height in accordance with dispersion study and risk analysis. This is subject to compliance to The Environment Protection Act/ Rules and their amendments from time to time.

6.3.7 Tank Roll Over

Under certain conditions, "roll over" of the liquid in the LNG tank can occur resulting in the rapid evolution of a large quantity of vapour with the potential to over pressurise the tank. Stratification can occur in an LNG tank if the density of the liquid cargo charged to the tank is significantly different from the left-over LNG in the tank. Inlet piping must be designed to avoid stratification of LNG. This can be done by having top and bottom fill lines to inject denser / lighter LNG at the top / bottom. Mixing of in tank LNG by providing re-circulation facility may be considered. Mixing may also be done by providing distribution holes along the fill line extending to the bottom. Temperature sensors are put to monitor the temperature of the liquid throughout the

liquid height at regular intervals. Provision for density measurement on tank shall be provided for the entire height of the tank.
All LNG tank systems shall be designed for both top and bottom filling unless other process means are provided to mitigate stratification.

The boil-off due to a roll-over shall be calculated appropriately. Alternately, the flow rate during roll-over shall be conservatively taken as 100 times of the boil-off rate.

For taking care of over pressurisation due to roll over, one of the following options shall be provided.

a) Sizing of inner tank pressure relief valves releasing to flare in case the flare system is designed for such loads.
b) Sizing of inner tank pressure relief valves releasing to atmosphere.
c) Rupture disc or equivalent device to be provided on the tank with isolation valve (lock open condition) releasing to atmosphere.

- Means to check rupture disc integrity should be provided. Fragments of the rupture disc should not fall into the tank.
- Failure of the rupture disc shall trip all the boil-off gas compressors automatically.
- Following failure, the rupture disc replacement should be possible during operation.

6.3.8 Over-Fill of Inner Tank

a) Two independent type level measuring instruments shall be provided. The level instrument shall be equipped to provide remote reading and high-level alarm signals in the control room. In addition, an independent transmitter for high level alarm and high-high level alarm with cut off shall be provided. The high-high level should be hard wired directly to close the liquid inlet valves to the tank.
b) The tank shall not be provided with overflow arrangement.

6.3.9 DYKE

Dyke is not required for full containment or membrane containment tanks with pre-stressed concrete wall.
Dyke shall be provided for the following types of LNG storage tanks.

- Single containment tank
- Double containment with metallic outer tank
- Full containment with metallic outer tank
- Membrane tank with metallic outer tank

a) The containment volume of the dyke shall be equivalent to 110% capacity of the largest tank within the dyke and considering that the trajectory of a leak at the upper liquid level does not overshoot the edge of the dyke.
b) No restriction on the dyke height is envisaged.
c) High volume foam generators shall be provided for the dykes.

d) Where more than one storage tanks are installed in a common dyke, each tank foundation shall be capable of with-standing contact with LNG or shall be protected against contact with an accumulation of LNG that might endanger structural integrity.

6.3.10 OTHER SAFETY CONSIDERATIONS

i) Where refrigerated storage tanks are located near process plants with a likelihood of exploding process equipment, the impact of flying object on the tank, one 4" valve travelling at 160 kmph (object of 50 kg weight with a speed of 45 m/sec) shall be considered.

ii) For the tank located within the flight path of an airport, the impact of a small aircraft or component shall be taken care of.

iii) Impact of explosion wave due to major leak from a nearby natural gas pipeline or a major spill of LNG may also be considered.

iv) Earthquakes: The risk level is determined on the basis of the seismic classification of the location. The data pertaining to the seismic activity level having been ascertained, the structure is to be designed based on IS-1893 and other relevant codes.

v) Rainfall run-off from the tank roof should be directed to a collecting system around the outer edge of the roof. Collected rainwater should be carried by a drainage piping system that directs the rainwater away from any LNG spill carrying surfaces and to graded drainage areas that are beyond (outside) the ring road.

6.3.11 Nozzles (also refer 4.5.1)

There shall be no penetrations of the primary and secondary container base or shell walls for LNG tanks to ensure fluid tightness.

In addition to the nozzles used for regular operations like liquid inlet, pump outlet, vapour outlet and instrument connections, the following provision shall also be provided:

i) Nitrogen connections for:

- inertisation of inner tank
- outer tank and insulating material.

ii) Chill down connections for the inner tank.

iii) Depressurisation and purging of the in-tank pump column.

6.4 Instrumentation and process control for tanks:

The instrumentation shall be suitable for the temperature at which LNG is stored. All instrumentation shall be designed for replacement or repair under tank operating conditions in a hazardous gas zone area. Instrumentation for storage facilities shall be designed in such a way that the system attains fail-safe condition in case of power or instrument air failure.

The Level instrumentation for ESD function shall be separate and independent of the device installed for monitoring.

6.4.1 Level: LNG containers shall be equipped with two independent liquid level gauging devices to monitor tank levels.

Each system shall have High and High-High Level alarms.

Local Level indication should be available at grade level apart from remote indication in the control room.

Gauging devices shall be designed and installed so that they can be replaced without taking the storage container out of operation.
Refer Para 6.4.6. Density variation shall be considered in the selection of gauging devices.

6.4.2 Pressure: The storage tank shall be provided with pressure transmitters to continuously monitor and control pressure with an indication in the control room. Indication in field at grade level should also be provided.

Instrument for detecting high pressure shall be independent of the tank pressure monitoring instrument.

The sequence for over pressure control and protection shall be as follows:

1. High pressure alarms
2. Increasing the BOG system to the full load.
3. Further increase in pressure shall be controlled by release to flare.
4. Further increase in pressure shall be controlled by closing of automated inlet valve.
5. The final over pressure protection shall be tank design pressure and then PSV.

All the above pressure control and actuation shall be on independent pressure transmitters.

Independent pressure transmitters shall be provided for low pressure detection that will trip the boil- off gas compressors.

In the event of continued drop in tank pressure even after tripping of BOG compressor, two layers of protection against vacuum shall be provided.

1. Automatic admission of natural gas from outside source into the tank vapour space (From vaporiser outlet). The tripping of the pumps or LNG recirculation shall be envisaged after admission of natural gas and before admission of nitrogen for protection against vacuum.
2. In the unlikely event of above action not being sufficient, a set of vacuum breakers installed will admit air into the space between the suspended deck and outer roof to prevent permanent damage to the tank.

Independent pressure transmitters shall be provided for the natural gas and nitrogen gas admission for vacuum protection.

6.4.3 Temperature: As LNG is a product of varied compositions, it would be necessary to measure temperature of liquid and vapour over the full tank height; the sensors shall be located at 2 meter intervals or every 10% interval of the tank height, whichever is less.

Measuring and recording the formation of layers of liquid with different temperatures should warn the operator of a possible roll over phenomenon.

In addition, for monitoring of the initial chill down operation, temperature elements are required to be provided at tank base, shell of primary container and may be for secondary containers.

The temperature elements must be provided at various heights and at various locations of the base to ensure monitoring and proper chilling of the tanks.

6.4.4 **Gas Detectors**: Automatic gas detection system for monitoring leakage of LNG shall be installed. Adequate number of gas alarm sensors shall be placed on the tank roof in the vicinity of roof nozzles and places where the possibility of gas or liquid release exists including below elevated tanks. The facility shall be equipped with emergency shutdown system. The ESD should be able to operate remotely/ locally. The need for any automatic actuation of ESD may be assessed based on risk perceptions.

6.4.5 **Flame/ Leak Detectors (See para 6.3.6 also):** Monitoring leaks through the primary container in double containment systems shall be provided by one of the following means:

1. Temperature measurement sensors in annular space.
2. Gas detection
3. Differential pressure measurement

The arrangement shall have redundancy to prevent spurious alarms.

Tank external leak / spillage detection shall be installed at every location where leaks are credible. These detectors may activate appropriate process shutdowns or isolation, activate remote operated fire protection systems or initiate emergency actions by operators.

The following leak detection devices shall be considered:

1. Low temperature sensors for LNG spills.
2. Flammable gas detection of IR type. Battery limit fences shall have open path type detectors.
3. Flame detectors of the UV/IR type
4. Heat temperature detectors for protection of tank relief valve fires and activation of tail pipe extinguishing packages, if provided.
5. Smoke detectors of the ionisation type
6. CCTV systems in unmanned areas and unloading Jetty, capable of detecting vapour clouds, fitted with motion sensor alarms.

Communication system between field operators, Jetty terminal and pipeline dispatching centre shall be made available.

6.4.6 **Density Meters:**
Density meters shall be provided on the storage tanks to check the homogeneity of LNG.
The density of LNG in the Tank shall be monitored at all levels and analysis performed to alert the operator of any density layering.

6.4.7 A provision in the tank for endoscopic inspection (through insertion of camera) shall also be considered. This will be helpful to know the health of the tank in the absence of visual inspection of the tank.

6.4.8 An Uninterruptible Power Supply (UPS), with battery back-up shall be provided to all critical instrumentation control and safety (F&G) systems so that plant may be kept safe in case of emergencies.

7.0 INSULATION

7.1 CONTAINER INSULATION

7.1.1 Any exposed insulation shall be

- noncombustible
- contain or inherently shall be a vapor barrier.
- moisture free
- resist dislodgment of fire water

7.1.2 Where an outer shell is used to retain loose insulation, the shell shall be constructed of steel or concrete. Exposed weather proofing shall have a flame spread rating not greater than 25.

7.1.3 The space between the inner tank and outer tank shall contain insulation that is compatible with LNG and natural gas. The insulation shall be such that a fire external to the outer tank cannot cause deterioration to the insulation thermal conductivity by means such as melting or settling and reduction of performance of primary containment due to damage of any component of the insulation system.

7.1.4 The load bearing bottom insulation shall be designed and installed in such a manner that cracking from thermal and mechanical stresses does not jeopardize the integrity of the container.

7.1.5 Material used between the inner and outer tank only shall not be required to meet the combustibility requirements, provided the material and design of the installation comply with all the following:

i) The flame spread rating of the material shall not exceed 25, and the material shall not support continued progressive combustion in air.

ii) The material shall be of such composition that surfaces that would be exposed by cutting through the material on any plane shall have a flame spread rating not greater than 25 and shall not support continued progressive combustion.

iii) The combustion properties of material shall not increase/ change significantly as a result of long-term exposure to LNG or natural gas at the anticipated service pressure and temperature.

iv) The material in the installed condition shall be capable of being purged with natural gas or inert gas (Nitrogen). The natural gas remaining after purging shall not be significant and shall not increase the combustibility of the material.

v) The LNG tanks shall be adequately insulated in order to minimise the boil off gas generation due to heat leak from ambient. The extent of insulation depends on boil off considerations for which the storage tank is designed. Proper insulation shall be ensured in tank base, tank shell, tank roof, suspended deck, etc.

7.2 PIPING INSULATION

Piping systems shall be insulated to minimize energy loss , protect against condensation /frost and for personnel protection.

7.2.1 Insulating materials shall be specified in accordance with relevant codes and standards considering boil-off limitations and control of surface condensation.

7.2.2 To avoid water vapor ingress into pipe work insulation, an efficient vapor barrier shall be provided and placed around the insulation material. It shall be designed to remain gas tight even after undergoing the anticipated differential movements between the pipe and the various products that make up the insulation system.

7.2.3 Insulation materials should be waterproof and have an effective vapour barrier, coating and cladding.

Monitoring and inspection of piping for corrosion damage under insulation shall be done.

7.3 INSPECTION

To ensure the integrity and safe operation of cryogenic LNG pipelines throughout their lifecycle, non-intrusive techniques (AET, GWUT, PECT, DTS, infrared thermography or any other latest available technologies), in-service inspection methods should be considered.

8.0 FIRE PROTECTION, SAFETY AND EMERGENCY SYSTEMS

8.1 GENERAL

8.1.1 Fire protection shall be provided for all LNG facilities. The extent of such protection shall be determined by an evaluation based upon sound fire protection engineering principles, analysis of local conditions, hazards within the facility and exposure to or from other property. The evaluation shall determine as a minimum:

i) The type, quantity, and location of equipment necessary for the detection and control of fire, leak and spill of LNG, flammable refrigerants or flammable gases and all potential fire non-process and electrical fire.

ii) The methods necessary for protection of the equipment and structures from the effects of the fire exposure.

iii) Fire protection system (refer section 8.5)

iv) Fire extinguishing and other fire control equipment

v) The equipment and process systems to be operated with the emergency shutdown (ESD) system, including analysis of subsystems, if any, and the need for depressurizing specific vessels or equipment or piping during a fire emergency or hazardous release.

vi) The type and location of sensors necessary for automatic operation of the emergency shutdown (ESD) systems or its subsystems.

vii) The availability and duties of individual plant personnel and the availability of external response personnel operating an emergency.

viii) The protective equipment and special training necessary by the individual plant personnel for their respective emergency duties.

8.1.2 A detailed emergency procedure manual shall be prepared to cover the potential emergency conditions. Such procedure shall include but not necessarily be limited to the following:

a) Shutdown or isolation of various equipment in full or partial and other applicable steps to ensure that the escape of gas or liquid is promptly cut off or reduced as much as possible.
b) Use of fire protection facilities.
c) Notification to public authorities.
d) First aid.
e) Duties of personnel.
f) Communication procedure in case of emergency.

8.1.3 An updated emergency procedure manual shall always be available in the operating control room and fire control room.

8.1.4 All personnel shall be trained in their respective duties contained in the emergency manual. Those personnel responsible for the use of fire protection or other prime emergency equipment shall be trained in the use of equipment. Refresher programmes for training of all personnel shall be conducted at least on annual basis.

8.2 IGNITION SOURCE CONTROL

8.2.1 Smoking within the LNG terminal is not allowed. Spark potential items like lighters, mobile phone, camera, etc. are not allowed. However, intrinsically safe devices may be allowed.

8.2.2 Work permit system shall be carried out as per the latest version of OISD-STD-105.

8.2.3 Vehicle and other mobile equipment that constitute potential ignition sources shall be prohibited within impounding areas or within 50 ft (15 m) of containers or equipment containing LNG, flammable liquids or flammable refrigerants except when specifically authorised and under constant supervision.

8.2.4 Any vehicle or other mobile equipment without PESO approved flame arrestor at the exhaust shall not be permitted within the plant areas.

8.2.5 Electrical equipment, switches, plugs, wiring etc. used within the hazardous area of process plant must be of flameproof type as per OISD-STD-113.

8.2.6 To avoid formation and accumulation of static charge, all the equipment, storage tank, pipeline containing LNG should be properly bonded and earthed. Continuity of earthing and bonding should be checked as per OISD-RP-110 on regular intervals and records shall be maintained.

8.3 EMERGENCY SHUTDOWN SYSTEMS

8.3.1 Each LNG facility shall incorporate an emergency shutdown (ESD) system that which when operated:

- Isolates or shuts off a source of hazardous fluids.
- Shuts down equipment, operation of which, if continued may add to an emergency.

8.3.2 When equipment shutdown results in an additional hazard or substantial mechanical damage to the equipment, the shut-down of such equipment or its auxiliaries shall be omitted from the ESD system, provided that continuous release of flammable or combustible fluid are controlled.

8.3.3 Vessel containing liquids that are subjected to metal overheating and catastrophic failure from fire exposure and not otherwise protected shall be depressurised by the ESD system.

8.3.4 Initiation of ESD system shall be either manual, automatic, or both manual and automatic, depending upon result of evaluation performed in accordance with fire protection facilities. Manual actuation of ESD system or critical equipment shall be located in an area accessible in the event of an emergency and shall be at least 15 m away from the equipment and should be marked with their designated function and explicitly visible during day and night.

8.3.5 The emergency shutdown system (s) (ESD) shall be of failsafe design and shall be installed, positioned, or protected to ensure ease of operation during an emergency or failure of the normal control system. If the ESD is not failsafe, all components located within 15 meters of the controlled equipment shall either be:

i) Installed or located to prevent exposure to fire, or

ii) Protected to withstand fire exposure for a minimum duration of 10 minutes.

8.3.6 The scope and configuration of ESD systems and components should be developed during the facility design. The emergency shutdown system shall consider process safety as well as leakage of gas, fire, smoke as well as cold detection and linear detection. Depending on the extent of severity, the level of shut down is required to be graded and considered. This could be by way of section isolation or total complex shut down.

8.3.7 Communication and interlock shall be provided between ship and terminal control room. Provision shall be given in the jetty for the above facility. During unloading operation, the terminal operator shall take control of the unloading. In addition to automatic shutdown system (ESD) the terminal operator shall be in a position to initiate manual shut down of unloading operations from a safe distance.

8.4 FIRE AND LEAK DETECTION SYSTEM

8.4.1 Those areas including enclosed buildings that have a potential for the presence of flammable gas, or any other fire hazard shall be monitored.

8.4.2 Continuously monitored low temperature sensors or flammable gas detection systems shall sound an alarm at the plant site and at a constantly attended location. Flammable gas detection systems shall initiate this alarm at 20 percent as well as 40 percent of lower flammable limit of the gas or vapour being monitored.

8.4.3 The Fire detectors shall initiate an audio and visual alarm at the plant site and at a constantly attended location.

8.4.4 Fire detectors shall activate appropriate portions of the emergency shutdown system with a voting system.

8.4.5 The detection system determined shall be designed, installed and maintained in accordance with dispersion study and relevant OISD standards.

8.4.6 All hazardous zones shall be provided with gas detection and shall be hooked up with the ESD system.

8.5 FIRE PROTECTION SYSTEM FOR LNG TERMINAL

The primary source of fire and explosion hazard is from a leak or spill from the LNG storage or transfer systems.

a. The fire protection facilities for the LNG terminal shall be designed to fight two major fires simultaneously anywhere in the LNG terminal including the jetty area.

b. Firefighting system of the port/jetty area shall be designed as per OISD-STD-156 latest version.

8.5.1 GENERAL CONSIDERATIONS

The size of LNG plant, pressure and temperature conditions, size of storage, plant location and terrain determine the basic fire protection need. Layout of an installation shall also be done in accordance with OISD-STD-118 to ensure adequate firefighting access, means of escape in case of fire and also segregation of facilities so that the adjacent facilities are not endangered during a fire.

Material of construction for infrastructure facilities shall conform to National Building Code (NBC)/ statutory regulations.

The following fire protection facilities shall be provided depending upon the nature of the installation and risk involved.

- Fire Water System
- Foam System
- Clean Agent Fire Protection system
- Carbon Dioxide System
- Dry Chemical Extinguishing System
- Detection and Alarm system
- Communication System
- Portable firefighting equipment
- Mobile firefighting equipment
- First Aid Fire Fighting Equipment.

8.5.2 DESIGN CRITERIA

The following shall be the basic design criteria for a fire protection system.

i) Facilities shall be designed on the basis that city fire water supply is not available close to the installation.

ii) Fire protection facilities shall be designed to fight two major fires simultaneously anywhere in the installation.

iii) LNG tank farms and other areas of installation where hydrocarbons are handled shall be fully covered by hydrant System.

iv) Water spray requirement with mode of operation to be considered in line with provisions of OISD-STD-173 for electrical installation.

Note: In case additional facilities other than LNG are installed or planning to be installed, relevant OISD standards shall be followed.

8.5.3 LOADING / UNLOADING GANTRY

LNG loading/ unloading Tank Truck (TT) gantries shall be provided with water spray system.

Automatic fixed water spray system is provided in TT gantry, the gantry may be divided into suitable number of segments and three largest segments operating at a time shall be considered as single risk for calculating the water requirement.

Accordingly, a provision shall be made to actuate the water spray system from a safe approachable central location i.e. affected zone and adjoining zones.

8.5.4 FIRE WATER SYSTEM

Based on the site requirement, water shall be used for fire extinguishment, fire control, cooling of equipment and protection of equipment as well as personnel from heat radiation. For these purposes water in appropriate form should be used such as straight jet, water fog, water mist, water curtain, water spray, deluge/ sprinkler, for foam making etc.

Fire water system shall comprise of fire water storage, fire water pumps and distribution piping network along with hydrants and monitors, as the main components.

8.5.4.1 FIREWATER FLOW RATE

Two of the largest flow rates calculated for different sections as shown below shall be added and that shall be taken as design flow rate. An example for calculating design major fire water flow rate is given in Annexure-IV.

Fire water flow rate for tank farm shall be aggregate of the following:

i) Water flow calculated for cooling all tank situated in the same dyke area at a rate of 3 lpm/m^2 of tank shell and roof area.

ii) Water flow rate requirements for firefighting in other major areas shall be calculated based on criteria in terms of lpm/m^2

iii) Fire water flow rate for supplementary stream, shall be based on using 4 single hydrant outlets and 1 HVLR simultaneously. Capacity of each hydrant outlet as 36 m^3/hr and of HVLR as 228 m^3/hr.

iv) The supplementary water stream requirement is in addition to design flow rates.

Note: Double Wall Storage Tank (DWST) on fire having concrete outer shell shall be considered in single fire zone if other DWST having concrete outer shell is located more than 30m.

8.5.4.2 HEADER PRESSURE

The fire water system shall be designed for a minimum residual pressure of 7.0 kg/cm^2 g at the hydraulically remotest point of application at grade level and minimum 5.25 kg/cm^2 g at landing valve at elevated structure on designed flow rate at that point.

8.5.4.3 STORAGE & MAKE-UP WATER

8.5.4.3.1 FIREWATER STORAGE

Water for firefighting shall be stored in any easily accessible surface or underground lined reservoir or above ground tanks of steel, concrete or masonry. The fire water storage should be located in line with OISD-STD-118 so as to avoid any damage in case of fire/explosion. The effective capacity of the reservoir above the level of suction point shall be minimum 4 hours aggregate working capacity of main pumps (excluding standby pumps). Where rate of makeup water supply is 50% or more, this storage capacity can be reduced to 3 hours aggregate working capacity of main pumps.

Storage reservoir shall be in two equal interconnected compartments to facilitate cleaning and repairs. In case of aboveground steel tanks there shall be minimum two tanks each having 50 % of required capacity.

Large natural reservoirs having water capacity exceeding 10 times the aggregate fire water requirement can be left unlined.

In addition to fire water storage envisaged as above, emergency water supply in the event of depletion of water storage shall be considered. Such water supplies can be connected from cooling water supply header and/or treated effluent discharge headers.

Fire water supply shall be from fresh water source such as river, tube well or lake. Where fresh water source is not easily available, fire water supply can be sea water or other acceptable source like treated effluent water. In case sea water or treated effluent water is used for firefighting purposes, the material of the pipe selected shall be suitable for the service.

8.5.4.3.2 MAKE UP WATER

Suitable provisions shall be kept for makeup firewater during firefighting time.

8.5.4.4 FIREWATER PUMPS

Firewater pumps shall be used exclusively for firefighting purposes.

8.5.4.4.1 TYPE OF PUMPS

Fire water pumps shall be of the following type:

i) Electric motor driven centrifugal pumps

ii) Diesel engine driven centrifugal pumps

The pumps shall be horizontal centrifugal type or vertical turbine submersible pumps.

Each pump shall be capable of discharging 150% of its rated capacity at a minimum of 65% of the rated head. The shut-off head shall not exceed 120% of rated head, for both horizontal pumps and 140% in case of vertical turbine type pumps.

The number of diesel driven pumps shall be a minimum of 50% of the total number of pumps (inclusive of standby pumps). A minimum of 50% of the total flow requirement should be available through diesel driven pumps all the time.

8.5.4.4.2 CAPACITY OF MAIN PUMPS

The capacity and number of main fire water pumps shall be fixed based on design fire water rate, worked out on the basis of design criteria as per section 8.5.4.5. The capacity of each pump shall not be less than 400 m^3/hr.

All pumps shall be identical with respect to capacity and head characteristics.

8.5.4.4.3 STANDBY PUMPS

i) At least one standby fire water pump shall be provided up to 2 nos. of main pumps. For main pumps 3 nos. and above, Min. 2 nos. of standby pumps of same type, capacity & head as main pumps shall be provided. Fire water pumps shall be equal capacity and head.

ii) In cases where two sets of firewater storage and pumps are provided, the number of pumps at each location shall be according to hydraulic analysis of piping network.

8.5.4.4.4 JOCKEY PUMPS

The fire water network shall be kept pressurized by jockey pump at all times. At least one standby fire water jockey pump shall be provided. The capacity of the pump shall be sufficient to maintain system pressure in the event of leakages from valves etc.

Its head shall be higher than the main fire water pumps. Auto cut-in / cut-off facility should be provided for jockey pumps

8.5.4.4.5 POWER SUPPLY FOR FIRE WATER PUMPS

i) The power supply to the electric driven pumps should be distributed from a switchboard having two separate incomers such that failure of any one incomer source does not disrupt the power supply to all the electric motor driven pumps.

ii) A direct feeder dedicated only to fire water pumps shall be laid from the sub-station to ensure a reliable power supply. The cables for firewater pumps should be laid in a manner that all the cables are not susceptible to damage all at once.

iii) The diesel engines shall be quick starting type with the help of push buttons located near the pumps, or at remote location.

iv) Each diesel engine shall have an independent fuel tank adequately sized for minimum 6 hours of continuous running of the pump.

v) Main fire water pumps shall start automatically and sequentially with pressure transmitter/ pressure switch on fire water mains. The system shall ensure auto-start of the standby pump in case a pump in sequence fails to take start.

8.5.4.4.6 LOCATION OF PUMPS

The location and inter-distances for firewater pump house shall follow OISD-STD-118.

8.5.4.5 DISTRIBUTION NETWORK

8.5.4.5.1 LOOPING & MAINTAINABILITY

The fire water network shall be laid in closed loops as far as possible to ensure multi-directional flow in the system. Isolation valves shall be provided in the network to enable isolation of any section of the network without affecting the flow in the rest. The isolation valves shall be located near the loop junctions. Additional valves shall be provided in the segments where the length of the segment exceeds 300 metres.

For ease of maintenance, firewater pumps should be segregated in two groups by providing an isolation valve on common discharge header of pumps.

In case of saline water, flushing connections with isolation valves should be provided at suitable locations in the firewater ring main.

For branch piping, an isolation valve shall be provided at the take-off point.

Permanent connection shall not be taken from fire water line / system for purposes other than fire protection/ fire prevention.

8.5.4.5.2 CRITERIA FOR ABOVE / UNDERGROUND NETWORK

The firewater network piping should normally be laid above ground at a height of at least 300 mm above finished ground level. Fire water mains shall not pass through buildings or dyke areas, foundations of equipment or structures. However, the fire water network piping shall be laid below ground level at the following places. Pipes made of composite material shall be laid underground.

i) Road crossings.

ii) Places where the above ground piping is likely to cause obstruction to operation and vehicle movement and get damaged mechanically.

iii) Where frost condition warrants, the ring main system shall be laid underground beneath the frost layer.

iv) Pipes made of composite material shall be laid underground only.

8.5.4.5.3 PROTECTION FOR UNDERGROUND PIPELINES

Where the pipes are laid underground, the following protections shall be provided:

i) The fire water main shall have at least one-meter earth cushion in open ground and 1.5 metre earth cushion under the roads. In the case of crane movement areas, pipes should be protected with concrete/ steel encasement.

ii) The mains shall be provided with protection against soil corrosion by suitable coating/wrapping. Where pipes are coming out/going in from/into the ground, the underground anticorrosion coating on the pipe will continue for a length of at least 300 mm above ground.

iii) Pipe supports under the pipeline shall be suitable for soil conditions.

iv) In concrete paved areas, removable paver blocks should be provided over the buried fire water lines for ease of maintenance without cutting pavement.

8.5.4.5.4 PROTECTION FOR ABOVE GROUND PIPELINES

Where the pipes are laid above ground, the following protection shall be provided:

i) The firewater mains shall be laid on independent sleepers by the side of road. These shall not be laid along with process piping on common sleepers.

ii) The mains shall be supported at regular intervals not exceeding 6 m. For pipeline size less than 150 mm, support interval shall not exceed 3 m.

iii) The system for above ground portion shall be analyzed for flexibility against thermal expansion and necessary expansion loops shall be provided wherever called for.

8.5.4.5.5 HYDRAULIC ANALYSIS & SIZING OF FIREWATER NETWORK

i) The hydraulic analysis of the network shall be done. Also, whenever fire water demand increases due to addition of plant & facilities or extensive extension of network, fresh hydraulic analysis shall be carried out.

ii) Fire water distribution ring main shall be sized for 120% of the design water rate. Design flow rates shall be distributed at nodal points to give the most realistic way of water requirements in an emergency. Several combinations of flow requirements shall be assumed for the design of the network.

iii) The velocity of water shall not exceed 5 m/s in fire water ring main.

8.5.4.5.6 FIRE HYDRANTS & FIXED WATER MONITORS – DETAILS

i) Fire water hydrants/ fixed water monitors shall be provided on the fire water network. Each of these connections shall be provided with independent isolation valves.

ii) Hydrants / Monitors shall be located along roadside berms for easy accessibility as far as possible. Hydrants/ Monitors shall not be positioned on dyke wall.

iii) Hydrants / Monitors shall be located with branch connections and not directly over main header for easy accessibility and to avoid corrosion of main header due to dripping from hydrants/accumulation of water.

iv) Hydrant/ Monitors should be painted in line with OISD-GDN-115.

8.5.4.6 HYDRANTS

i) Hydrants shall be located keeping in view the fire hazards at different sections of the premises to be protected and to give most effective service. At least one hydrant post shall be provided for every 30 m of external wall measurement or perimeter of unit battery limit in case of hazardous areas. Hydrants protecting utilities and non-plant buildings can be spaced at 45 m intervals. The horizontal range and coverage of hydrants with hose connections shall not be considered more than 45 m.

ii) The hydrants shall be located at a minimum distance of 15 m from the periphery of storage tank or hazardous equipment under protection. For process plant location of hydrants shall be decided based on coverage of all areas. In the case of buildings, this distance shall not be less than 5 m and more than 15 m from the face of building.

iii) Double headed (4" standpost with two independent Type-A landing valves) shall be used. All hydrant outlets shall be situated at a workable height of about 1.2 meters above ground level. In addition, fire hydrant post with four separate landing valves (Type A) on 6" stand post should be considered at suitable location around process/ tank farm area.

iv) Provision of hydrants within buildings shall be in accordance with Standard IS Standard-3844/ NBC.

8.5.4.7 MONITORS

i) The requirement of monitors shall be established based on hazard involved, layout considerations and where it is not possible to approach the higher levels. A minimum of 2 monitors shall be provided for the protection of each such area.

ii) Monitors shall be located to direct water on the object as well as to provide water shield to firemen approaching a fire. The monitors, including HVLRM, shall be installed at a minimum of 15 m away from the equipment or facilities to be protected. However, the distance of monitors shall not exceed the performance range of the monitor from the hazard to be protected.

iii) Field adjustable variable flow monitors with suitable capacity shall be installed at critical locations. These shall be in compliance to the national or international standard.

8.5.4.8 DRY/WET RISERS

Dry/ Wet Risers with hydrants should be provided on each floor of technological structures and multistorey buildings.

8.5.4.9 FIXED REMOTE / MANUAL OPERATED HIGH VOLUME LONG RANGE WATER OR WATER CUM FOAM MONITORS

i) Minimum 2 no. of remote / manual operated high-volume long-range water monitors (Minimum capacity 1000 GPM & above) shall be variable flow type with flow adjustable manually in the field shall be provided in areas like LNG Tank farm area, inaccessible areas such as compressor house, etc.

ii) The basic water cum foam monitors shall be complying to the National or International Standard. The electrical or hydraulic remote-control mechanism shall be in line with Hazardous Area Classification.

iii) The location of HVLRM is to be planned in such a way that the very purpose of these monitors is served, and the throw of the monitors is safely delivered at the aimed object. The distance of monitors shall not exceed the performance range of the monitor from the hazard to be protected.

8.5.4.10 HOSE BOXES OR STATIONS

Provision of hose boxes or stations should be given at critical locations for housing hoses and nozzles.

8.5.4.11 MATERIAL SPECIFICATIONS

All the materials required for firewater system using fresh water shall be of approved type as indicated below and in line with OISD-GDN-115.

i) **Pipes**: Carbon Steel as per IS:3589/ IS:1239/IS:1978 or Composite materials as per API 15LR/API 15 HR or its equivalent shall be used.

In case saline water/ treated effluent water is used, the fire water main of steel pipes shall be, internally cement mortar lined or glass reinforced epoxy coated or made of pipe material suitable for the quality of water. The material selection for the fire main line shall consider the quality or impurities in the water. Alternatively, pipes made of composite materials shall be used. Cast iron pipes shall not be used for fire water services.

ii) **Isolation Valves:** Cast Steel valves shall be used in all areas including unit areas, offsite and fire water pump stations.

Isolation valves having open/ close indication shall be gate valves more than 16” should be provided with gear mechanism.

iii) All fire equipment shall comply to the National or International Standard requirements.

8.5.4.12 FIXED WATER SPRAY SYSTEM

a) General

It is a fixed pipe system connected to a reliable source of water supply and equipped with water spray nozzles for specific water discharge and distribution over the surface of area to be protected. The piping system is connected to the Fire water network water supply through an automatically or manually actuated valve which initiates the flow of water.

i) Fixed water spray system should be provided in high hazard areas where immediate application of water is required.

ii) Water supply patterns and their densities shall be selected according to need. Fire water spray system for exposure protection shall be designed to operate before the possible failures of any containers of flammable liquids or gases due to temperature rise. The system shall, therefore, be designed to discharge effective water spray within shortest possible time.

b) Water Spray Application Rates

i) For the storage tanks, water sprays shall be provided on the tank shell including the roof and the appurtenances on the tank. For single containment tanks, water application rate for the tank roof and walls shall be calculated using method detailed in Appendix 5 of IP Model Code of Safe Practice Part 9. The water application rate on the appurtenances shall be 10.2 lpm/ m2 as per this code. For double/full containment metallic tanks, the water application rate for the tank roof and walls shall be 3 lpm/m2, required for cooling the outer shell of tanks adjoining to the one on fire. Water spray for shell of pre-stressed concrete double/full containment LNG tanks is not considered necessary. However, water spray shall be provided on roof for all the LNG tanks regardless of the construction material used.

ii) The water application rate applicable to other equipment shall be as follows:

Vessels, structural members

Piping & valves manifolds	: 10.2 lpm/ m^2
Pumps	: 20.4 lpm/ m^2

iii) The water spray shall be divided into subsystems to provide protection to the different sections of the tank, i.e., one system to cover each segment of vertical wall, one to cover the dome roof and the tank appurtenances. The roof section should be provided with duplicate 100% risers preferably on the other side of the tank.

iv) The deluge valves on the water spray systems on the tanks as well as the pumps, compressors, vessels etc. shall be actuated automatically through a fire detection system installed around the facilities with provisions of manual actuation from control room or locally at site.

8.5.4.13 FIXED WATER SPRINKLER SYSTEM

i) A fixed water sprinkler system is a fixed pipe tailor made system to which sprinklers with fusible bulbs are attached. Each sprinkler riser/ system includes a controlling valve and a device for actuating an alarm for the operation of the system. The system is usually activated by heat from a fire and discharges water over the fire area automatically.

ii) The water for the sprinkler system shall be tapped from the plant fire hydrant system, the design of which should include the flow requirement of the largest sprinkler installation.

iii) Minimum Design density and assumed area of operation of fixed water sprinkler system shall be in accordance with hazard area occupancies classification given in IS 15105 or NFPA 13.

iv) The design flow for sprinkler installation would depend on the type of hazard and height of piled storage. The designed water flow shall be restricted to a minimum of 100 m^3/hr and to a maximum 200 m^3/hr. The design flow rate for other areas shall be taken as 10.2 lpm/ m^2 of the area protected by sprinkler installation, subject to a minimum of 150 m^3/hr and a maximum of 400 m^3/ hr. Higher water application rates should be used if called for depending on risk involved.

8.5.5 FOAM SYSTEM

For combating large hydrocarbon fire particularly in a contained area like storage tank, foam has proved useful for its inherent blanketing ability, heat resistance and security against burn-back. Efficient and effective foam delivery system is a vital tool for its usefulness in controlling fire. The process of adding or injecting the foam to water is called proportioning. The mixture of water and foam compound (foam solution) is then mixed with air in a foam maker for onward transmission to burning surface.

8.5.5.1 HIGH EXPANSION FOAM

For combating large LNG fires particularly in a contained area like dyke area, foam has proved useful for its inherent blanketing ability, heat resistance and security against burn-back. Aqueous Film Forming Foam (AFFF) compound is technically superior and compatible with other firefighting agents.

Efficient and effective foam delivery system is a vital tool for its usefulness in controlling the fire.

The process of adding or injecting the foam to water is called proportioning. The mixture of water and foam compound (foam solution) is then mixed with air in a foam generator for onward transmission to burning surface.

For single containment tanks, double containment tanks, full containment tank having metallic outer tank and membrane tanks having metallic outer tank, high expansion foam systems shall be provided as per NFPA 11A. Water turbine powered high expansion foam generators shall be located on the impounding area around the storage tanks. Foam units comprising storage facilities and pumps shall be provided in a safe area removed from the protected risk and shall be accessible in an emergency.

Portable high expansion foam generators should be considered, suitable for coupling to hydrant hose lines for isolated LNG spills.

8.5.5.2 FOAM CONVEYING SYSTEMS

The system consists of an adequate water supply, supply of foam concentrate, suitable proportioning equipment, a proper piping system, and discharge devices designed to adequately distribute the foam over the hazard.

Conventional systems are of the open outlet type in which foam discharges from all outlets at the same time, covering the entire hazard within the confines of the system. There are three types of systems:

(i) Fixed

(ii) Semifixed

(iii) Mobile

8.5.5.3 FIXED FOAM SYSTEM

Fixed foam conveying system comprises of fixed piping for water supply at adequate pressure, foam concentrate tank, eductor, suitable proportioning equipment for drawing foam concentrate and making foam solution, fixed piping system for onward conveying to foam generator for making foam.

8.5.5.4 SEMIFIXED FOAM SYSTEM

Semi-fixed foam system gets supply of foam solution through the mobile foam tender. A fixed piping system connected to high expansion foam generator conveys foam to the surface of dyke area.

8.5.5.5 MOBILE FOAM SYSTEM

Mobile system includes foam producing unit mounted on wheels which can be self-propelled or towed by a vehicle. These units supply foam through high expansion foam generator to the burning surface.

8.5.5.6 DURATION OF FOAM DISCHARGE

Spill fire protection - 30 minutes.

8.5.5.7 FOAM QUANTITY REQUIREMENT

The aggregate quantity of foam solution for a single largest dyke area should be calculated for a minimum period of 30 minutes. The quantity of foam compound required should be calculated based on 3% or 6% concentrate.

A typical example showing calculation of foam compound requirement is given at Annexure-V.

8.5.5.8 FOAM COMPOUND STORAGE

i) Foam compound should be stored as explained in IS-4989:2006/UL-162.

ii) Type of foam compound used can be protein or fluoroprotein or AFFF. Minimum life of foam compound shall be taken as per manufacturer's data.

~~iii)~~ Foam compound shall be tested periodically in line with OISD-STD-142 to ensure its quality and the deteriorated quantity replaced.

iv) Quantity of foam compound equal to 100% of requirement as calculated should be stored in the installation.

8.5.6 CLEAN AGENT BASED PROTECTION SYSTEM FOR PROCESS CONTROL ROOM AND SRR

Clean Agent based fire suppression system should be provided in line with OISD-STD-163.

Selection of Clean Agent and design of Fire protection system shall follow the Standard on "Clean Agent Extinguishing systems NFPA Standard 2001 (latest edition) or equivalent including its safety guidelines with respect to "Hazards to Personnel", electrical clearance and environmental factors in line with environmental considerations of Kyoto & Montreal protocols and latest MoEFCC guidelines.

The gas-based (clean agent) extinguishing system should be provided in control rooms which are manned continuously (24x7 basis) and where a suitably designed smoke detection and alarm system is available. Clean Agent System along with all equipment, software for hydraulic calculations and corresponding design concentration of clean agent shall be approved by BIS/UL/FM/VdS/LPCB. Cylinder along with cylinder valve assembly shall also be PESO approved."

QUANTITY AND STORAGE OF CLEAN AGENT

Each hazard area to be protected by the protection system shall have an independent system.

The time needed to obtain the gas for replacement to restore the systems shall be considered as a governing factor in determining the reserve supply needed. 100% standby charge of clean agent containers shall be considered for each protected hazard.

Storage containers shall be located as near as possible to hazard area but shall not be exposed to fire.

Storage containers shall be carefully located so that they are not subjected to mechanical, chemical or other damage.

All the components of the system shall be capable of withstanding the heat of fire and severe weather conditions.

8.5.7 BATTERY FIRE

From consumer electronics and electric vehicles to industrial equipment and energy storage systems, lithium-ion batteries can be found everywhere. Due to the materials utilized to manufacture these batteries, the scene of a lithium-ion battery fire can best be compared to a hazmat (hazardous material) incident. Although extinguishing fire is a primary concern, flammability, explosivity and toxicity must be addressed to ensure proper protection and prevent re-ignition.

In a cell or battery, ignition occurs when lithium-ion batteries enter a chemical process called thermal runaway. Cells produce heat and emit explosive and toxic gases, developing a self-propelling loop that leads to thermal runaway propagation. This chain reaction will continue to impact adjacent cells until thermal runaway propagation is halted. Confined spaces such as server rooms, automobiles, aircraft, spacecrafts, shipping containers, energy storage systems and more can pose an increased risk.

During the battery fire, a flammable electrolyte used within cells catches fire. Thus, fire starts inside a cell which makes it difficult to reach. All the conventional firefighting media, including plain water, works at the macro / surface level to remove heat. In the case of battery fire, due to deep seated nature of fire, the effectiveness of conventional firefighting media is very less and thus they are required in humongous quantity to achieve extinguishment.

In many battery fires, one of the byproducts of the oxidation (burning) chemical process is oxygen, thus the fire is fed by creating its own oxygen. Hence, any firefighting media / techniques that works on a principle of oxygen depletion find limited success.

For fire protection of Li ion battery, Encapsulator Agent (EA) technology should be used in line with NFPA 18A.

8.5.8 CARBON DIOXIDE SYSTEMS

Fixed CO_2 systems shall be provided in Turbo generator enclosure, Gas turbine enclosure etc. as per OISD STD 173. A fixed CO_2 systems should be designed and installed in accordance with NFPA-12.

Before the CO_2 flooding system is operated, persons in the confined area shall be evacuated.

Carbon dioxide is an odorless and colourless inert gas having a proven fire extinguishing property. When applied in a proper manner and in proper quantities on a fire hazard, it extinguishes fire by cutting off the oxygen and creating an inert atmosphere around the hazard. If the inert atmosphere is maintained for a reasonable time, the possibility of flash back also is reduced.

Carbon dioxide is best applied to a fire hazard through a fixed system consisting of CO_2 storage, distribution piping and discharge nozzles.

8.5.9 DRY CHEMICAL EXTINGUISHING SYSTEM

The extinguishing system comprises supplying the dry chemical agent, manually or automatically, through a distribution system onto or into the protected hazard.

Fixed dry chemical powder or nitrogen snuffing systems shall be provided for each relief valve outlet of the LNG storage tanks. Each set shall provide two shots of dry chemicals in the event of ignition during venting.

8.5.9.1 RECOMMENDED USE:

A dry chemical powder extinguishing system can effectively be used on following hazards.

- Electrical hazard such as transformers or oil circuit breakers.
- Combustible solids having burning characteristic like naphthalene or pit which melts while on fire.
- Class 'A','B','C' & 'D' fire using multipurpose dry chemical. Requirement for each item should be finalised while deciding design basis.

8.5.9.2 SYSTEM DESIGN:

Basic requirement of designing the dry chemical extinguishing system is to provide for sufficient quantity and rate of discharge depending upon the hazard.

The system consists of dry chemical powder and expellant gas container assemblies of capacity sufficient for given hazard with distribution piping and discharge nozzles. System can be actuated manually or automatically on visual or automatic means of detection. Alarm and indication shall be provided to show that the system has operated, and a personnel response is needed.

Personnel safety shall include training, warning signs, discharge alarm, respiratory protection and prompt evacuation of personnel.

Following types of systems can be provided to protect a hazard:

a. Total flooding system

b. Local application system

c. Hand hose line system, and

d. `Pre-engineered system

(Refer NFPA-17 for limitations & precautions for use of dry chemical and for system design)

8.5.10 MODERNIZATION PLAN FOR FIREFIGHTING EQUIPMENT

As a part of modernization of firefighting equipment, following may be explored in addition to the OISD-GDN-115 based on its suitability to hazards, long term maintainability, reliability and certification.

a. Hydro-chem Nozzle

The nozzle is designed to throw liquid (foam solution / water) and DCP simultaneously. The liquid forms a hollow jet and the DCP is inside the hollow space. Application wise DCP provides quick knock down of flames and the liquid provides cooling. Also, DCP can be thrown at a longer distance than conventional DCP hand nozzle or monitor.

b. Fire extinguisher of HDPE / Composite material body

This will ensure less weight and longer shelf life. This is also included in IS 15683:2018.

8.5.11 FIRST-AID FIRE FIGHTING & OTHER EQUIPMENT

i. Portable or wheeled fire extinguishers suitable for gas fires, preferably of the dry chemical type shall be made available at strategic locations.

ii. Fixed fire extinguishing and other fire control systems that may be appropriate for the protection of specific hazards, are to be provided.

8.5.11.1 CRITERIA TO DETERMINE THE QUANTITY NEEDED:

Portable firefighting equipment shall be provided in LNG plant as indicated in the following table

	Description	Norms/criteria to determine the quantity needed
Conventional firefighting equipment		
i)	Dry chemical powder (DCP)* fire extinguishers - 9 kg capacity: IS:15683/ UL299 (Stored pressure type or Cartridge type)	While selecting the Extinguisher, due consideration should be given on the factors like flow rate, discharge time and throw in line with IS: 2190 / UL711. Extinguisher to be located in process area, Compressor houses, tank truck/ tank loading areas, substations, Workshops, laboratory, power station buildings etc. The number should be determined based on the max. traveling distance of 15 M in the above areas. At least one fire extinguisher shall be provided for every 250 m^2 of hazardous operating area. Fire extinguisher should also be considered at elevated location such as technical structure There shall be not less than two extinguishers at one designated location e.g. Compressor house.
ii)	Dry chemical powder fire extinguishers 25/50/75 kg capacity: IS:10658/ UL299 (in addition to 9kg DCP requirement)	The extinguishers with the selection criteria, viz. flow rate, discharge time and throw mentioned as above, to be located in critical operating areas. At least one fire extinguisher should be provided for every 750 m^2 of hazardous operating area.
iii)	CO_2 extinguishers 4.5/6.5/9.0/22.5 kg capacity (IS:16018/ IS 15683/UL154)	To be in substations, power stations, office buildings and control room. The number should be determined based on the maximum traveling distance of 15 metre. At least one fire extinguisher shall be provided for every 250 m^2 of hazardous operating area. There shall not be less than 2 nos. extinguishers at one designated location e.g. control room.
iv)	Portable clean agent extinguishers	This should be as an alternative to CO_2 extinguisher. To be located in control rooms, computer rooms, laboratories and office buildings.
v)	Portable water-cum-foam monitor.	Minimum 2 no. for LNG Plant
vi)	Portable Encapsulator agent extinguisher	To be located in battery rooms.
vii)	Rubber hose reel (25mm)	To be located in Process unit battery limits and other process areas for quenching of incipient fires.

Note : 1) In place of 9 kg, 6 Kg DCP extinguisher shall be provided at elevated locations.

2) For substations, power stations; the number and area for placement of extinguishers should be decided as per OISD STD-173 requirements.

8.5.11.2 OTHER EMERGENCY EQUIPMENT:

Additionally, the following items shall also be provided.

i) Thermal imaging camera: As an aid to the fireman during firefighting operation to locate the seat of the fire and to facilitate search and rescue operation in smoky area e.g. during cable gallery.

ii) Personal Protective Equipment required during Fire Fighting like Water gel-based blanket, Fire Proximity Suit, Self-contained breathing apparatus, Airline breathing apparatus, Safety Helmets, Fire Helmets, Stretcher, First Aid box, Rubber hand gloves, canister mask etc.

iii) Other equipment like Portable Gas detectors, Explosive meter, Oxygen meter, Hand operated siren, Red/Green Flag for fire drill, Safe Walk roof top ladder, emergency lighting, intrinsically safe portable megaphone, various leak plugging gadgets, oil dispersants and oil adsorbents, lifting jacks (for rescue of trapped workers), etc.

iv) Water curtains should be considered to mitigate gas releases, dilute the LNG vapour and facilitate its dispersion.

Note: The number of units required for these should be decided on case-to-case basis.

8.5.11.3 MOBILE FIRE FIGHTING EQUIPMENT:

8.5.11.3.1 Fire Tenders:

The exact number of fire tenders shall be higher of the items (a) or (b):

a) The quantities firmed up in each case based on two simultaneous major fires taking into consideration the size, location of the plant and statutory requirements.

b) The quantities indicated below.

  i) One foam tender should have foam tank capacity of 3000/3600 litre and the pump capacity of minimum 4000 lpm at 8.5 kg/cm^2.

  ii) One DCP tender having 2 vessels of 2000 kg capacity each with Nitrogen as expellant gas.

  Alternately, a minimum of 2 nos. of multipurpose fire tender should be provided.

A monitor should have a variable throw of 15/25/40 kg/sec. The throw of the monitor shall be 40 to 50 M for the DCP charge.

Note: Multipurpose firefighting tender should be used as a single self-sufficient unit of having capacity of discharging foam, water/ water mist & DCP and thus performing multiple functions effectively, individually or together, saving power and time in combating a fire. Such a multipurpose firefighting tender should be used in lieu of one foam tender/ DCP tender.

8.5.11.3.2 Other Firefighting Equipment

Following, other firefighting equipment shall be provided:

i) Emergency rescue equipment like cutters, expanders, inflatable lifting bags, leak pads, protective clothing, trolley mounted BA set

ii) Fire Hoses: IS 636: Type I or Synthetic hose of Type III.

The hose length shall be calculated as follows:

iii) For installation with hydrants up to 100: - One 15 m hose length/hydrant.

  i) For installation with more than 100 hydrants:

  - One 15 m hose length/hydrant, for the first 100 hydrants; and,
  - One 15 m hose length for every 10 hydrants above 100.

ii) Fire jeep (s) with two-way radio communication facility and towing facility shall be available all the time.

iii) One ambulance fitted with medical aid and suitable arrangements.

iv) Other accessories, foam making branch pipes, nozzles, etc. as per requirements.

v) 4" or higher diameter hoses of suitable length for feeding large capacity monitors wherever required.

8.5.12 FIRE PROOFING

Vessels, equipment which are likely to be exposed to LNG fire radiation, should be provided with suitable fireproofing or water deluge for the duration of hazard.

Structures that are likely to be exposed to LNG fire radiation shall be provided with passive fire protection (PFP) in the form of suitable fireproofing for the duration of the hazard. Fireproofing shall be executed as per OISD-STD-164.

Safety critical instruments, cables, etc. that are likely to be exposed to LNG fire radiation shall be required to follow OISD-STD-164. Fireproofing shall be used to protect equipment, typically: ESD valves, safety critical equipment, vessels containing quantities of liquid hydrocarbon and structural supports, which on failure would escalate the incident and/or endanger the activities of emergency response personnel. Equipment which can receive thermal radiation, in excess of that defined below for a sufficient period to cause failure, shall be provided with fireproofing protection. The fireproofing shall provide protection for the duration of the hazard event but shall, as a minimum, provide 90 min protection.

Fireproofing should be designed and executed in accordance with as per OISD-STD-164.

Typical applications of PFP are:

Buildings:

- Fire barriers or doors between enclosures.
- Protection of escape routes in buildings

LNG facilities:

- Protection of critical structural members in process areas
- Protection of critical equipment (e.g., ESD valves and actuators)
- Protection of critical control cabling
- Protection of process equipment
- Protection of tanks
- Protection of vessels and vessel supports.
- Containment pits
- Run off channels.

Embrittlement Protection: Equipment and structures shall be protected by insulation or appropriate metallurgy selection against cold shock and failure due to a spill of LNG.

8.5.13 COMMUNICATION SYSTEM

Effective communication is an essential element in the fire protection system of any plant. The following communication systems shall be provided in the LNG plants.

i) TELEPHONE

The fire station control room shall be provided with 2 nos. of internal telephones which are exclusively meant for receiving fire/ emergency calls only. These phones should have facilities for incoming calls only. For general communication a separate telephone shall be provided.

The fire station shall also have a fixed landline for contacting mutual aid parties.

ii) PUBLIC ADDRESS SYSTEM

The public address system shall be connected to all control rooms, administration building (all floors), all departmental heads, security etc. Telephone exchange shall control and take care of this system.

iii) FIRE SIRENS

The fire siren/s shall be located suitably to cover the whole area with the operational control in the fire station control room. These shall be tested at least once in a week to keep them in working condition.

Fire siren code should be as follows:

a) Emergency Level I – The siren code for Level I shall be decided by entity depending upon the size and complexity of installation (preferably entity may choose a siren code different from Level-II).

b) Emergency Level II - A wailing siren for two minutes.

c) Emergency Level III - Same type of siren as in the case of Level II but shall be sounded for three times at the interval of one minute that is to say. (wailing siren 2 minutes + gap 1 minute + wailing siren 2 minutes + gap 1 minute + wailing siren 2 minutes) and the total duration of disaster siren shall be eight minutes.

d) ALL CLEAR (For fire): Straight run siren for two minutes.

e) TEST: Straight run siren for two minutes at frequency at least once a week.

iv) WALKIE-TALKIE / WIRELESS

All the Fire Tenders shall be provided with a walkie-talkie/ wireless system which will help in communicating with the people in case the other system fails. Besides, key personnel coordinating emergency operations should also be provided with walkie-talkie.

v) Intrinsically safe mobile phones may be used as alternate communication mode.

vi) FIRE & GAS ALARM SYSTEM

The fire alarm systems include manual call points (break glass), automatic gas/smoke/heat detectors, release & inhibit switches for fire suppressing clean agent and conventional or micro-processor-based data gathering panels viz. central fire alarm panel, mimic panels & associated equipment.

Manual Call Points shall be provided at suitable locations like access points, approach roads, walkways etc. to cover the critical areas. These manual call points activate the audio-visual alarm in the Central fire alarm panel installed in fire station and in the repeater panel installed in the respective area control room(s). The location of these points shall be conspicuously marked on the annunciation panel for proper identification. These manual call points should also have suitably wired telephone handset to facilitate communication with respective area control room and fire station.

8.5.14 PERSONNEL SAFETY

i) Personnel shall be advised of the serious danger from frostbite that can result upon contact with LNG or cold refrigerant. Suitable protective clothing and equipment shall be made available.

ii) Presence of personnel in the vicinity of Emergency Release Coupling (ERC) during cool-down/ bulk unloading should be restricted so as to avoid any injuries due to accidental release of ERC.

iii) Those employees who will be involved in emergency activities shall be equipped with the necessary clothing and equipment.

iv) Self-contained breathing apparatus shall be provided for those employees who may be required to enter an atmosphere that could be injurious to health during an emergency.

v) A portable flammable gas indicator (personal gas monitoring device) shall be readily available because LNG and hydrocarbon refrigerants within the process equipment are usually not odorized, and the sense of smell cannot be relied upon to detect their presence.

8.5.15 INSPECTION & TESTING OF FIRE PROTECTION SYSTEM:

The fire protection equipment shall be kept in good operating condition all the time and the firefighting system shall be periodically tested for proper functioning and logged for record and corrective actions. In addition to routine daily checks/maintenance, the following periodic inspection/testing shall be ensured.

8.5.15.1 FIRE WATER TANK/ RESERVOIR /FOAM TANKS

i. Above ground fire water tanks should be inspected externally & internally as per OISD-STD- 129.

ii. The water reservoir shall be inspected periodically, not more than 5 years, to ensure the integrity of the reservoir and there is no seepage and leakage from the reservoir. During such inspection, any accumulation of mud, sand, weeds, and other undergrowth, which can reduce the capacity of the reservoir and obstruct free flow of water, shall be recorded and the required remedial action shall be taken.

iii. The foam tanks shall be inspected every 3 years externally and shall undergo internal inspection every 10 years.

8.5.15.2 FIRE WATER PUMPS

i. Fire water pumps should be tested in line with OISD-STD-142.

ii. Once a month each pump should be checked for the shut off pressure and auto start operation. Observations should be logged.

iii. Once in six months each pump shall be checked for performance. This can be done by opening the required number of hydrants/monitors depending on the capacity of the pump and by verifying that the discharge pressure, flow and the motor load are in conformance with the design parameters. For flow measurement, suitable devices like ultrasonic instruments can be considered.

8.5.15.3 FIRE WATER RING MAIN

i. The ring main should be checked once a year for leaks etc. by operating one or more pumps with the hydrant points kept closed as required to get the maximum operating pressure.

ii. The ring main, hydrants, monitors, valves should be visually inspected every month for any pilferage, defects and damage as per OISD-STD-142.

iii. All fire main valves should be checked for operation and lubricated once in six months for fresh water and once in three months for saline/ ETP water.

iv. Thickness survey & inspection of firewater header should be done once in three years in line with OISD-STD-142.

v. Segment - wise flushing of main header should be done once a year.

8.5.15.4 FIRE WATER SPRAY/ SPRINKLER SYSTEM

i. Fixed water-cooling spray/sprinkler system shall be tested in line with OISD-STD-142.

ii. The spray system in the LNG plant should be visually inspected every three months and should be tested at least once in every six months in line with OISD-STD-142. The operation of remote operated valves should be checked for functionality once in three months.

8.5.15.5 FIXED / SEMI-FIXED FOAM SYSTEM

i. Foam system should be tested once in 12 months. This shall include the testing of foam maker/ generator/ chamber.

ii. Piping should be flushed with water after testing foam system.

iii. Thickness survey & inspection of foam system piping should be done in line with OISD-STD-142.

8.5.15.6 CLEAN AGENT BASED EXTINGUISHING SYSTEM

The systems shall be checked in line with OISD-STD-142 once a year.

8.5.15.7 MOBILE FIRE FIGHTING EQUIPMENT AND ACCESSORIES

i. Foam tenders should be tested at least once a week. This should include running pump and foam generation equipment.

ii. All other mobile equipment should be checked, serviced and periodically tested under operating conditions, at least once a month.

iii. Trailer mounted pumps should be test run at least once a week.

iv. All the fire hoses shall be service pressure tested annually at maximum working pressure of fire water network/Fire fighting vehicle.

v. DCP tender should be visually inspected every week.

vi. Records shall be maintained of all maintenance, testing and remedial/ corrective actions taken wherever necessary.

8.5.15.8 DCP/ CO_2/ FOAM TYPE FIRE EXTINGUISHERS:

Inspection and testing frequency and procedure should be in line with OISD-STD-142.

8.5.15.9 COMMUNICATION SYSTEM

Fire sirens shall be tested at least once a week. Visual inspection of manual call points once in a month & functionality testing of manual call points once in three months, walkie-talkies every week and other communication systems as per manufacturer’s guidelines.

8.5.15.10 DETECTORS

The operability of all types of detectors should be tested once in every three months. Calibration of gas detectors using test gas should be done once in every six months or as per manufacturer's specification, whichever is earlier.

8.5.16 MUTUAL AID

LNG plants should have written mutual aid agreements with neighbouring industries fully detailing the responsibilities of the members of the scheme, the procedures to be adopted, the minimum number of equipment and manpower and minimum quantity of consumables to be exchanged/loaned in line with OISD-STD-227.

A chart showing mutual aid arrangement shall be exhibited prominently at least in the fire station and Disaster Control Room.

8.5.17 FIRE EMERGENCY PROCEDURES

Each installation shall prepare a detailed "fire emergency procedures" or emergency response plan manual outlining the actions to be taken by each personnel during a major incident for use by the organization and this manual shall be available to all personnel in the installation.

The fire emergency procedures including firefighting plan should be prepared for fighting fires in and around tanks, product loading gantries, separators, electrical fire, warehouse and building fires etc.

8.5.18 FIRE STATION / CONTROL ROOM

i. The Fire Station Control Room is of critical importance as it is the main coordinating centre between the emergency site and response crew. Important activities of the control room are communication, mobilization, up-keeping and maintenance. The location and construction of the control room should therefore be suitable for these activities.

ii. Fire station(s) equipped with state of art infrastructure, alarm and communication system and critical resources shall be provided at strategic locations for quick mobilization during any emergency. Entity to define the response time and ensure that all hazardous locations are within the defined response time and shall not exceed 5 minutes.

iii. Based on the size and complexity of the entity, the number of fire stations, emergency responders and resources shall be kept available to mitigate two simultaneous emergencies at the complex.

iv. A reliable communication system is a must for supporting effective fire service dept. operations. The following minimum equipment must be available in the Control Room, (1) Telephones (2) Wireless sets / walkie-talkie (with a dedicated frequency) (3) two independent/ separate means of communication with neighboring industries/civil Fire Brigade (4) Fire Alarm system with central control in fire stations.

8.5.18.1 LOCATION

Fire stations should be located at minimum risk area. It should be spaced at a safe distance from hazardous areas. For details refer OISD-STD-118.

When planning for a new fire station, adequate land should be provided for parking and maneuvering of fire appliances. Also, access and exits to the building should not be obstructed by other vehicles.

The fire station control room should be close to the fire vehicles parking bay & fire appliances and should have a good view of fire vehicles parked.

Additional fire-post should be considered during the expansion for improving the response time.

8.5.18.2 GENERAL

a) The fire station should have adequate overhead storage tanks for foam compound storage, so that during emergency refilling is not delayed.

b) Fire station control room should have portable emergency lights in addition to critical lighting.

c) Display of fire water network pressure should be available in the fire station control room.

d) In addition to the normal lighting, each installation shall be equipped with emergency (AC) and critical (DC) lighting. Emergency lighting shall enable the operators to carry out safe shut-down of the plant, to gain access and permit ready identification of firefighting facilities such as fire water pumps, fire alarm stations etc.

e) Critical lighting, sourced from 220V or 110V DC system shall enable safe evacuation of operating personnel and shall be employed along escape route, assembly point and critical installations such as first aid centre, control rooms, manned sub-stations, fire water pump house etc. as per OISD-STD-173. The healthiness of the lighting system shall be ensured by periodical inspection and records to be maintained.

f) Personnel in LNG terminal shall be trained as per OISD-STD-154. Records of such training shall be maintained. Firefighting skill upgradation / refresher training shall be given periodically.

g) A mock fire drill should be conducted as per OISD-STD-227 to keep the firefighting team trained and alert.

h) For LNG plants not equipped with adequate fire training facilities, skill up gradation/ refresher training shall be periodically conducted at reputed institutes having live module firefighting training facilities.

8.6 SAFETY IN ROAD TRANSPORTATION

Transporting Liquefied Natural Gas (LNG) by road involves unique challenges due to its cryogenic nature and potential hazards like leaks, rollovers, and pressure buildup. Strict safety measures shall be ensured for secure transportation. Transportation of LNG by road is regulated by PESO and The Motor Vehicle Act. Entities involved in LNG transportation are mandated to develop and implement an ERDMP for handling emergencies. Containers and tank trucks shall be fabricated in accordance with the applicable guidelines specified.

8.6.1 DESIGN - GENERAL REQUIREMENTS

The safety relief valves provided on the inner vessel of the LNG transport tank shall be sized to meet most stringent condition of simultaneous occurrence of loss of vacuum and external fire and the combined capacity of the safety valves shall be sufficient to limit the pressure in the vessel to the test pressure -

i) The transport vessel shall be designed and constructed as per the ASME Boiler and Pressure Vessel Code Section VIII Div I pressure vessel code, EN13530 or equivalent code approved by PESO and also to meet the requirements of ISO 20421 and the design temperature of the vessel, piping and valves shall be such that it is suitable for requirement sustaining cold shock caused by a loading of liquid Nitrogen into the vessel during its testing and commissioning.

ii) Each vessel shall have adequate insulation that will prevent the vessel pressure from activating the relief valve set pressure within the specified holding time when the vessel is loaded with LNG at the design condition of –

(a) Specified temperature and pressure of the LNG,

(b) The exposure of the vessel to the average ambient temperature of 40°C.

iii) The outer vessel or jacket of the cryogenic vessel for transportation of LNG shall be made of no other material other than steel.

iv) No Aluminium valve or fitting external to the wetted outer vessel shall be installed on LNG transportation vessels and each transportation vessel shall consist of a suitably supported welded inner vessel enclosed within an outer shell with vacuum insulation between the two.

8.6.2 STRUCTURAL INTEGRITY:

i) The design and construction of each vessel used for transportation of LNG shall be in accordance with ASME Boiler and Pressure Vessel Code Section VIII Div. 1, EN13530 or any other code approved by Chief Controller and the vessel design shall include calculation of stress due to design pressure, the weight of lading, the weight of structure supported by the vessel wall, and the effect of the temperature gradients resulting from lading and ambient temperature extremes.

ii) In order to account for stresses due to impact in an accident, the design calculation of the vessel shell and heads shall include the load resulting from the design pressure in combination with the dynamic pressure resulting from a longitudinal deceleration of 2g. For this loading condition, the stress value used shall not exceed 75% of the yield strength of the material of construction.

iii) The fittings and accessories mounted on the vessel shall be protected in such a way that damage caused by overturning cannot impair operational integrity and this protection may take the form of cylindrical profile of the vessel, of strengthening rings, protective canopies or transverse or longitudinal members so shaped that effective protection is given.

iv) The welding of the appurtenances to the vessel wall shall be made of attachment of the mounting pad so that there shall be no adverse effect upon the loading retention integrity of the vessel.

8.6.3 PRESSURE RELIEF DEVICES, PIPING, VALVES AND FITTINGS

i) Hoses shall be suitable for the service and shall be designed for a bursting pressure of at least five times working pressure.

ii) If a threaded pipe is used, the pipe and fitting shall be Schedule 80 or higher rating.

iii) Each hose coupling shall be designed for a pressure of at least 120% of the hose design pressure and there shall be no leakage when connected.

iv) Piping shall be protected from damage due to thermal expansion and contraction, jarring and vibration and slip joints shall not be used.

v) Each valve shall be suitable for the vessel design pressure at the vessel design service temperature.

vi) All fittings shall be rated for the maximum vessel pressure and suitable for the coldest temperature to which they will be subjected in actual service.

vii) When a pressure building coil is used on the vessel, the vapor connection to that coil shall be provided with a valve or check valve as close to the vessel shell as practicable to stop flow in case of damage to the coil and the liquid connection to the coil shall also be provided with a valve.

viii) Each vessel shall be rated for its holding time, the holding time being the time as determined by testing that will elapse from loading until the pressure of the contents, under equilibrium conditions reaches the level of the lowest pressure relief valve setting.

ix) All the discharge lines of relief valves, vent valve, bleed valves etc., shall be connected to a vent stack which shall vent at a safe height.

x) Bursting discs shall not be used on the LNG transport vessels.

xi) The outer vessel shall be protected by any accidental accumulation of pressure in the annular space by using a relief plate or plug or a bursting disc and the relief device shall function at a pressure not exceeding the internal design pressure of the outer tank, the external design pressure of the inner tank, or 25 psi whichever is less.

8.6.4 ACCIDENT DAMAGE PROTECTION

i) All valves, fittings, pressure relief devices and other accessories to the vessels, which are not isolated from the vessel by closed intervening shut off valves or check valves shall be installed within the motor vehicle framework or within a suitable collision resistant guard or housing and appropriate ventilation, shall be provided and each pressure relief device shall be protected so that in the event of the upset of the vehicle on to a hard surface, the device's opening will not be prevented and its discharge will not be restricted and the threaded end connection safety valves are preferred in stainless steel body construction.

ii) Each protective device or housing and its attachment to the vehicle structure shall be designed to withstand static loading in any direction that it may be loaded as a result of front, rear, side or side wise collision or the overturn of the vehicle and all the valves of tank shall be at rear inside on operation box (cabinet) of suitable size and does not project out of tank diameter and the thickness of cabinet shall be minimum 3mm.

8.6.5 REAR END PROTECTION:

i) Rear end vessel protection devices shall consist of at least one rear bumper designed to protect the transport vessel and piping in the event of a rear end collision and the rear end vessel protection device shall be designed so that it transmits the force of the collision directly to the chassis of the vehicle. The rear end vessel protection device and its attachments to the chassis shall be designed to withstand a load equal to twice the weight of the loaded cargo vessel and attachments, using a safety factor of four based on the tensile strength of the materials used with such load being applied horizontally and parallel to the major axis of the transport vessel.

ii) Every part of the loaded transport vessel and any associated valve, pipe, and enclosure or protected fitting or structure shall be at least 35.5 cm above ground level.

8.6.6 DISCHARGE CONTROL DEVICES:

i) Each liquid filling and liquid discharge line shall be provided with a shut off valve located as close to the vessel as practicable and unless this valve is manually operable at the valve, the line shall also have a manual shut off valve.

ii) Each liquid filling and liquid discharge line shall be provided with an on vehicle remotely controlled self-closing shutoff valve.

iii) Each control valve shall be of fail-safe design and spring- based.

iv) Each remotely controlled shut off valve shall be provided with on vehicle remote means of automatic

v) closure, both mechanical and thermal.

vi) Each remotely controlled shut off valve shall be provided with on-vehicle remote means automatic closure, both mechanical and thermal. One means may be used to close more than one remotely controlled valve and remote means of automatic closure shall be installed at the ends of the tanker farthest away from the loading or unloading connection area.

8.6.7 SHEAR SECTION:

Unless the valve is located in a rear cabinet forward of and protected by the bumper, the design and installation of each valve, damage to which could result in loss of liquid or vapor shall incorporate a shear

section or breakage groove adjacent to and outboard of the valve and the shear section or breakage groove shall yield or break under strain without damage to the valve that would allow the loss of liquid or vapour.

8.6.8 SUPPORTS AND ANCHORING:

In case, the transport tanker vehicle is designed and constructed in such manner that the vessel is not wholly supported by the vehicle frame, the transport vessel shall be supported by external cradles or load rings and the design calculations for the supports and load bearing vessel and the support attachments shall include beam stress, shear stress, torsion stress, bending moment and acceleration stress for the loaded vehicle as a unit, using a safety factor of four based on the tensile strength of the material and static loading that uses the weight of the transport vessel and its attachments when filled to the design weight of the loading. Minimum static loadings shall be maximum of the following individually, namely:

i) Vertically downward of two (2)

ii) Vertically upward of one and half (1.5)

iii) Longitudinally of one and half (1.5) and

iv) Laterally of one and half (1.5)

8.6.9 GAUGING DEVICES:

i) Liquid level gauging devices:

ii) The vessel shall have one liquid level device that provides a continuous level indication ranging from full to empty and that is maintainable or replaceable without taking the vessel out of service.

iii) Pressure gauges:

iv) Each vessel shall be equipped with a pressure gauge connected to the vessel at a point above the maximum liquid level that has a permanent mark indicating the maximum allowable working pressure of the tanker and the pressure gauge shall be housed in a canopy of the tanker.

8.6.10 GUIDELINES FOR SAFE ROAD TRANSPORTATION:

In order to handle emergencies that may arise due to incidents involving LNG Tank Trucks, it is essential that a comprehensive Emergency Response Plan (ERP) is in place and is clearly understood and practiced by all concerned users so that the emergencies, if any, can be handled in a systematic manner with minimal response time in accordance with the prescribed procedures. The ERP shall be reviewed and updated from time to time in line with the prevailing regulations.

a) The entity shall ensure installation and proper functioning of ABS (Anti-Lock Braking System), VTS (Vehicle Tracking System), Speed Governor, Front/Rear cameras, Anti-collision devices etc. Moreover, in-transit surveillance of Tank Truck shall be maintained, and deviations (if any) should be investigated for possible causes.
b) Entities shall ensure the availability of an alternate driver/ co-driver/ helper throughout the duration of the journey. This requirement shall be included in the transportation contract agreement.
c) To address concerns regarding driver/ crew fatigue, entities should take necessary steps, such as identifying suitable halts along the route, to enable adequate rest and refreshment of drivers/ crew etc.
d) The vehicle should be inspected for integrity checks at regular intervals, and records should be maintained. The entity may develop a checklist for such inspections, following guidelines similar to OISD-GDN-161.
e) Further, entities shall develop comprehensive Journey Management Plan (JMP) which shall include the following:
    i. Identified suitable halts en-route within a predetermined time frame to prevent diver/ crew fatigue
    ii. Identify blind spots & accident-prone areas and precautions to be taken thereof.
    iii. Climate/ weather forecast for the route.
    iv. TREM card and other foreseeable hazards enroute.

The above shall be disseminated to drivers/ crew of each tank truck with appropriate records.

f) In accordance with Motor Transport Workers Act, 1961, restrictions on the hours of work for any person engaged in operating a Road Transport Vehicle shall be ensured. Further, a valid driving license with endorsement for hazardous goods transport shall be ensured.
g) The entity shall ensure medical fitness of driver/ crew at least on annual basis.
h) The driver/ co-driver/ helper shall be trained on
    i. hazardous goods transportation
    ii. defensive driving
    iii. awareness of all hazards associated with LNG
    iv. basic emergency/ firefighting handling and first-aid
    v. Journey Route Management aspects

Training records shall be maintained and verified before entry of truck for loading.

i) Emergency actions to be taken in case of accidents shall be available with the crew all the time.

j) Upon receipt of information about the incident, LNG installation/ LNG facility/ Receiver of consignment/ Transporter, etc. shall convey the information to all concerned agencies using the quickest mode of communication and ensure assistance & adequate resources for rescue.
k) Rescue teams mobilized should take utmost care in tackling the situation.

9 SAFETY REQUIREMENTS FOR SMALL SCALE LNG FACILITIES:

The requirements for the installation, design, fabrication, and siting of small-scale LNG facilities or plants with individual vessels or tanks of 1000 m^3 water capacity or less constructed in accordance with the ASME Boiler and Pressure Vessel Code shall be considered as follows in this section.

The maximum aggregate capacity of such LNG facility or installation shall not exceed 4000 m^3 water capacity.

9.1 INSTALLATION, DESIGN & CONTAINMENT

Site preparation shall include provisions for retention of spilled LNG, within the limits of facility, and for surface water drainage.

9.1.1 Impoundment or dyke areas, topography, or other methods to direct LNG spills to a safe location and to prevent LNG spills from entering water drains, sewers, waterways, or any closed-top channel shall be used.

9.1.2 Compressors, CNG cascades, odorizers, flammable liquid or storage vessels etc. shall not be located inside the impounding area.

9.1.3 The ambient vaporizers and remotely heated vaporizers may be located inside impounding area.

9.1.4 The impounding system for LNG storage vessel shall have a minimum volumetric liquid capacity of-

(a) 110% of maximum liquid capacity of vessel for an impoundment serving a single vessel.

(b) 110% of maximum liquid capacity of the largest vessel serving for more than one vessel.

9.1.5 Impounding areas shall be designed or equipped to clear rain or other water.

9.1.6 Where automatically controlled sump pumps are used, they shall be equipped with an automatic cut off device that prevents their operation when exposed to LNG temperatures.

9.1.7 The height of the impoundment wall shall be adequate to contain spillage of any LNG and the dyke wall height of 0.6 meter to 1 meter is recommended from the dyke floor level and the height of the foundation of the vessel shall be minimum 0.4 meter or designed in such a way to prevent exposure of carbon steel material to the spilled LNG.

9.1.8 All-weather accessibility to the site for emergency services equipment shall be provided.

9.1.9 A clear space of at least 0.9 meters shall be provided for access to all isolation valves serving multiple vessels and the isolation valve of LNG vessel piping shall be as close to outer vessel as possible.

9.1.10 LNG vessels, cold boxes, piping and pipe supports and other cryogenic apparatus installed within dyke shall be designed and constructed in a manner to prevent damage to these structures and equipment due to freezing or frost heaving in the soil.

9.1.11 Adequate flameproof lighting arrangement shall be done for facilities transferring LNG during night.

9.1.12 Electrical grounding and bonding shall be provided.

9.1.13 Layout shall ensure unobstructed access and exit of the consumers and supply vehicles at all times.

9.1.14 Entrance, exit and paving shall be arranged in a manner, so as to minimise the risk of collision.

9.1.15 The operating personnel shall have an unobstructed overall view of the facilities both from the sales room and from the delivery area.

9.1.16 The designated tanker unloading location shall be so located that it does not hinder other traffic and at the same time permits truck to be in drive out position for allowing it to come out of the premises easily in case of an emergency.

9.1.17 Crash or impact barriers shall be installed to protect vulnerable equipment against accidents involving vehicle movements.

9.1.18 The storage area which includes the pumps, and the related piping shall be suitably segregated from the rest of the premises and located in a manner that it is away from the area frequented by public during their movement within the station and also from the path of vehicles entering and leaving the premises.

9.1.19 The minimum distance from edge of impoundment or vessel drainage system serving aboveground tanks larger than 3.8 m^3 to buildings and property lines that can be built upon and between vessels shall be in accordance with Table 9.1. For LNG containers of 3.8 m^3 and smaller, a minimum distance of 3.0 m from buildings and the line of adjoining property shall be ensured.

9.1.20 Buried and underground containers shall be provided with means to prevent the 0°C isotherm from penetrating the soil. Where heating systems are used, they shall be installed such that any heating element or temperature sensor used for control can be replaced.

9.1.21 All buried or mounded components in contact with the soil shall be constructed from material resistant to soil corrosion or protected to minimize corrosion.

9.1.22 LNG containers of greater than 0.5 m^3 capacity shall not be located in buildings.

9.1.23 LNG tanks and their associated equipment shall not be located where exposed to failure of overhead electric power lines operating at over 600 volts.

Table 9.1- Distances from Impoundment Areas to Property Lines That Can be Built Upon

Water Capacity of the vessel (m^3)	**Minimum distance from edge of impoundment or vessel drainage system to property line (m)**	**Minimum distance between storage vessels (m)**
3.8 – 7.6	4.6	1.5
≥7.6 – 68.1	7.6	1.5
≥68.1 – 114	15	1.5
≥114 – 265	23	

≥265 – 379	30.5	1/4 of the sum of the diameters of adjacent containers [1.5 m minimum]
≥379 – 454	38	
≥454 – 757	61	
≥757 – 4000	91.4	

9.2. LNG STORAGE – GENERAL DESIGN REQUIREMENTS:

9.2.1 The vessel meant for storage of LNG including piping between inner and outer vessel shall be designed in accordance with ASME Boiler and Pressure Vessel Code Section VIII Div I / EN13458 / ASME: B.31.3, process piping or equivalent code.

9.2.2 The inner vessel shall be designed for the most critical combination of loading resulting from internal pressure, liquid heads and the inner vessel supports system shall be designed for shipping, seismic, and operating loads.

9.2.3 The support system to accommodate the expansion and contraction of the inner tank shall be designed so that the resulting stresses imparted to the inner and outer tanks are within allowable limits as per design.

9.2.4 The outer vessel shall be equipped with a relief or other device to release internal pressure and shall have discharge area of at least 0.34 mm^2/lit of the water capacity of the inner vessel but not exceeding 2000 cm^2 and have pressure setting not exceeding 25 psi.

9.2.5 Thermal barriers shall be provided to prevent the outer tank from falling below its design temperature.

9.2.6 Those parts of LNG vessels which come in contact with LNG and all materials used in contact with LNG or cold LNG vapor shall be physically and chemically compatible with LNG and intended for service at –162°C.

9.2.7 All piping that is a part of LNG vessel including all piping internal to the vessel, within void space, and external piping connected to the vessel up to the first circumferential external joint of the piping shall be in accordance with ASME Boiler and Pressure Vessel Code, Section VIII or ASME B 31.3 or equivalent.

9.2.8 LNG vessels shall be designed to accommodate both top and bottom filling unless other positive means are provided to prevent stratification.

9.2.9 Any portion of the outer surface area of an LNG vessel that could accidentally be exposed to low temperatures resulting from the leakage of LNG or cold vapor from flanges, valves etc., shall be intended for such temperatures or protected from the effects of such exposure.

9.2.10 Seismic loads shall be considered in the design of the LNG vessel support systems.

9.2.11 LNG vessels foundations shall be designed and constructed in accordance with recognised structural engineering practices and prior to the start of design and construction of the foundation, a subsurface investigation shall be conducted by a qualified soil engineer to determine the stratigraphy and physical properties of the soils underlying the site.

9.2.12 The design of saddles and legs for the LNG vessel shall include erection loads, wind loads and thermal loads.

9.2.13 Foundation and support shall have a fire resistance rating of not less than 02 hours. If insulation is used to achieve the fire resistance rating of at least 02 hours, it shall be resistant to dislodgement by hose streams.

9.2.14 LNG storage vessels installed in areas subject to flooding shall be secured to prevent the release of LNG or floatation of the vessel in the event of a flood.

9.2.15 After acceptance tests are completed, there shall be no field welding on the LNG vessels. Where repairs or modifications incorporating welding are required, they shall comply with the code or standard under which the vessel was fabricated.

9.3 INSTRUMENTATION

9.3.1 Each LNG double walled vessel shall have at least 2 numbers of safety relief valves capable of achieving the required relief capacity on standalone basis and shall be sized to relieve the flow capacity determined for the largest single contingency or any reasonable and probable combination of contingencies and shall include conditions resulting from operational upset, vapor displacement and flash vaporization.

9.3.2 Vacuum-relieving devices shall be installed if the vessel can be exposed to a vacuum condition in excess of that for which the vessel is designed.

9.3.3 Relief devices shall be vented directly to the atmosphere and each safety relief valve for LNG vessel shall be able to be isolated from the vessel for maintenance or other purposes by means of a manual full opening stop valve or a flow diverter valve.

9.3.4 Safety relief valve shall be designed and installed to prevent any accumulation of water, or other foreign matter at its end and shall discharge vertically upward.

The minimum pressure relieving capacity in kg/hr shall not be less than 3% of the full tank contents in 24 hours.

9.3.5 All liquid connections to the LNG vessel except relief valve and instrumentation connection shall be equipped with automatic fail-safe product retention valves.

9.3.6 The automatic shut off valves shall be designed to close on occurrence of any of the following conditions, namely

i) fire detection.

ii) uncontrolled flow of LNG from vessel.

iii) manually and remotely operated.

9.3.7 Such automatic shutoff valves that require excessive time to operate during emergency, i.e., sizes exceeding 200 mm, shall be pneumatically operated and also have means of manual operation.

9.3.8 Instrumentation control devices for LNG facilities shall be designed so that, in the event of power or instrument air failure, the system will go into a failsafe condition that can be maintained until the operators can take action to reactivate or secure the system.

9.3.9 Level Gauging

9.3.9.1 LNG vessels shall be equipped with liquid level gauging devices as follows:

i) Vessels smaller than 3.8 m^3 shall be equipped with either a fixed length dip tube or other level devices.

ii) Vessels of 3.8 m^3 through 114 m^3 shall have a minimum of one liquid level device that provides a continuous level indication ranging from full to empty.

iii) Vessels larger than 114 m^3 shall have two independent devices that compensate for liquid density variations.

9.3.9.2 Gauging devices shall be designed and installed so that they can be replaced without taking the vessel out of service.

9.3.9.3 Each vessel greater than 114 m^3 shall be provided with two independent high liquid level alarms, which shall be permitted to be part of the liquid level gauging devices.

9.3.9.4 LNG vessels shall be equipped with an excess flow cutoff device, which shall be separate from all gauges.

9.3.10 Pressure Gauging and Control

9.3.10.1 Each vessel shall be equipped with a minimum of two independent pressure gauging devices connected to the vessel at a point above the maximum liquid level that has a permanent mark indicating the maximum allowable working pressure (MAWP) of the vessel.

9.3.10.2 Vacuum-jacketed equipment shall be equipped with instruments or connections for checking the pressure in the annular space.

9.3.10.3 Safety relief valves shall be sized to include conditions resulting from operational upset, vapor displacement, and flash vaporization resulting from pump recirculation and fire.

9.3.11 Temperature Indicators

9.3.11.1 Temperature monitoring devices shall be provided in field erected vessels to assist in controlling temperatures when the vessel is placed into service or as a method of checking and calibrating liquid level gauges.

9.3.11.2 Where the potential risk of damage to piping and components downstream of heat exchangers exists due to temperature limitations, indication shall be provided to monitor outlet temperatures.

9.3.11.3 Temperature monitoring and alarm systems shall be provided where foundations supporting cryogenic vessels and equipment could be adversely affected by freezing or frost heaving of the ground.

9.3.12 Inner vessel pressure relief valves shall have a manual full opening stop valve to isolate it from the vessel.

9.3.13 The installation of pressure relief valves shall allow each relief valve to be isolated individually for testing or maintenance while maintaining the full relief capacities.

9.3.14 Where only one pressure relief valve is required, either a full-port opening three-way valve used under the pressure relief valve and its required spare or individual valves beneath each pressure relief valve, shall be installed.

9.3.15 Stop valves under individual safety relief valves shall be locked or sealed when opened and shall not be opened or closed except by an authorized person.

9.4 EQUIPMENT

9.4.1 Pumps and compressors employed in LNG source shall be provided with a pressure relieving device on the discharge to limit the pressure to the maximum safe working pressure of the casing and downstream piping and equipment.

9.4.2 Each pump shall be provided with adequate vent, relief valve, or both that will prevent over-pressurising the pump casing during the maximum possible rate of cool down.

9.4.3 The discharge valve of each vaporizer and the piping components and relief valves installed upstream of each vaporizer outlet valve/ spec break flange shall be designed for operation at LNG temperatures.

9.4.4 Two inlet valves shall be provided to isolate an idle, manifolded vaporizer to prevent leakage of LNG into the vaporizer and a safe means of disposing of the LNG or gas that can accumulate between the valves shall be provided in case the vaporizers are of size having inlets more than 50 mm diameter.

9.4.5 The ambient air vaporizers shall be installed inside the containment area.

9.4.6 Where the heated vaporizer is located 15 meter or more from the heat source, the remote shutoff location shall be at least 15 meter from the vaporizer.

9.4.7 Where the heated vaporizer is located less than 15 meter from the heat source, it shall have an automatic shut off valve in the LNG liquid line located at least 3 meter from the vaporizer and shall close when either of the following occurs:

(i) Loss of line pressure (excess flow)

(ii) The occurrence of a fire is detected.

(iii) Low temperature in the downstream of the vaporizer

(iv) Manual ESD trip

9.4.8 Any ambient vaporizer or a heated vaporizer installed within 15 meter of an LNG vessel shall be equipped with an automatic shutoff valve in the LNG liquid line. The automatic shutoff valve shall be located at least 3 meter from the ambient or heated vaporizer and shall close in either of the following situations, namely:

(i) Loss of line pressure (excess flow).

(ii) Abnormal temperature sensed in the immediate vicinity of the vaporizer (fire)

(iii) Low temperature in the downstream of the vaporizer.

(iv) Activation of Emergency Shut Down System (ESD).

9.4.9 If the facility is attended, manual operation of the automatic shutoff valve shall be from a point at least 15 meter from the vaporizer, in addition to the requirement as specified in 9.4.8.

9.4.10 The above conditions shall be applicable for LNG vaporizers for purposes other than pressure building coils or LNG to CNG (LCNG) systems.

9.4.11 A distance of minimum 1 meter shall be maintained between vaporizers.

9.4.12 Vaporizers shall be designed, fabricated and inspected as per the requirements of ASME Boiler and Pressure Vessel Code, Section VIII Div. 1 or any other equivalent code.

9.4.13 The material of construction of LNG vaporizers is recommended as austenitic stainless-steel piping with outer aluminium fins.

9.4.14 Manifold vaporizers shall be provided with both inlet and discharge block valves for each set of vaporizer.

9.4.15 Each set of vaporizer shall be provided with a safety relief valve(s) sized in accordance with the following requirements, namely:

a) Ambient vaporizers–relief valve capacity shall allow discharge equal or greater than 150% of the rated vaporizer natural gas flow capacity without allowing the pressure to rise 10% above the vaporizer maximum allowable working pressure.

b) Relief valves on heated vaporizers – same as sub-clause (a) above, however, it shall be located such that they are not subjected to temperatures exceeding 60°C during normal operation.

9.4.16 Automation shall be provided to prevent the discharge of either LNG or vaporizer gas into a distribution system at the temperature either above or below the design temperature of the send out system.

9.4.17 Vaporizers shall be provided with outlet temperature monitors.

9.5 PIPING SYSTEM

9.5.1 All piping system and components shall be designed –

a) To accommodate the effects of thermal cycling fatigue to which the systems shall be subjected.

b) To provide for expansion and contraction of piping and piping joints due to temperature changes.

9.5.2 Piping material including gaskets and thread compounds shall be compatible throughout the range of temperature to which they are subjected.

9.5.3 The valves provided in the installation shall be of extended bonnet type with packing seals in a position that prevents leakage or malfunction due to freezing.

9.5.4 Shut-off valves shall be provided for all vessel connections except connections for liquid level alarms and connections that are blind flanged or plugged.

9.5.5 All the piping section between the two valves where the liquid may be trapped shall have the thermal relief valve.

9.5.6 Aluminium shall be used only downstream of a product retention valve in vaporizer service.

9.5.7 Isolation valves shall be installed so that each transfer system can be isolated at its extremities and where power-operated isolation valves are installed. An analysis shall be made to determine the closure time so that it does not produce a hydraulic shock capable of causing line or equipment failure.

9.5.8 Adequate check valves shall be provided to prevent backflow and shall be located as close as practical to the point of connection to any system from which backflow might occur.

9.5.9 In addition to a locally mounted device for shutdown of the pump or compressor drive, a readily accessible, remotely located device shall be provided at least 7.5 meters away from the equipment to shut down the pump in an emergency.

9.5.10 The truck unloading area shall be of sufficient size to accommodate the vehicles without excessive movement or turning.

9.5.11 Transfer piping, pumps, and compressors shall be located or protected by barriers so that they are safe from damage by vehicle movements.

9.5.12 Isolation valves and bleed connections shall be provided at the unloading manifold for both liquid and vapour return lines so that hoses and arms can be blocked off, drained of liquid, and depressurized before disconnecting and bleeds or vents shall discharge in a safe area.

9.6 EMERGENCY SHUT DOWN SYSTEM

9.6.1 Each LNG facility shall incorporate an ESD system that when operated isolates or shuts off sources of LNG and shuts down equipment that add or sustain an emergency if continued to operate.

9.6.2 The ESD system shall be of a failsafe design and shall be installed, located, or protected from becoming inoperative during an emergency or failure at the normal control system.

9.6.3 Initiation of the ESD system shall be manual, automatic, or both manual and automatic. Manual actuators shall be located in an area accessible in an emergency and at least 15 meters away from the equipment they serve and shall be distinctly marked with their designated function.

9.7 FIRE PROTECTION FACILITIES

9.7.1 General

Fire protection shall be provided for all LNG facilities. The extent of such protection shall be determined by an assessment based on hazards within the facility, local conditions, fire protection principles, and the affect to or from other property in the vicinity. The fire protection assessment shall be conducted, and fire protection equipment shall be ready in place before commissioning the facility. Such assessment study shall include the following:

- The type, quantity, and location of equipment necessary for the detection and control of fires, leaks, and spills of LNG and other hazardous fluids.
- Protection of the equipment and structures from the effects of fire exposure.
- Fire water system requirements.
- Requirements for fire-extinguishing and other fire control equipment
- The equipment and processes to be incorporated within the emergency shutdown (ESD) system, including analysis of sub-systems.
- The type and location of detectors necessary to initiate automatic activation of the ESD system or its sub-systems.
- Emergency Response Plan & Preparedness during an emergency.
- The personal protective equipment, special training, and qualification needed by individual plant personnel for their respective emergency duties.
- Requirements for any other hazards

9.7.2 Each LNG storage facility shall be provided with flammable gas detectors, which shall activate visual and audible alarms at the plant site and at constantly attended location, if the facility is not attended continuously.

9.7.3 Flammable gas detection system shall activate an audible and a visual alarm at level not higher than 20% of the LEL of the gas being monitored.

9.7.4 Fire detectors shall activate an alarm at the plant site and at a constantly attended location if the plant site is not attended continuously and if determined by an evaluation that it is necessary then fire and gas detectors shall be permitted to activate the ESD system.

9.7.5 A fire water supply and delivery system shall be provided based on the risk assessment. However, as a minimum a fire water storage tank of 10% of LNG storage tank capacity or 5000 litre whichever is higher with pumping facilities and hose reel system away from the LNG storage facility shall be provided for dispersion of LNG vapor cloud in case of leakage or spillage.

9.7.6 Portable or wheeled fire extinguishers of DCP and/or foam type shall be available at strategic locations and shall be provided and maintained in accordance with NFPA 10.

9.7.7 Fire extinguishers provided in the LNG installations shall be based on high expansion foam and dry chemical powder.

9.8 PERSONAL PROTECTION

9.8.1 Every person operating the equipment in the LNG installation shall be equipped with the following personal protective equipment, namely:

(i) Suitable goggles for protection of eyes from LNG spray,

(ii) Hand gloves suitable for cryogenic liquid handling,

(iii) Protective apron,

(iv) Safety Shoes.

9.9 OPERATIONS & MAINTENANCE

9.9.1 In addition to a locally mounted device for shutdown of the pump or compressor drive, a readily accessible, remotely located device shall be provided a minimum of 7.5 m away from the equipment to shut down the pump or compressor in an emergency.

9.9.2 Remotely located pumps and compressors used for loading or unloading trucks shall be provided with controls to stop their operation that are located at the loading or unloading area and at the pump or compressor site.

9.9.3 Signal lights shall be provided at the loading or unloading area to indicate whether a remotely located pump or compressor used for loading or unloading is idle or in operation.

9.9.4 Each facility shall have written maintenance procedures based on experience, knowledge of similar facilities, and conditions under which they will be maintained.

9.9.5 Each facility operator shall carry out periodic inspection, tests, or both, as required on every component and its support system in service in its facility, to verify that it is maintained in accordance with the equipment manufacturer's recommendations.

9.9.6 Each operation manual for a facility that transfers LNG from or to a truck shall contain procedures for loading or unloading, including the following:

i. Prior to connecting a truck, the vehicle shall be checked, and the brakes set, the derailer or switch properly positioned, and warning signs or lights placed as required.
ii. The warning signs or lights shall not be removed or reset until the transfer is completed and the car disconnected.
iii. Unless required for transfer operations, truck vehicle engines shall be shut off.
iv. Brakes shall be set, and wheels checked prior to connecting for unloading or loading.
v. The engine shall not be started until the truck has been disconnected and any released vapors have dissipated.
vi. Prior to loading LNG into a truck that is not in exclusive LNG service, a test shall be made to determine the oxygen content.

vii. If a truck in exclusive LNG service does not contain a positive pressure, it shall be tested for oxygen content.

viii. If the oxygen content in either case exceeds 2 percent by volume, the truck shall not be loaded until it has been purged to below 2 percent oxygen by volume.

9.10 EMERGENCY PROCEDURES

9.10.1 Each facility shall have a written manual of emergency procedures included in the operations manual that shall include the types of emergencies that are anticipated from an operating malfunction, structural collapse of part of the facility, personnel error, forces of nature, and activities carried on adjacent to the facility, including the following:

i. Procedures for responding to controllable emergencies, including notification of personnel and the use of equipment that is appropriate for handling of the emergency and the shutdown or isolation of various portions of the equipment and other applicable steps to ensure.

ii. that the escape of gas or liquid is promptly cut off or reduced as much as possible

iii. Procedures for recognizing an uncontrollable emergency and for taking action to ensure that harm to the personnel at the facility and to the public is minimized.

iv. Procedures for the prompt notification of the emergency to the appropriate local officials, including the possible need to evacuate persons from the vicinity of the facility.

v. Procedures for coordinating with local officials in the preparation of an emergency evacuation plan that sets forth the steps necessary to protect the public in the event of an emergency.

9.10.2 A comprehensive Emergency Response Plan considering the potential scenarios shall be developed.

9.10.3 All fire-control access routes within an LNG facility shall be maintained and kept unobstructed in all weather.

9.11 DISPENSATION

LNG dispensing facilities shall comply with national and international standards for safety, efficiency, and environmental protection during LNG dispensation. The type of dispenser used for dispensing LNG shall conform to specifications and be of a type approved by PESO. Facilities dispensing LNG shall meet the safety requirements as per the latest version of the SMPV (Unfired) Rules.

10 SHIP TANKER RECEIVING FACILITIES AND MINIMUM PORT FACILITIES AND BERTHING CONDITIONS:

The design of ship tanker receiving, and port facilities does not form part of this standard. However, these paragraphs have been included for the general guidance and information of the users of this standard. For detailed study, reference should be made to information paper no. 14 - Site Selection and Design for LNG Ports and Jetties by SIGTTO (Society of International Gas Tankers and Terminal Operators Ltd.) and SIGTTO publication on 'LNG Operations in Port Areas – recommendations for the Management of Operational Risk Attaching to Liquefied Gas tankers and Terminal Operations in Port areas'.

10.1 PORT DESIGN

i) **Approach Channels.** Harbour channels should be of uniform cross-sectional depth and have a minimum width, equal to five times the beam of the largest ship where only one way traffic at a time is envisaged. For two-way simultaneous traffic movement, the minimum width of the harbour channel should be equal to seven times the beam of the largest ship.

ii) **Turning Circles.** Turning circles should have a minimum diameter of twice the overall length of the largest ship to be received where current effect is minimal. Where turning circles are located in areas of current, diameters should be increased by the anticipated drift.

iii) **Tug Power.** Available tug power, expressed in terms of effective Bollard pull, should be sufficient to overcome the maximum wind force generated on the largest ship using the terminal, under the maximum wind speed permitted for harbour manoeuvres and with the LNG carrier's engines out of action.

iv) **Traffic Control.** A Vessel Traffic Service (VTS) System should be a port requirement, and this should be able to monitor and direct the movement of all ships coming within the operating area of LNG carriers.

v) **Operating Limits.** Operating criteria for maximum wind speed, wave height and current should be established for each terminal and port approach. Such limits should match LNG carrier size manoeuvring constraints and tug power.

vi) **Speed Limits.** Speed limits should be set for areas in the port approach presenting either collision or grounding risks. These limits should apply not only of LNG carriers but also to any surrounding traffic.

vii) **Jetty Location** A jetty shall be earmarked for LNG unloading/loading and if other liquefied gases and petroleum products are also proposed to be handled at the same jetty, risk assessment shall be carried out for consideration in design basis as well as the minimum inter distance requirements between various facilities at jetty.

The LNG unloading jetty location and layout shall ensure that the centre of jetty head to centre of jetty head distance between two adjacent jetties/ berths or port facilities is determined based on site specific mooring, navigation simulation and safety studies carried out to ensure ship / vessel maneuverability, tug operations and safe operations at either one of the jetty/ berth or at both the jetties/ berths simultaneously.

For initial planning, centre of jetty head to centre of jetty head distance between two adjacent jetties or births should be considered as 500 m or more. In case such a distance is not available, detailed site-specific mooring, navigation simulation and safety studies mentioned above shall be carried out as part of initial planning (Feasibility Report/ Detailed Feasibility Report) stage itself prior to obtaining statutory clearances like environmental clearance etc.

The maximum credible LNG spill and its estimated gas-cloud range should be carefully established for the jetty area. As a minimum, a process safety zone of 250m radius centered on the loading arms to any adjoining marine/port facility shall be ensured

Salient parameters for LNG Port site selection and port safety to be considered in the above mentioned site-specific studies are given in Annexure-I.

viii) **Other Port/Marine Conditions**

Minimum depth of the sea (water depth) required to accommodate the large LNG tankers is of the order of 13 - 14 metres w.r.t. the CD (i.e. Chart Datum).

The sizes of the tankers, which are currently in use, vary from 40,000 m^3 to 266,000 m^3 having compartments varying between 3-5 numbers. This LNG from the ship should be unloaded by minimum two liquid and one vapour arm depending on the unloading rate and spill considerations. One of the unloading arms should be designed for the use of both liquid and vapour.

All these unloading arms are provided with an emergency shut-off arrangement (ESD valves and ERS) which shall operate in case of any emergency so that no liquid/vapour is passed through to the LNG ship.

Basic criteria needs to be looked into its approach channel to the terminal, traffic control and the berthing facilities apart from interface with the shore for communication, mooring system etc.

Each gas terminal requires a unique risk management system and mere application of guidelines/criteria may be insufficient. The approach channel specification turning circle radius, power required by tug, operating limits, need to be looked into before arriving at the final specification for port facilities.

The water current in the turning areas should be less than 1 knot and the wind speed with the assistance of tug would be in the order of 25 knots. The design criteria for mooring is

i) Wind speed 60 knots from any direction.
ii) Water current 3 knots.

Wind speed on the berth is to the extent of 27 knots however the unloading operations can take place upto 30 - 35 knots and the arms to be disconnected at 35 knots wind speed subject to manufacturer’s specification.

The vessels lie alongside the jetty and the discharge operation takes place at 1 to 1.5 metre height of the bay. With specialised mooring arrangements, the height may go up to 2 metres. The ship can be discharged during the day or night if managed effectively.

The traffic control management is to ensure that no credibility risk of breach in cargo containment and such limits are synergies with vessel size and manoeuvring. The ships shall be berthed normally with head out to sail off in the event of any emergency.

The design policies and standards need to take care of port safety and environment in regard to

i) Coastline assessment
ii) Environmental impact assessment
iii) Risk conditions.
iv) Policy development.

The security of supply is based on the physical assessment, sparing philosophy, reliability of equipment's and operation.

The normal distance between the jetty and shore is between 25 to 500 metres.

The design safety and commercial criteria for ship operations generally require that a ship should be unloaded in 12 hours so that it can turn around in 24 hours. With larger ships the unloading rates shall increase which in turn will result in larger unloading lines on the jetty and into the terminal. The development in the LNG ship design has reduced the boil-off from 0.25% of the contents per day to a value of 0.1%. Generally, the boil-off is used for the following:

a) In the boiler for generating the steam which in turn is being used for power generation,

b) As a fuel in dual fuel for diesel electric propulsion.

c) Small Re-liquefaction plant

10.2 SAFETY

a) The LNG Jetty provision shall have emergency escape routes from a fire or liquid spill as per OCIMF guidelines. All such emergency escape routes shall be distinctly marked and displayed.
b) Jetty elevated monitor streams shall be capable of reaching the fire hazard area (ship manifold and loading arms).
c) Life buoys should be provided with self-ignition lights. All life-saving appliances/ equipment should be regularly inspected and maintained in constant state of readiness.

11.0 ABBREVIATIONS

AET	Acoustic Emission Testing
API	American Petroleum Institute
BOG	Boil Off Gas
CCTV	Closed Circuit Television
DBE	Design Basis Earthquake
DTS	Distributed Temperature Sensing
ERC	Emergency Release Coupler
ERS	Emergency Release System
ERDMP	Emergency Response and Disaster Management Plan
ERP	Emergency Response Plan
ESD	Emergency Shutdown
FSRU	Floating Storage and Regasification Unit
FSU	Floating Storage Unit
GWUT	Guided Wave Ultrasonic Testing
IFV	Intermediate Fluid Vaporizers
IHTF	Intermediate Heat Transfer Fluid
LNG	Liquified Natural Gas
MCE	Maximum Considered Earthquake
MCP	Manual Call Point
MSDS	Material Safety Data sheet

OBE	Operating Basis Earthquake
OEM	Original Equipment Manufacturer
ORV	Open Rack Vaporizer
PECT	Pulse Eddy Current Testing
PERC	Powered Emergency Release Coupling
PESO	Petroleum & Explosive Safety Organisation
PNGRB	Petroleum & Natural Gas Regulatory Board
PCV	Pressure Control valve
QRA	Quantitative Risk Analysis
SCV	Submerged Combustion Vaporiser
SD	Spill Detector
SIGTTO	Society of International Gas Tanker and Terminal Operators
SMPV	Static and Mobile Pressure Vessels (Unfired)
SOP	Standard operating Procedure
SSE	Safe Shutdown Earthquake
T4S	PNGRB (Technical Standards and Specifications including Safety Standards for Liquified Natural Gas Facilities) Regulations, 2018.
TCP	Thermal Corner Protection
TSV	Thermal Safety Valve
TLF	Tank Lorry Filling
TT	Tank Truck
TW	Tank Wagon
VTS	Vessel Traffic Service

12.0 REFERENCES

(i) NFPA 59A: Standard for the Production, Storage, and Handling of Liquefied Natural Gas (LNG), 2023 Edition.

(ii) BS EN 1473: Installation and equipment for liquefied natural gas–Design of onshore installations, 2021 Edition.

(iii) NFPA 15: Standard for Water Spray Fixed Systems for Fire Protection.

(iv) API-STD-620: Design and Construction of Large, Welded, Low-pressure Storage Tanks.

(v) BS EN 14620: Design and manufacture of site built, vertical, cylindrical, flat bottomed steel tanks for the storage of refrigerated, liquefied gases with operating temperatures between 0^0C and -165^0C.

(vi) IS 1893 (Part 4): Criteria for earthquake resistant design of structures.

(vii) Site Selection and Design for LNG Ports and Jetties, Information Paper No.14 by The Society of International Gas Tanker and Terminal Operators.

(viii) International Safety Guide for Oil Tranter & Oil Terminals (ISGOTT).

(ix) Society of International Gas Tanker and Terminal Operators (SIGTTO).

(x) International Gas Carriers (IGC) Code.

(xi) Cryogenic storage of liquefied gases - Hydrocarbon Asia.

(xii) LNG receiving terminal design is different-DB Crawford and CA Durr, Hydrocarbon Processing

(xiii) OCIMF - Guidelines including Guidelines for Ship-to-Ship Transfer of LNG/ Liquefied Gases.

(xiv) Seismic Design Guidelines and Data.

(xv) EEMUA Publication No. 147 (The Engineering Equipment And Material Users Association)

(xvi) LNG Operations in Port Areas - 'Recommendations for the Management of Operations Risk Attaching to Liquefied Gas Tanker & Terminal Operations in Port Areas'. A publication by SIGTTO.

(xvii) EI (Energy Institute), UK, Model Code of Safe Practice Part-19

(xviii) OISD-STD-112: Safe Handling of Air Hydrocarbon Mixtures & Pyrophoric Substances.

(xix) OISD-STD-114: Safe Handling of Hazardous Chemicals.

(xx) OISD STD-116: Fire Protection Facilities for Petroleum Refineries and Oil/Gas Processing Plants.

(xxi) OISD STD-118: Layouts for Oil & Gas Installations.

(xxii) OISD-STD-155: Personnel Protective Equipment.

(xxiii) OISD-STD-156: Fire Protection Facilities for Ports Handling Hydrocarbons.

(xxiv) Site Selection and planning issues for new LNG marine terminals by HR Wallingford.

(xxv) Journal Paper on Reducing the catastrophe risk in coastal area: risk management at FSRU terminals by Filip Jovanovic, Mirano Hess.

13.0 ANNEXURES

Annexure-I

LNG PORTS-PARAMETERS FOR SITE SELECTION & PORT SAFETY

The site-specific mooring, navigation simulation and safety studies carried out for site selection of an LNG Port should consider the following parameters and provide suitable mitigation/ measures for addressing each of the parameters listed below, meeting the actual site-specific constraints.

1 THE PORT

1.1 Port Analysis

Speed restrictions for LNG carriers should be appropriate to limit grounding and collision damage.

1.2 Approach Channels and Turning Basins

Minimum depth of the sea (water depth) required to accommodate the large LNG tankers is of the order of 13 - 14 metres w.r.t. the CD (i.e. Chart Datum).
Under-keel clearances should be established in accordance with the sea-bed quality.
Approach channels should be of uniform cross-sectional depth and have a minimum width, equal to five times the beam of the largest ship where only one way traffic at a time is envisaged. For two-way simultaneous traffic movement, the minimum width of the harbour channel should be equal to seven times the beam of the largest ship.
Turning circles should have a minimum diameter of two times the overall length of the largest ship to be received where current effect is minimal. In areas of current, diameters shall be increased by the anticipated drift.
Short approach channels are preferable to long inshore routes which carry more numerous hazards
Traffic separation schemes should be established in approach routes.
Anchorages should be established at the port entrance and inshore, for the safe segregation of LNG carriers and to provide lay-by facilities in case, at the last moment, the berth proves unavailable.

1.3 Navigational Aids

Buoys to mark the width of navigable channels should be placed at suitable intervals.
Leading marks or lit beacons, to mark channel centre lines and to facilitate rounding channel bends, should be appropriately placed.
Electronic navigational aids, to supports navigation under adverse weather conditions, are needed in most ports.
Lit navigational aids should be provided to allow ship movements at night.

1.4 Port Services

Tugs should be made available and three to four are normally required giving 140 tonnes total bollard pull.
Mooring services are often required and these services should normally provide a minimum two boats, each having at least 400 horsepower.
Escort services comprising fast patrol craft, to clear approach channels, turning areas, jetty, etc. should be provided in busy port areas.
Firefighting services comprising specially equipped craft, or, one or more suitably equipped tugs should be provided.

1.5 Port Procedures

Traffic control or VTS systems should be strictly enforced to ensure safe harbour manoeuvring between the pilot boarding area and the jetty.
Speed limits should be introduced in appropriate parts of the port approach, not only for the LNG carriers but also for other ships.
Pilotage services should be required to provide pilots of high quality and experience. Pilot boarding areas should be at a suitable distance offshore.
Ship movements by nearby ships, when the LNG carriers is pumping cargo, should be disallowed.
Pilots and tugs should be immediately available in case the LNG carrier has to leave the jetty in an emergency.

1.6 Port Operating Limits
Environmental limits for wind, waves, and visibility should be set for ship manoeuvres and these should ensure adequate safe margins are available under all operating conditions.
Weather limits for port closure should be established.

1.7 Weather Warnings
Forecasting for long range purpose should be provided to give warning of severe storms, such as typhoons and cyclones.
Forecasting for short range purposes, such as those required for local storms and squalls should be made available.

2 THE JETTY

2.1 Jetty Location
Jetty location should be remote from populated areas and should also be well removed from other marine traffic and any existing port activity which may cause a hazard.
The distance between a moored LNG vessel and a passing ship (Through traffic only) should be at least 250m.
Jetty location should be such that the jetty operations are independent from nearby other port facilities with no or limited interfaces.
Jetty location should be such that the jetty is not in direct line of way of any passing ship specially near turning circles, curved approach channels etc.
The maximum credible spill and its estimated gas-cloud range should be carefully established for the jetty area. A safety zone of 250m radius centred on the loading arms shall be ensured.
River bends and narrow channels should not be considered as appropriate positions for LNG carriers jetties.
Breakwater, if required, based on the model studies and downtime analysis shall be provided.
Restrictions, such as low bridges, should not feature in the jetty approach.
Ignition sources should be excluded within a predetermined radius from the jetty manifold.

2.2 Jetty Layout
Mooring dolphin spacing between the outermost dolphins on the same jetty, should not be less than the ship's overall length Mooring dolphins should be situated about 50 meters inshore from the berthing face.
Mooring points should be suitably positioned and have suitable strength, for the environmental conditions. Quick-release hooks should be provided at all mooring points.
Breasting dolphin spacing should be designed to ensure that the parallel body of the ship is properly supported. Fendering for the dolphins, and for the berth face, should be to a suitable standard.

2.3 Jetty Equipment

Pipelines and pumps etc. should be designed to provide a rapid port turn round. Emergency Release Systems at the hard arms should be fitted in accordance with industry specifications. The ERS should be suited to both ship and shore by interlinking and a PERC should be fitted to each hard arm for emergency stoppage and quick release purposes.
Emergency shut-down valves should be fitted to both and shore pipelines and should form part of the ERS system. Powered emergency release couplings (PERCs) with flanking quick-acting valves should be fitted to the hard arm as part of the ERS system.

Plugs both on ship and shore to carry all ESD and communication signals should be standardised. Surge pressure control should be provided in LNG pipelines. Communications equipment (telephone, hot-line and radios) should be provided for ship/ shore use. Load monitors, to show the mooring force in each mooring line, should be fitted to quick release hooks. Gangways should be provided to give safe emergency access to or from the ship.

2.4 Basic Firefighting Facilities

Water curtain pumps and pipelines should be provided. Fixed Dry Powder systems should be provided. Gas detection monitors should be fitted at strategic locations. Fireproof material should be used for the construction of hard arms.

2.5 Jetty Procedures

On shore jetty safety zones, be effectively policed while the ship is alongside thus providing control over visitors and vehicles. Offshore safety zones should be effectively policed by a guard boat to limit the approach of small craft. Passing ships, close to the jetty, should be well established and tested. Contingency plans should be available in written form. Operating procedures should be available in written form. A Port Information/ Regulation Booklet should be provided for passing operational advice to the ship.

Annexure-II

GENERAL GUIDELINES FOR FSRU/ FSU FACILITIES

Floating storage and regasification units (FSRUs) provide solutions to meet the energy demands of an ever-changing global market. Floating regasification is a flexible, cost-effective way to receive and process shipments of LNG. Floating regasification is increasingly being used to meet natural gas demand or as a temporary solution until onshore regasification facilities are built.

2.1 Floating regasification involves the use of a specialized vessel often referred to as an FSRU, which is capable of transporting, storing, and regasifying LNG onboard. An FSRU can be purpose-built or be converted from a conventional LNG vessel by installing a regasification plant.

2.2 Regasification plants are installed to vaporize the liquid gas onboard a vessel. The gas may be delivered to the shore side facility via various arrangements proven internationally, such as through an offshore buoy piping system or vessel manifold system to facility.

2.3 These guidelines provide for ensuring safety and security measures of the floating terminals for the import of LNG/ RLNG and should be read in conjunction with IMO Regulations.

2.4 FSRU terminals are adapted to special local conditions and legal regulations and are subject to the national standards of the country. Guidelines issued by DG Shipping for the safety, security, and environmental protection to Indian FSRUs/ FRUs, and non-Indian FSRUs/ FRUs while operating in Indian waters shall be in force for strict compliance.

2.5 FSRUs shall comply with the following conventions, codes, and regulations of the International Maritime Organisation (IMO)

i. Requirements of International Convention for the Safety of Life at Sea (SOLAS) for safety, stability, and firefighting systems.

ii. The terminal shall comply to the International Convention for the prevention of pollution from ships (MARPOL). All potential effects on the environment including but not limited to emissions of pollutants to atmosphere, sea water discharges, sewage treatment system to treat and discharge waste, noise emissions shall meet the MARPOL standards.

iii. Ports handling FSRU/ FSU should adhere to the safety provisions & regulations related to the handling, storage and transportation of LNG as stipulated in the International Gas Carrier Code (IGC Code).

iv. International Ship and Port Facility Security Code (ISPS) shall be in place for protecting both the ships and the ports or terminals.

v. International Safety Management (ISM) & International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW) are also applicable.

vi. In addition to the above conventions, FSRU ships must comply with all other conventions in the maritime world.

2.6 Additionally, below mentioned guidelines should be considered as a minimum safety requirement to maintain the integrity of an LNG FSRU/ FSU throughout its operation:

i. In case of a berthed FSRU/ FSU being used as a floating storage unit with LNG transfer across the jetty, the onshore part for land-based regasification systems is covered by this standard as applicable.

ii. A detailed risk assessment study shall evaluate the quantum of risk due to the continuous existence/ presence of the FSRU/ FSU berthed at Jetty. Based on the risk assessment study, fire protection facilities at jetty governed by OISD-STD-156 shall be additionally supplemented for the safety & security of the vessel & other jetty facilities. Fire protection facilities at jetty shall be governed by OISD-STD-156 and recommendations of the risk assessment study.

iii. Relevant OISD standards shall be followed if the FSRUs/ FSUs berthed at offshore transports RLNG to the Onshore Receiving Facility (ORF) through subsea pipeline.

iv. A well adopted system of coastal zone management and risk management should be in place to reduce the risk of disasters in the coastal area.

v. Regular monitoring of Mooring and Berthing Systems shall be ensured.

vi. It is necessary to have Comprehensive Emergency Response Plans shall be developed to address potential accidents, spills, or other incidents that may occur on the jetty.

vii. Reassessment of the adequacy of safeguards applied for FSRUs/ FSUs must be ensured periodically.

viii. The following safeguards shall be considered to mitigate any potential risk on FSRU ships:
- safety measures for gas supply systems, pipelines, tanks
- regasification and distribution
- availability of medical, fire, police services and other services
- availability of a sufficient number of adequate tugs
- warning and alarm systems
- the possibility of rapid, efficient cessation of cargo operations
- safety aspects of FSRU ship design (double hull, fireproof, etc.)
- safety distance from other port areas and facilities, residential, commercial, industrial facilities and the like
- pollution prevent measures in the event of an LNG spill
- measures in case of fire, explosion
- measures in the event of a collision with an LNG ship on arrival
- measures for termination and commencement of cargo operations
- evacuation plan
- security systems of the FSRU ship and the surrounding area of the terminal (protective fences, security guards, CCTV, etc.)

2.7 Risk assessment study related to the import of liquefied natural gas through the FSRU terminal, must also include the incoming LNG tankers, other coastal infrastructure and the terminal, location and local meteorological and oceanographic conditions, traffic in the waters and the surrounding area. In addition, the following four categories of factors influencing risk assessment shall be considered:
- location, specific local conditions, LNG imports, importance for the region
- potential threats, potential accidents, possible accidents, critical coastal infrastructure and FSRU ship, incoming LNG tankers
- risk management objectives, identification of consequences to be avoided (e.g. injuries or damage to property), importance of LNG supply.
- possibilities of protective mechanisms, safety measures, warning and alarm systems, reaction measures in case of accident

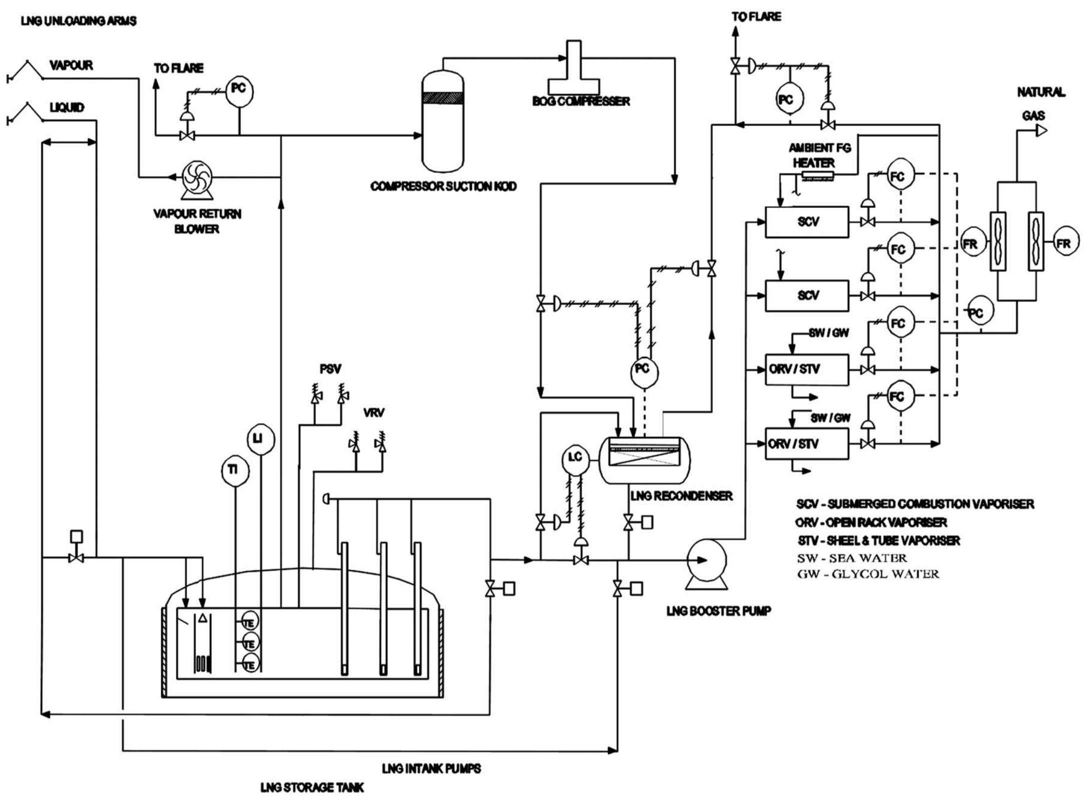
The above-mentioned categories must be considered when determining whether the implemented safety measures can successfully achieve the desired level of risk management for FSRU/ FSU projects in a particular area.

2.8 LNG Carrier to FSRU/ FSU transfer of LNG:
- Flexible hoses shall not be used for the transfer of LNG between large LNG carriers and jetty based FSRU/ FSUs.

- In case of FSRU/ FSUs stationed in the sea (not attached to jetty) flexible cryogenic hoses conforming to EN 1474-2 or equivalent standards can be used for transfer of LNG from LNG carriers to FSRU/ FSUs.

Annexure-III

SCHEMATIC DRAWINGS



TYPICAL SCHEMATIC FLOW DIAGRAM FOR LNG TERMINAL

Vacuum Relief valve is designed to allow air ingress in the extreme case of vacuum formation in the tank.

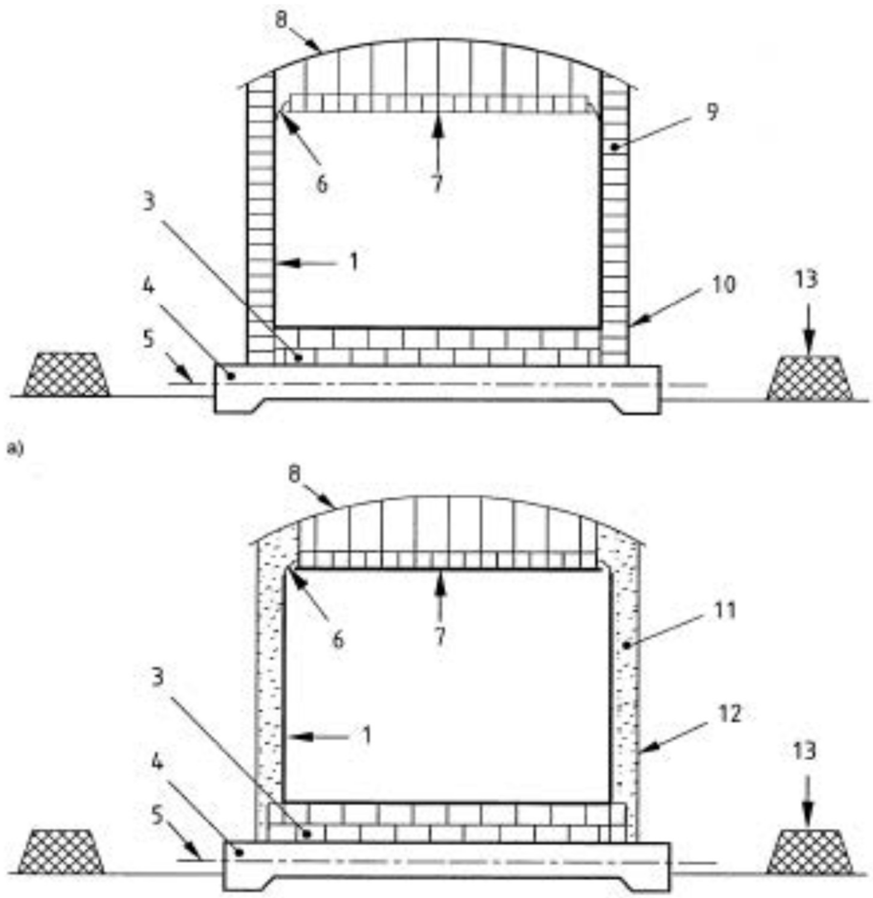


Figure-1: Single Containment Tanks

1	primary container (steel)	8	roof(steel)
3	bottom insulation	9	external shell insulation
4	foundation	10	external water vapour barrier
5	foundation heating system	11	loose fill insulation
6	flexible insulating seal	12	outer steel shell (not capable of containing liquid)
7	suspended roof (insulated)	13	bund wall

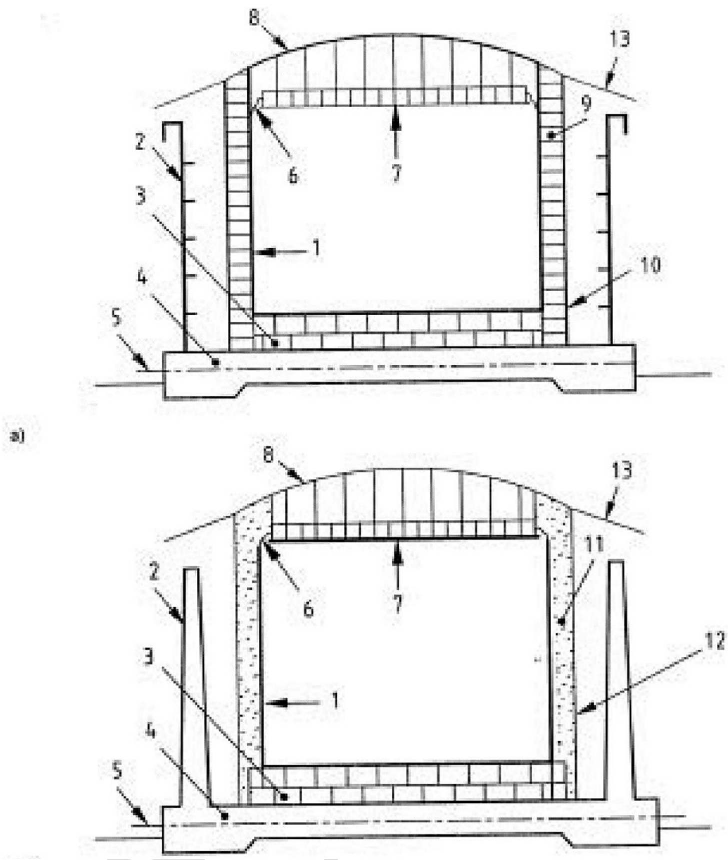


Figure-2: Double Containment Tanks

1 primary container (steel)
2 secondary container (steel or concrete)
3 bottom insulation
4 foundation
5 foundation heating system
6 flexible insulating seal
7 suspended roof (insulated)
8 roof(steel)
9 external insulation
10 external water vapour barrier
11 loose fill insulation
12 outer steel shell (not capable of containing liquid)
13 cover (rain shield)

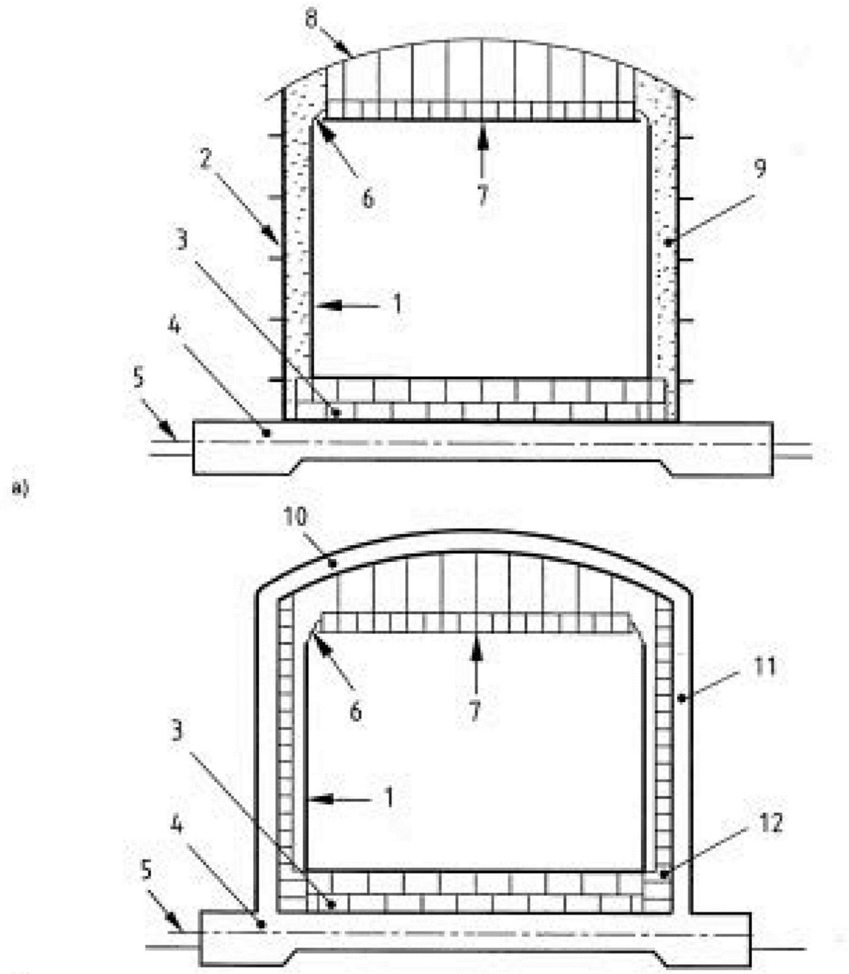


Figure-3: Full Containment Tanks

1	primary container (steel)
2	secondary container (steel)
3	bottom insulation
4	foundation
5	foundation heating system
6	flexible insulating seal
7	suspended roof (insulated)
8	roof (steel)
9	loose fill insulation
10	concrete roof
11	pre-stressed concrete outer tank (secondary container)
12	insulation on inside of pre-stressed concrete outer tank

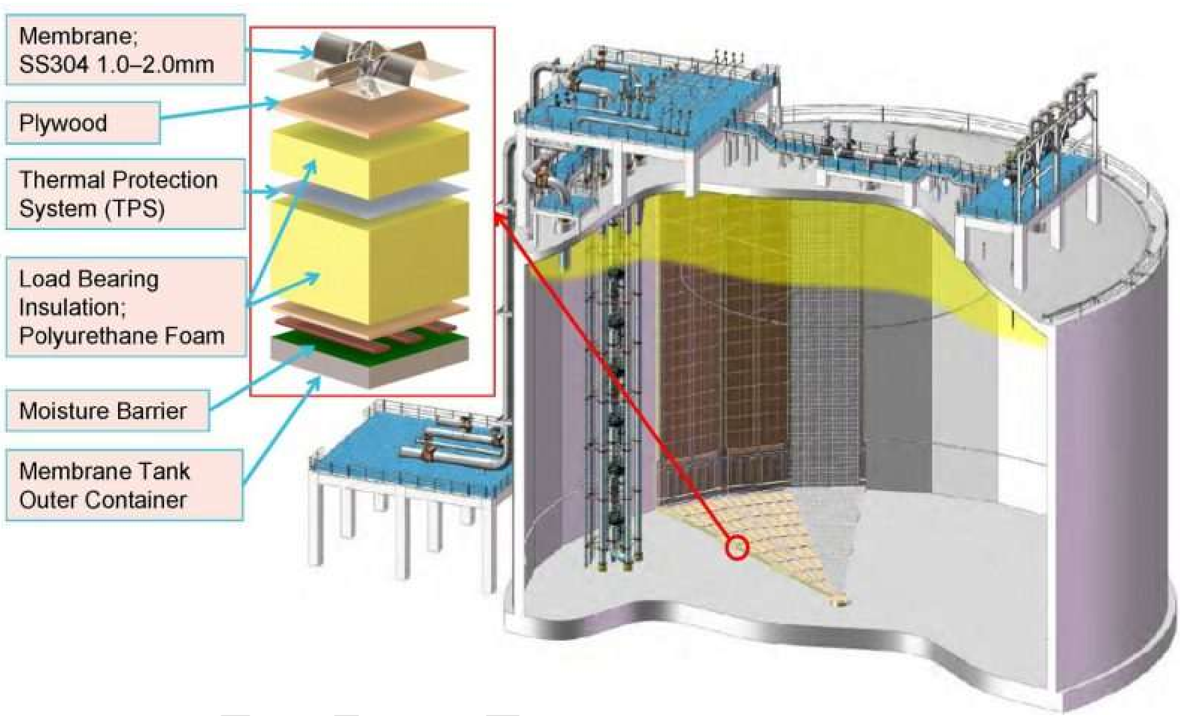


Figure-4: Membrane Tank

Annexure-IV

FIRE WATER DEMAND CALCULATION

1.0 **DESIGN BASIS**

1.1 Fire water flow rate shall be calculated based on two major fire contingencies simultaneously.

1.2 Fire water flow rate for proposed terminal shall be calculated based on the following:

1.3 LNG Tanks situated in the same dyke area:

1.3.1 Fire water flow rate for LNG tank roof and wall shall be at a rate of 3.0 lpm/m².

1.3.2 The water application rate on the appurtenances shall be at a rate of 10.2 lpm/m².

1.3.3 The water densities applicable to other equipment shall be as follows:

Vessels, structural members piping & valves manifolds: 10.2 lpm/m².

1.4 Pumps and BOG Compressors house including lube oil skid etc.: 20.4 lpm/m².

1.5 Fire water flow rate for road bulk loading/ unloading gantry area: 10.2 lpm/m².

1.6 Fire water flow rate for supplementary stream, shall be based on using 4 single hydrant outlets and 1 HVLR (1000 GPM) simultaneously. The capacity of each hydrant outlet is 36 m³/hr and of each monitor 228 m³/hr shall be considered at a pressure of 7 kg/cm²g.

The supplementary water stream requirement is in addition to design flow rates as calculated in 1.3 to 1.5 mentioned above.

Note:

Double Wall Storage Tank (DWST) on fire having concrete outer shell shall be considered in single fire zone if other DWST having concrete outer shell is located more than 30m.

2.0 **FIRE WATER FLOW RATE CALCULATION**

2.1. **FIRE ZONE-1: LNG TANKS** (Roof Type- Metallic & Shell Type- Metallic)

LNG Tanks situated in the same dyke area

Size (D X H) & No. of Tank: 45.0 m Dia X 21.5 m Height X 2 Nos.

Dome Height (h): 8m

Q_1 = (Tank Surface Area + Tank Dome Roof Area) X Water Application Rate

= [Π X D X H + Π X (h^2 +r^2)] X Water Application Rate

= [Π X 45.0 X 21.5 + Π X (8^2+22.5^2)] X 3.0 lpm

= (3039.5+1791.5) X 3 lpm

= 14493 lpm

$= \frac{14493 \times 60}{1000}$ m³/hr.

= 869.6 m³/hr.

Q_2 = (Pump Operating platform area + Instrument platform area + PSV & VSV Operating platform area + Vertical riser area) X Water Application Rate

= (275+25+30+50) X 10.2 lpm

= 3876 lpm

$= \frac{3876 \times 60}{1000}$ m³/hr.

= 232.6 m^3/hr.

Total cooling water rate Q_A = 869.6+232.6 = 1102.2 m^3/hr.
Cooling water required for another tank

Q_B = 1X1102.2 m^3/hr

= 1102.2 m^3/hr

Q_C = 372 m^3/hr (supplementary hose stream)

Total cooling water rate Q = Q_A +Q_B + Q_c = 1102.2+1102.2+372= 2576.4 m^3/hr.

2.2. **FIRE ZONE-2: LNG TANKS (**roof type - concrete & shell type – concrete)
Size (D X H) & No. of Tank: 45.0 m Dia X 21.5 m Height
Dome Height (h): 8 m

Q_1 = (Effective tank Dome Roof Area*) X Water Application Rate

= [(Π X (h^2 +r^2)) – Structural Platform area] X Water Application Rate

= [(Π/X (8^2+22.5^2)) – (275+25+30)] X 3.0 lpm

= (1791.5 – 330) X 3 lpm

= 4384.5 lpm

= $\frac{4384.5 \text{ X } 60}{1000}$ m^3/hr.

= 263.1 m^3/hr.

[(*Effective tank Dome Roof Area = Tank Dome Roof Area – Structural Platform Area (Pump Operating Platform area + Instrument platform area + PSV & VSV Operating platform area)]

Q_2 = (Pump Operating platform area + Instrument platform area + PSV & VSV Operating platform area + Vertical riser area) X Water Application Rate

= (275+25+30+50) X 10.2 lpm

= 3876 lpm

= $\frac{3876 \text{ X } 60}{1000}$ m^3/hr.

= 232.6 m^3/hr.

Total cooling water rate Q_A = 263.1+232.6 = 495.7 m^3/hr.

Q_B= 372 m^3/hr (supplementary hose stream)

Total cooling water rate Q = Q_A +Q_B = 495.7+372= 867.7 m^3/hr.

2.3. **FIRE ZONE-3:** LNG Process Area (HP LNG pump, BOG Recondenser, BOG compressor, Booster compressor, STV, fuel gas heater, BOG suction knockout drain drum, BOG suction knockout drum, metering area etc.)
[For fire zoning criteria, minimum 16m distance to be considered in between two fire zones]

2.3.1 Fire Zone-3a: HP pump, BOG Recondenser, STV & Booster compressor

Q_{3a}= 400 m^3/hr.

Q_{3a}= 372 m^3/hr (supplementary hose stream)

Total cooling water rate (Q) = 400 + 372 = 772 m^3/hr.

2.3.2 Fire Zone-3b: BOG compressor, BOG suction knockout drain drum & BOG suction knockout drum

Q_{3b}= 500 m^3/hr.

Q_{3b}= 372 m^3/hr (supplementary hose stream)

Total cooling water rate (Q) = 500 + 372 = 872 m^3/hr.

2.3.3 Fire Zone-3c: fuel gas heater & metering area

Q_{3c}= 100 m^3/hr.

Q_{3c}= 372 m^3/hr (supplementary hose stream)

Total cooling water rate (Q) = 100 + 372 = 472 m^3/hr.

Largest cooling water rate in Fire zone-3: 872 m^3/hr.

2.4. **FIRE ZONE-4: LOADING/UNLOADING GANTRY**

(i) Size of Shed w.r.t. Zone- 44m X 15m

Water application rate- 10.2 lpm/m^2

Q_{4a}= Surface Area X Water Application Rate

= 44X 15 X 10.2 lpm

= 6732 lpm

$= \frac{6732 \times 60}{1000}$ m^3/hr

= 403.9 m^3/hr

Q_{4b}= 372 m^3/hr (supplementary hose stream)

Total cooling water rate (Q) = 403.9 + 372 = 775.9 m^3/hr.

3.0 **FIRE WATER-FLOW RATES FOR FOUR CASES**

Fire water flow rate as per OISD w.r.t. two major fires simultaneously (i.e. FIRE ZONE-1 & FIRE ZONE-2): 2576.4 + 867.7 = 3444.1 m^3/hr **(Say 3600 m3/hr.)**

Note: If firewater demand in other areas (e.g., Jetty) requires a higher water flow rate, two major fires shall be considered accordingly.

4.0 **FIRE WATER PUMP CAPACITY AND STORAGE REQUIREMENT w.r.t. TWO MAJOR FIRE CONTINGENCIES SIMULTANEOUSLY FOR TERMINAL**

Fire water pumps shall be diesel/electric engine driven horizontal centrifugal pumps. The pumps shall be located in a covered pump house of RCC construction. The pump details are given below:

Design flow rate = 3600 m^3/hr.
No. of working pump required = 4 x 900 m^3/hr @ 100 m head
No. of stand-by pumps required = 2 x 900 m^3/hr @ 100 m head

Main Fire Water Pumps

Total No. of pumps	:	6
Working/Drive	:	4 (2 DE + 2 E)
Standby/Drive	:	2 (1 DE + 1 E)
Rated Discharge Capacity	:	900 m^3/hr. each

Head	:	100 m
Jockey Pumps w.r.t. (3% of Design flow rate as per OISD)		
Total No. of pumps	:	2
Working/Drive	:	1 (E)
Standby/Drive	:	1 (E)
Type of drive	:	Electric (with emergency power backup)
Rated Discharge Capacity	:	108 m^3/hr. (Say 110 m^3/hr.) each (3% of design flow rate)
Head	:	110 m

Each Diesel engine shall have an independent fuel tank for 6 hours continuous running of engine. Whereas electric driven jockey pumps shall have emergency power backup.

5.0 FIRE WATER STORAGE

Fire water storage shall be fully dedicated to fire service. The capacity of fire water storage is 4 hours aggregate working capacity of pumps.

Fire water storage required = 4 x 3600 = 14400 m^3 plus 20% contingency extra fire water for losses to be considered i.e. 14400 x1.2 = 17280 m^3

Annexure-V

FOAM DEMAND CALCULATION

1.0 DESIGN BASIS

1.1 Foam demand shall be calculated based on two major fire contingencies simultaneously for this plant.

1.2 Foam demand for proposed terminal shall be calculated based on the following:

1.3 LNG Tanks situated in the dyke area.

1.4 Discharge duration:15 min.

1.5 Max. submergence time: 3 min.

1.6 Depth of foam: 2 feet.

1.7 Compensation for leakage: 1

1.8 Compensation for normal foam leakage: 1.15

1.9 Min. Rate of Discharge (R) = (V/T) X C_N X C_L

where

R= Rate of Discharge in ft^3/min.

V= Submerge volume in ft^3

T= Submerge Time in Minute

C_N = Compensation for normal foam leakage.

C_L = Compensation for leakage

2.0 FOAM DEMAND CALCULATION

Impounding Area = 6970 m^2 = 75024.5 ft^2

V = 75024.5 X 2 = 150049 ft^3

Min. Rate of Discharge (R) = (V/T) X C_N X C_L

(150049/3) X 1.15 X 1 = 57518.8 ft^3/min.

Required nos. of Foam Generator considering capacity of 19600 ft^3/min @ 7 bar

= 57518.8/ 19600 = 2.93 (Say 3)

Foam solution w.r.t. expansion ration (560:1)

= (19600/560) X 3 = 105 ft^3/min.

= 105 X 28.316 = 2973.18 lpm

Foam storage considering 3% foam concentration for 30 min. duration

= 2973.18 X .03 X 30 = 2675.9 liter

= 2675.9 X 1.3 (For losses) = 3478.6 liter (Say 3.5 m^3)

Foam storage requirement w.r.t. two major fire contingencies simultaneously for terminal in addition to foam tank located near foam generator = 3.5 X 2 = 7 m^3

Total 2 no. of foam storage tank @ 7m^3

Annexure-VI

MAJOR CHANGES IN REVISED EDITION

OISD-STD-194 was taken for revision in the month of November-2022 through the functional committee (FC) constituted by OISD. Most of the FC meetings were held in physical mode only. Major changes being proposed by the committee are as mentioned below:

- Name of the standard has been revised to **"STANDARD FOR HANDLING, STORAGE AND REGASIFICATION OF LIQUEFIED NATURAL GAS (LNG)"**
- Scope has been modified to include safety requirements for small-scale LNG facilities.
  Safety requirements for small scale LNG facilities for unloading, storage & re-gasification are included as a separate chapter in this standard.
- Requirements for firefighting facilities at LNG terminals have been incorporated.
- Safety requirements for road transportation and LNG dispensation have been included.
- General safety guidelines for FSRU/ FSU facilities are included as Annexure-II.
- LNG loading facilities and transportation of LNG by means of trucks including ISO containers via. roadways, railways, seaways, including inland waterways etc. to the consumer's end is subject to regulatory approvals.
- Provision for unloading/ loading of other liquefied gases and petroleum products based on risk assessment is subject to regulatory approvals.

Annexure-VII

LIST OF PREVIOUS FUNCTIONAL COMMITTEE(S)

1st Functional Committee
(August 2000)
On
Standard For The Storage And Handling Of Liquefied Natural Gas (LNG)

LIST OF MEMBERS

S. No.	Name	Organisation	Position in Committee
1	Sh. R. Rajaraman	EIL	Leader
2	Sh. V.S. Sadana	OISD	Coordinator
3	Sh. R.K. Ghosh	ONGC	Member
4	Sh. N. Haran	BPCL	Member
5	Sh. M V R Someswarudu	GAIL	Member
6	Sh. Lakshman Venugopal	HPCL	Member
7	Sh. C. Chattopadhyay	IOCL	Member

2nd Functional Committee
(April 2016)
On
Storage And Handling of Liquefied Natural Gas (LNG)

LIST OF MEMBERS

S. No	Name	Organization	Status
1	Shri L K Vijh	Engineers India Limited.	Leader
2	Shri M M Ahuja	Petronet LNG Limited	Member
3	Shri Peeyush Nigam	Essar Oil Limited	Member
4	Shri S Kannan	Reliance Industries Limited	Member
5	Shri N Jagdeeshwar	GAIL (India) Limited	Member
6	Shri Sunil Gupta	Indian Oil Corp. Limited	Member
7	Shri Dhirendra Kumar	Shell Hazira LNG. Pvt. Limited	Member
8	Shri G G Tandon	BPCL Mumbai Refinery	Member
9	Shri Jomy Sebastian	BPCL Kochi Refinery	Member
10	Shri Y P Gulati	OISD	Member-Coordinator